Statistics for Applications

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Lecture 1: Introduction to Statistics

Statistics turns questions about a random phenomenon into questions about the parameters of a model, and answers them from data. This lecture contrasts statistics with probability, then sets up a first statistical experiment and its modeling assumptions.

1.1 Randomness, probability, and statistics

Three classical uses of statistics share one feature.

- *Hydrology.* Dikes in the Netherlands (10th century) must be high enough for most floods but not too expensive.
- *Insurance.* Given a driving record, car, and coverage, what is a fair premium?
- *Clinical trials.* A drug is tested on 100 patients; 56 are cured and 44 show no improvement. Is the drug effective?

The common feature is randomness, and the associated questions are about averages (a *fair* premium is an expected cost), quantifying chance (a dike higher than *most* floods), and significance and variability (are 56 cures more than chance?).

Probability studies randomness when the random process is completely known, as for dice, cards, roulette, or fair coins.

Example: *Two dice games.* Roll one fair die. Alice gets \$1 if the number of dots is at most 3, and Bob gets \$2 if it is at most 2. Their expected gains are

$$1 \cdot \tfrac{3}{6} = \tfrac{1}{2} \qquad \text{and} \qquad 2 \cdot \tfrac{2}{6} = \tfrac{2}{3},$$

so Bob's position is better. Now roll two dice, choose $z \in \{2, \dots, 12\}$, and win \$100 if the sum equals z. The expected gain is $100\,\mathbb{P}(X + Y = z)$, which is maximized by the most likely sum $z = 7$, with $\mathbb{P}(X + Y = 7) = 6/36$. Both answers follow from the known model: each face has probability $1/6$ and the dice are independent.

Statistics handles processes whose parameters are unknown and must be estimated from data. Sometimes the randomness is real (a randomly chosen student, a biased coin, measurement error). More often the phenomenon is deterministic but too complex, and statistical modeling writes

$$\text{complicated process “=” simple process + random noise.}$$

The noise collects everything we do not understand. Good modeling chooses a plausible simple

process *and* noise distribution, which requires domain knowledge: a model must keep the systematically important effects (sex, in a study of a breast cancer drug) and should depend on only a few parameters. A wrong model causes model error; accepting some model error is the price of extracting anything from data.

Probability goes from the truth to the data: given the distribution, it describes what samples will look like. Statistics goes backwards, from one observed data set to the distribution that generated it.

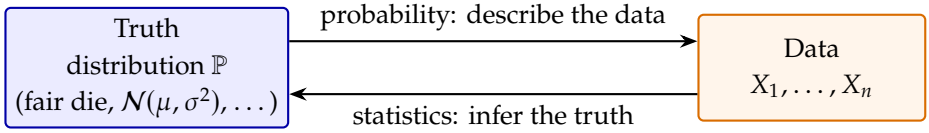


Figure 1: Probability and statistics run in opposite directions between the data-generating distribution and the observed data.

Example: *The same drug, two directions. Probability:* previous studies showed that the drug is 80% effective. For a study on 100 patients, on average 80 will be cured, and at least 65 will be cured with probability 99.99%. *Statistics:* we observe that 78 of 100 patients were cured. We can then be 95% confident that the drug is effective on between 69.88% and 86.11% of patients; these endpoints are $0.78 \pm 1.96\sqrt{0.78 \cdot 0.22/100}$.

The probabilistic statement is about all possible data sets drawn from a known model. The statistical one starts from a single data set, infers the model with error bars, and only then makes (less precise) predictions.

1.2 A first statistical experiment

A study in *Nature* (2003) claims that "a neonatal right-side preference makes a surprising romantic reappearance later in life." To make this a mathematical question, let p denote the proportion of couples that turn their head to the right when kissing. Observe n kissing couples, record 1 for RIGHT and 0 for LEFT, and estimate p by the proportion $\hat{p}$ of RIGHT. The study observed $n = 124$ couples, 80 of them turning right, so

$$\hat{p} = \frac{80}{124} = 64.5\%.$$

The data were collected from strangers in public places. Photographs from an image search would not do: they are mostly staged stock images and are not representative of the population of couples about which the claim is made.

Is 64.5% convincingly larger than 50%? With the data RIGHT, RIGHT, LEFT ($n = 3$) we would get 66.7%, which is even farther from 50% but clearly unconvincing. The sample size must be large enough, and how large depends on the unknown p. Half of 124 is 62; for this sample size, standard methods conclude a right-side preference from about 72 right turns at confidence 95%, 75 at 99%,

and 80 at 99.9%. Quantifying the accuracy of $\hat{p}$ requires a model.

Formally, for $i = 1, \ldots, n$ let $R_i = 1$ if the ith couple turns to the right and $R_i = 0$ otherwise. The estimator of p is the sample average

$$\hat{p} = \bar{R}_n = \frac{1}{n} \sum_{i=1}^{n} R_i.$$

For this question only the number of ones matters, not their order; the bar always denotes an average ($\bar{X}_n$, $\bar{Y}_n$, and so on).

Definition 1.1 [Estimator and estimate]. An estimator is a function of the random observations, and is therefore itself a random variable. An estimate is its realization: the number obtained by plugging in the observed data.

Here $\bar{R}_n$ is the estimator and 64.5% the estimate. Since an estimator is random, its accuracy has two aspects. Its distribution should be concentrated, so that a new sample gives a similar value; averaging only two couples would fluctuate much more than averaging 124. It should also concentrate near p: the constant estimator "72%, whatever the data" never fluctuates but is useless. The first aspect is measured by the variance, the second by the bias.

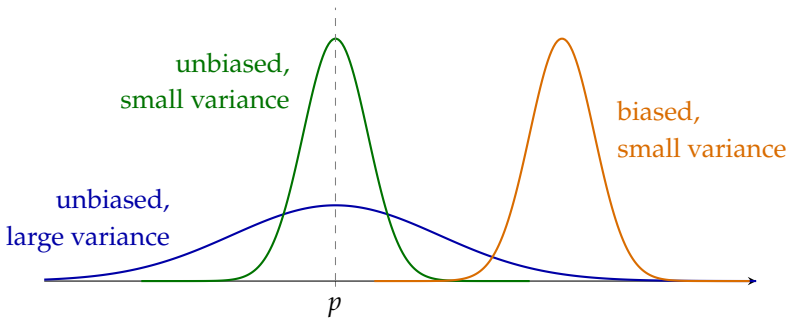


Figure 2: Schematic densities of three estimators of p. A good estimator has a narrow distribution centered near p.

1.3 Modeling assumptions

A model consists of assumptions on the observations $R_1, \ldots, R_n$ that let us draw conclusions:

1. each R_i is a random variable;
2. each R_i is Bernoulli with parameter p;
3. $R_1, \ldots, R_n$ are mutually independent.

Randomness models lack of information: with perfect information about the kissers (including what goes on in their minds), physics or sociology would predict the outcome. *Bernoulli* costs nothing: a

random variable with values in $\{0, 1\}$ is necessarily Bernoulli. The real assumption is the *common parameter*. Each couple could have its own $R_i \sim \text{Ber}(p_i)$, but then the only estimate of p_i would be R_i itself, equal to 0 or 1. Assuming $p_i = p$ for all i (a homogeneous population) is what allows pooling information across observations; the conclusion is then a statement about the population, not about any individual. *Independence* is reasonable because the couples were observed at different places and times and did not coordinate. Assumptions 2 and 3 together say that the R_i are I.I.D., independent and identically distributed.

1.4 Replacing expectations by averages

Example: *Exam grades.* Fifteen grades from a large probability class are modeled as $X_1, \dots, X_{15}$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$. This is a modeling convenience, not a physical law; it even gives positive, though negligible, probability to negative scores. Since $\mu = \mathbb{E}[X]$, a natural estimator of μ is the average score

$$\hat{\mu} = \bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

For the variance, $\sigma^2 = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big]$; replacing both expectations by averages gives

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} \big(X_i - \hat{\mu}\big)^2, \qquad \hat{\sigma} = \sqrt{\hat{\sigma}^2}.$$

Calculators divide by $n - 1$ instead of n to make $\hat{\sigma}^2$ unbiased.

This is the guiding principle of the course: *statistics replaces expectations by averages*, sometimes followed by further steps such as minimizing the resulting expression or plugging in other estimates.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 1, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 2: Introduction to Statistics (cont.)

The law of large numbers makes an average a consistent estimator, and the central limit theorem quantifies its fluctuations, which yields confidence intervals. Building intervals for functions of averages requires the modes of convergence, Slutsky's theorem, and the delta method.

2.1 The LLN, the CLT, and a first confidence interval

Let $X, X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$.

Theorem 2.1 [Laws of large numbers and central limit theorem]. *The weak and strong laws of large numbers state that*

$$\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow[n\to\infty]{\mathbb{P},\ a.s.} \mu.$$

The central limit theorem states that

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \xrightarrow[n\to\infty]{(d)} \mathcal{N}(0,1), \qquad \textit{equivalently} \qquad \sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow[n\to\infty]{(d)} \mathcal{N}(0,\sigma^2).$$

The normalization $\sqrt{n}/\sigma$ is one over the standard deviation $\sigma/\sqrt{n}$ of $\bar{X}_n$. In the kiss example of Lecture 1, the LLN gives $\bar{R}_n \to p$, so for large n the average is a good (consistent) estimator of p. The CLT refines this by quantifying how good it is. Let Φ be the CDF of $\mathcal{N}(0,1)$ and Φ_n the CDF of $\sqrt{n}(\bar{R}_n - p)/\sqrt{p(1-p)}$. The CLT says $\Phi_n(x) \approx \Phi(x)$ for large n, so for every $x > 0$,

$$\mathbb{P}\big[|\bar{R}_n - p| \geq x\big] \approx 2\left(1 - \Phi\left(\frac{x\sqrt{n}}{\sqrt{p(1-p)}}\right)\right).$$

For $\alpha \in (0,1)$ let $q_{\alpha/2}$ be the $(1-\alpha/2)$-quantile of $\mathcal{N}(0,1)$, so that $\mathbb{P}[|Z| > q_{\alpha/2}] = \alpha$ for $Z \sim \mathcal{N}(0,1)$. For example, $q_{0.025} \approx 1.96$.

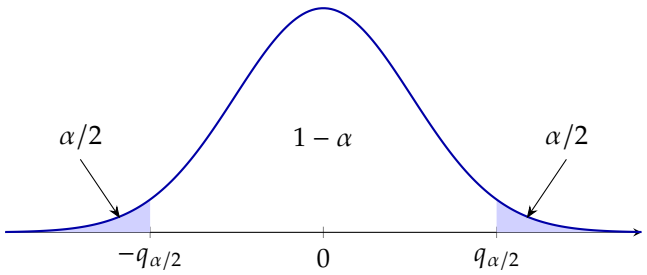


Figure 3: The standard normal density. Each shaded tail beyond $\pm q_{\alpha/2}$ has probability $\alpha/2$.

Then, with probability approximately $1-\alpha$ when n is large,

$$\bar{R}_n \in \left[p - \frac{q_{\alpha/2}\sqrt{p(1-p)}}{\sqrt{n}},\ p + \frac{q_{\alpha/2}\sqrt{p(1-p)}}{\sqrt{n}} \right].$$

The key point is that the limit $\mathcal{N}(0,1)$ of the standardized average does not depend on anything unknown. Such a distribution is called PIVOTAL: its quantiles can be read from a table or simulated. Had we not divided by $\sqrt{p(1-p)}$, the limiting variance would depend on p.

To turn this into a statement about p, we must solve for p. One could solve the quadratic inequality $n(\bar{R}_n - p)^2 \le q_{\alpha/2}^2\, p(1-p)$. A simpler route uses that $p(1-p) \le 1/4$ for every p. Then, roughly with probability at least $1-\alpha$, $\bar{R}_n \in [p - q_{\alpha/2}/(2\sqrt{n}),\ p + q_{\alpha/2}/(2\sqrt{n})]$. In other words, for large n the interval

$$\left[\bar{R}_n - \frac{q_{\alpha/2}}{2\sqrt{n}},\ \bar{R}_n + \frac{q_{\alpha/2}}{2\sqrt{n}} \right]$$

contains p with probability at least about $1-\alpha$. It is an ASYMPTOTIC CONFIDENCE INTERVAL for p.

Example: *Kisses.* With $\bar{R}_n = 0.645$, $n = 124$, and $\alpha = 5\%$,

$$\left[0.645 \pm \frac{1.96}{2\sqrt{124}} \right] = [0.56,\ 0.73],$$

which lies above 0.5. For the extreme case $n = 3$ with $\bar{R}_3 = 2/3$, the same formula gives $[0.10, 1.23]$, but there the CLT is not valid. For $n = 3$ one can compute exactly: $3\bar{R}_3 \sim \mathrm{Bin}(3, p)$, so $\bar{R}_3$ takes the values $0, \frac{1}{3}, \frac{2}{3}, 1$ with probabilities

$$(1-p)^3, \quad 3p(1-p)^2, \quad 3p^2(1-p), \quad p^3.$$

This distribution depends on p, so it is not pivotal; inverting it is possible but tedious.

2.2 Hoeffding's inequality

When n is not large, a non-asymptotic bound is available for bounded variables.

Theorem 2.2 [HOEFFDING'S INEQUALITY, I.I.D. CASE]. *Let $n \ge 1$ and let $X, X_1, \dots, X_n$ be i.i.d. with $X \in [a,b]$ almost surely, where $a < b$, and $\mu = \mathbb{E}[X]$. Then for all $\varepsilon > 0$,*

$$\mathbb{P}\left[|\bar{X}_n - \mu| \ge \varepsilon \right] \le 2e^{-\frac{2n\varepsilon^2}{(b-a)^2}}.$$

The tails of $\bar{X}_n$ are thus dominated by Gaussian-type tails, for every n and without any limit. For Bernoulli variables the sharpest interval is $[a,b] = [0,1]$, so $(b-a)^2 = 1$. Setting $2e^{-2n\varepsilon^2} = \alpha$ gives

$\varepsilon = \sqrt{\log(2/\alpha)/(2n)}$, and passing to the complement of the event bounded above,

$$\mathbb{P}\left[\bar{R}_n - \sqrt{\frac{\log(2/\alpha)}{2n}} \le p \le \bar{R}_n + \sqrt{\frac{\log(2/\alpha)}{2n}}\right] \ge 1 - \alpha$$

for every sample size n. Since $\sqrt{\log(2/\alpha)/(2n)} = \sqrt{2\log(2/\alpha)}/(2\sqrt{n})$, this interval replaces $q_{\alpha/2}$ in the CLT interval by $\sqrt{2\log(2/\alpha)}$, which is larger (about 2.72 versus 1.96 at $\alpha = 5\%$). Making fewer assumptions gives a wider, more conservative interval; asymptotics and independence are what allow sharper statements.

2.3 Modes of convergence

The CLT gives convergence in distribution, the weakest mode. It says nothing about the random variables themselves being close: if $X \sim \mathcal{N}(0,1)$, then X and $-X$ have the same distribution, although the realizations 1.2 and -1.2 are far apart. This weakness prevents combining limits, so the stronger modes are needed too.

Definition 2.3 [Modes of convergence]. Let $(T_n)_{n\ge1}$ be a sequence of random variables and T a random variable (possibly deterministic).

- Almost sure convergence: $T_n \xrightarrow{\text{a.s.}} T$ iff $\mathbb{P}\big[\{\omega : T_n(\omega) \to T(\omega)\}\big] = 1$.
- Convergence in probability: $T_n \xrightarrow{\mathbb{P}} T$ iff $\mathbb{P}[|T_n - T| \ge \varepsilon] \to 0$ for all $\varepsilon > 0$.
- Convergence in L^p ($p \ge 1$): $T_n \xrightarrow{L^p} T$ iff $\mathbb{E}[|T_n - T|^p] \to 0$.
- Convergence in distribution: $T_n \xrightarrow{(d)} T$ iff $\mathbb{P}[T_n \le x] \to \mathbb{P}[T \le x]$ for all $x \in \mathbb{R}$ at which the CDF of T is continuous.

These definitions extend to random vectors in $\mathbb{R}^d$, with norms in place of absolute values.

Here ω is the underlying random outcome and a random variable is a (measurable) function of it. Almost sure convergence holds for almost every realization; convergence in probability allows, for each n, a small probability of being far from the limit, which must vanish as $n \to \infty$. A mean-zero sequence whose variance tends to 0 converges to 0 in L^2.

Proposition 2.4 [Characterizations of convergence in distribution]. *The following are equivalent:*

(i) $T_n \xrightarrow{(d)} T$;

(ii) $\mathbb{E}[f(T_n)] \to \mathbb{E}[f(T)]$ *for every continuous and bounded* f;

(iii) $\mathbb{E}[e^{ixT_n}] \to \mathbb{E}[e^{ixT}]$ *for all* $x \in \mathbb{R}$.

The expectation in (iii) is the CHARACTERISTIC FUNCTION. The standard proof of the CLT uses (iii): the characteristic function of $\mathcal{N}(0,1)$ is $e^{-x^2/2}$, and this is what the second-order Taylor expansion of the characteristic function of a standardized average produces in the limit.

Proposition 2.5 [RELATIONS BETWEEN MODES].

- *If T_n converges a.s., it converges in probability, and the limits are equal a.s.*
- *If T_n converges in L^p, it converges in L^q for all $q \le p$ and in probability, and the limits are equal a.s.*
- *If T_n converges in probability, it converges in distribution.*
- Continuous mapping: *if f is continuous and $T_n \to T$ a.s., in probability, or in distribution, then $f(T_n) \to f(T)$ in the same mode.*

Convergence in distribution has no "limits are equal" clause: it describes only the limiting distribution, and any random variable with that distribution (for example X or $-X$) is a valid limit.

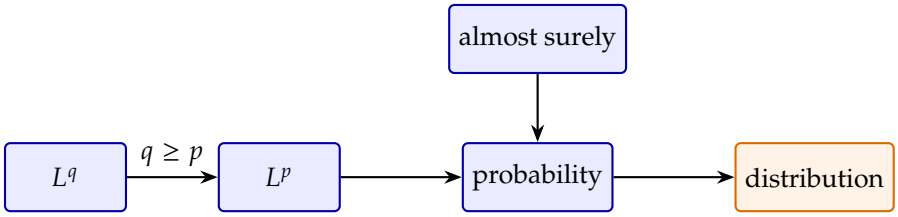


Figure 4: Implications between modes of convergence. Every mode implies convergence in distribution.

Limits and operations. If $U_n \to U$ and $V_n \to V$, both a.s. or both in probability, then in the same mode $U_n + V_n \to U + V$, $U_n V_n \to UV$, and, if $V \neq 0$ a.s., $U_n/V_n \to U/V$. The ratio follows from the product and continuous mapping with $x \mapsto 1/x$. In general these rules fail for convergence in distribution unless the pair (U_n, V_n) converges in distribution to (U, V). For instance, with $Z \sim \mathcal{N}(0,1)$, both $U_n = Z$ and $V_n = -Z$ converge in distribution to $\mathcal{N}(0,1)$, but $U_n + V_n = 0$, which does not have the distribution $\mathcal{N}(0,2)$ of a sum of two independent standard Gaussians.

Theorem 2.6 [SLUTSKY'S THEOREM]. *Let (X_n) and (Y_n) be sequences of random variables such that $X_n \xrightarrow{(d)} X$ and $Y_n \xrightarrow{\mathbb{P}} c$, where X is a random variable and c is a given real number. Then $(X_n, Y_n) \xrightarrow{(d)} (X, c)$. In particular*

$$X_n + Y_n \xrightarrow{(d)} X + c, \qquad X_n Y_n \xrightarrow{(d)} cX,$$

and, by continuous mapping, any continuous combination of X_n and Y_n converges in distribution to the same combination of X and c.

The limit of Y_n must be a deterministic constant; this is what makes the combination possible.

2.4 Example: inter-arrival times

We observe the times $T_1, \dots, T_n$ between arrivals of trains at a station and assume they are mutually independent exponential random variables with a common parameter $\lambda > 0$. The goal is to estimate λ.

Independence is plausible but not completely justified, as is often the case. The exponential family has the LACK OF MEMORY property

$$\mathbb{P}[T_1 > t + s \mid T_1 > t] = \mathbb{P}[T_1 > s] \qquad \text{for all } s, t \geq 0,$$

which is why it is widely used for waiting and service times, and $T_i > 0$ almost surely. A common parameter means the same average inter-arrival time, which is true only over a limited period (rush hour differs from 11pm).

The density of T_1 is $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$, with $\mathbb{E}[T_1] = 1/\lambda$ and $\mathrm{Var}(T_1) = 1/\lambda^2$. Replacing the expectation by an average, $\bar{T}_n = \frac{1}{n}\sum_i T_i$ is a natural estimator of $1/\lambda$, and

$$\hat{\lambda} := \frac{1}{\bar{T}_n}$$

is a natural estimator of λ. By the LLN, $\bar{T}_n \to 1/\lambda$ a.s. and in probability, so by continuous mapping (with $x \mapsto 1/x$, continuous away from 0) $\hat{\lambda} \to \lambda$ a.s. and in probability. By the CLT,

$$\sqrt{n}\left(\bar{T}_n - \frac{1}{\lambda}\right) \xrightarrow{(d)} \mathcal{N}(0, \lambda^{-2}).$$

Multiplying by λ, with probability approaching $1 - \alpha$ we have $|\sqrt{n}(\lambda \bar{T}_n - 1)| \leq q_{\alpha/2}$, that is,

$$\lambda \in \left[\frac{1 - q_{\alpha/2}/\sqrt{n}}{\bar{T}_n},\ \frac{1 + q_{\alpha/2}/\sqrt{n}}{\bar{T}_n}\right] = \left[\hat{\lambda} - \frac{q_{\alpha/2}\hat{\lambda}}{\sqrt{n}},\ \hat{\lambda} + \frac{q_{\alpha/2}\hat{\lambda}}{\sqrt{n}}\right]. \tag{2.1}$$

The width depends on $\bar{T}_n$ itself, since the average carries information about the variance as well as the mean. The CLT here concerns $\bar{T}_n$, not $\hat{\lambda}$, which is not an average. The delta method transfers it.

2.5 The delta method

Theorem 2.7 [DELTA METHOD]. *Let $(Z_n)_{n\geq 1}$ satisfy $\sqrt{n}(Z_n - \theta) \xrightarrow{(d)} \mathcal{N}(0, \sigma^2)$ for some $\theta \in \mathbb{R}$ and $\sigma^2 > 0$ (the sequence is ASYMPTOTICALLY NORMAL around θ). Let $g : \mathbb{R} \to \mathbb{R}$ be continuously differentiable at θ. Then $(g(Z_n))$ is also asymptotically normal, and*

$$\sqrt{n}\big(g(Z_n) - g(\theta)\big) \xrightarrow{(d)} \mathcal{N}\big(0,\ g'(\theta)^2\sigma^2\big).$$

Proof. By a first-order Taylor expansion (the mean value theorem), there is a point $\bar{\theta}_n$ between Z_n and θ with

$$\sqrt{n}\big(g(Z_n) - g(\theta)\big) = \sqrt{n}(Z_n - \theta)\, g'(\bar{\theta}_n).$$

The first factor converges in distribution to $\mathcal{N}(0, \sigma^2)$. It also forces $Z_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(Z_n - \theta) \to 0$ in probability, by Slutsky's theorem. Since $\bar{\theta}_n$ is sandwiched between Z_n and θ, $\bar{\theta}_n \to \theta$ in probability, and by continuity of g' at θ, $g'(\bar{\theta}_n) \to g'(\theta)$ in probability. The limit $g'(\theta)$ is a constant, so Slutsky's theorem gives convergence of the product to $g'(\theta)\,\mathcal{N}(0, \sigma^2) = \mathcal{N}(0, g'(\theta)^2\sigma^2)$. □

For the inter-arrival times, apply it with $Z_n = \bar{T}_n$, $\theta = 1/\lambda$, $\sigma^2 = \lambda^{-2}$, and $g(x) = 1/x$, so $g'(x) = -1/x^2$ and $g'(1/\lambda)^2\sigma^2 = \lambda^4 \cdot \lambda^{-2}$:

$$\sqrt{n}\big(\hat{\lambda} - \lambda\big) \xrightarrow{(d)} \mathcal{N}(0, \lambda^2).$$

Hence, for large n, $|\hat{\lambda} - \lambda| \le q_{\alpha/2}\lambda/\sqrt{n}$ with probability about $1 - \alpha$. The interval $[\hat{\lambda} \pm q_{\alpha/2}\lambda/\sqrt{n}]$ cannot serve as a confidence interval, because it depends on λ. There are two ways out.

Solve for λ. The inequality is equivalent to $\lambda(1 - q_{\alpha/2}/\sqrt{n}) \le \hat{\lambda} \le \lambda(1 + q_{\alpha/2}/\sqrt{n})$, that is,

$$\lambda \in \left[\hat{\lambda}\left(1 + \frac{q_{\alpha/2}}{\sqrt{n}}\right)^{-1},\ \hat{\lambda}\left(1 - \frac{q_{\alpha/2}}{\sqrt{n}}\right)^{-1}\right],$$

an asymptotic confidence interval for λ.

Slutsky. The delta method gives $\sqrt{n}(\hat{\lambda} - \lambda)/\lambda \xrightarrow{(d)} \mathcal{N}(0, 1)$, and $\hat{\lambda} \xrightarrow{\mathbb{P}} \lambda$, so $\lambda/\hat{\lambda} \xrightarrow{\mathbb{P}} 1$. Writing

$$\sqrt{n}\,\frac{\hat{\lambda} - \lambda}{\hat{\lambda}} = \sqrt{n}\,\frac{\hat{\lambda} - \lambda}{\lambda} \cdot \frac{\lambda}{\hat{\lambda}},$$

Slutsky's theorem shows that this also converges to $\mathcal{N}(0, 1)$. Hence $[\hat{\lambda} - q_{\alpha/2}\hat{\lambda}/\sqrt{n},\ \hat{\lambda} + q_{\alpha/2}\hat{\lambda}/\sqrt{n}]$ is another asymptotic confidence interval for λ; it coincides with the direct interval (2.1). In general, an unknown parameter in the normalization may be replaced by a consistent estimator, because the ratio of the two tends to 1 in probability.

Example: *Kisses revisited.* The bound $p(1 - p) \le 1/4$ was a problem-specific trick; no such bound exists for $\lambda \in (0, \infty)$. By Slutsky's theorem one may instead plug in $\bar{R}_n$ for p, obtaining the asymptotic confidence interval

$$\left[\bar{R}_n - \frac{q_{\alpha/2}\sqrt{\bar{R}_n(1 - \bar{R}_n)}}{\sqrt{n}},\ \bar{R}_n + \frac{q_{\alpha/2}\sqrt{\bar{R}_n(1 - \bar{R}_n)}}{\sqrt{n}}\right].$$

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 2, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 3: Parametric Inference

A statistical model restricts the distribution of the data to a family indexed by a parameter. This lecture formalizes models and identifiability, and defines estimators, consistency, bias, and quadratic risk.

3.1 The rationale behind statistical modeling

Let $X_1, \dots, X_n$ be n independent copies of X. The goal of statistics is to learn the distribution of X; identical distribution is what makes averaging over the sample informative. If $X \in \{0, 1\}$ this is easy: $X \sim \text{Ber}(p)$, and only p must be learned.

Example: *Siblings.* A partial data set of the number of siblings (including self) reported by college students reads $2, 3, 2, 4, 1, 3, 1, 1, 1, 1, 1, 2, 2, 3, 2, 2, 2, 3, 2, 1, 3, 1, 2, 3, \dots$ With no assumption beyond grouping large values, we must learn the PMF

x	1	2	3	4	5	6	≥ 7
$\mathbb{P}(X = x)$	p_1	p_2	p_3	p_4	p_5	p_6	$\sum_{i \geq 7} p_i$

which has 7 parameters. The empirical proportions converge to them by the LLN, but with only a couple dozen observations several are poorly estimated: no value above 4 appears, so $\hat{p}_5 = \hat{p}_6 = \hat{p}_{\geq 7} = 0$, although the corresponding population proportions are surely positive. Assuming instead $X \sim \text{Poiss}(\lambda)$, with $\mathbb{P}(X = x) = e^{-\lambda}\lambda^x / x!$ for $x = 0, 1, 2, \dots$, leaves one parameter to learn, and the fitted PMF is positive at every value. The model uses the frequent small values to inform the rare large ones.

Each assumption (a density, a smooth density, a piecewise constant density, a parametric family) reduces the space of candidate distributions. Modeling seeks a space small enough for averaging to be effective, yet large enough to contain a distribution close to the truth.

3.2 Statistical models

Definition 3.1 [Statistical model]. Let the observed outcome of a statistical experiment be a sample $X_1, \dots, X_n$ of i.i.d. random variables in a measurable space E (usually $E \subseteq \mathbb{R}$), with common distribution $\mathbb{P}$. A statistical model associated to the experiment is a pair

$$\big(E, (\mathbb{P}_\theta)_{\theta \in \Theta}\big),$$

where E is the sample space, $(\mathbb{P}_\theta)_{\theta \in \Theta}$ is a family of probability measures on E, and Θ is any set, called the parameter set.

The sample space should be the smallest set in which a generic observation X takes its values. Each θ determines the distribution completely.

Definition 3.2 [Well specified, true parameter, parametric**].** The model is well specified if $\mathbb{P} = \mathbb{P}_\theta$ for some $\theta \in \Theta$. This θ is the true parameter, often written θ^*; it is unknown. The model is parametric if $\Theta \subseteq \mathbb{R}^d$ for some $d \geq 1$.

Well-specification is a strong assumption ("all models are wrong, but some are useful"). The aim is to estimate θ, which then gives all probabilities of the data, or to check properties of θ that have a meaning of their own, such as $\theta > 2$ or $\theta \neq 1/2$. In regression, for instance, a coefficient measures the sensitivity of a response to an input, and whether it is zero decides whether that input matters. A nonparametric model has infinitely many parameters: the siblings PMF without the cap at 7, or the family of all distributions with a density f, where the unknown is a function (histograms are the basic nonparametric tool). In this course $\Theta \subseteq \mathbb{R}^d$, and most often $d = 1$.

Example: *Statistical models.*

1. n Bernoulli trials: $\big(\{0,1\}, (\mathrm{Ber}(p))_{p\in(0,1)}\big)$.
2. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{Exp}(\lambda)$, $\lambda > 0$ unknown: $\big((0,\infty), (\mathrm{Exp}(\lambda))_{\lambda>0}\big)$.
3. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{Poiss}(\lambda)$, $\lambda > 0$ unknown: $\big(\mathbb{N}, (\mathrm{Poiss}(\lambda))_{\lambda>0}\big)$, where $\mathbb{N}$ includes 0.
4. Siblings capped at 7: $E = \{1, \ldots, 7\}$, $\mathbb{P}_p(X = k) = p_k$, with $\Theta = \{p \in \mathbb{R}^7 : p_k \geq 0,\ \sum_k p_k = 1\}$. Named families such as Poisson are shorthand for such explicit descriptions.
5. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ unknown: $\big(\mathbb{R}, (\mathcal{N}(\mu, \sigma^2))_{(\mu,\sigma^2)\in\mathbb{R}\times(0,\infty)}\big)$.
6. Gaussian with known variance, as for a measuring device whose error σ^2 is calibrated by the manufacturer: $\big(\mathbb{R}, (\mathcal{N}(\mu, \sigma^2))_{\mu\in\mathbb{R}}\big)$ with σ^2 fixed.
7. Censored data. Only $Y = \mathbf{1}\{X > 5\}$ is observed, where $X \sim \mathrm{Exp}(\lambda)$ is a waiting time; the indicator $\mathbf{1}\{A\}$ equals 1 if A is true and 0 otherwise. Then Y is Bernoulli with

$$\mathbb{P}(Y = 1) = \int_5^\infty \lambda e^{-\lambda x}\, dx = e^{-5\lambda},$$

   and parametrizing by the quantity of interest gives $\big(\{0,1\}, (\mathrm{Ber}(e^{-5\lambda}))_{\lambda>0}\big)$. Censoring is common in surveys of income and in survival analysis.
8. $X \sim \mathrm{Unif}[0, \theta]$, $\theta > 0$: $\big([0,\infty), (\mathrm{Unif}[0,\theta])_{\theta>0}\big)$. Writing $E = [0, \theta]$ makes no sense, because θ is not a fixed number but ranges over the whole family. If $\theta \leq 1$ is known in advance, $E = [0, 1]$ and $\Theta = (0, 1]$.

3.3 Identification

Even infinitely many observations determine at best the distribution $\mathbb{P}_\theta$. If two parameters give the same distribution, the data cannot tell them apart.

Definition 3.3 [Identified parameter]. The parameter θ is identified if the map $\theta \in \Theta \mapsto \mathbb{P}_\theta$ is injective, that is, $\theta \neq \theta' \Rightarrow \mathbb{P}_\theta \neq \mathbb{P}_{\theta'}$; equivalently, $\mathbb{P}_\theta = \mathbb{P}_{\theta'} \Rightarrow \theta = \theta'$.

The parameters in the Bernoulli, exponential, Poisson, and Gaussian examples above are identified.

Example: *A sign of a Gaussian.* Let $X_i = \mathbf{1}\{Y_i \geq 0\}$, where $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ are unobserved, with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ unknown. Then X_i is Bernoulli with parameter

$$\mathbb{P}(Y_i \geq 0) = \mathbb{P}\left(\frac{Y_i - \mu}{\sigma} \geq -\frac{\mu}{\sigma}\right) = 1 - \Phi\left(-\frac{\mu}{\sigma}\right) = \Phi\left(\frac{\mu}{\sigma}\right),$$

where Φ is the standard normal CDF. This depends only on μ/σ: replacing (μ, σ) by $(c\mu, c\sigma)$ for any $c > 0$ leaves the distribution of X_i unchanged. So μ and σ^2 are not identified, but $\theta = \mu/\sigma$ is, and an identified model is $\big(\{0,1\}, (\mathrm{Ber}(\Phi(\theta)))_{\theta \in \mathbb{R}}\big)$. The normalized mean μ/σ compares the mean with the size of the fluctuations.

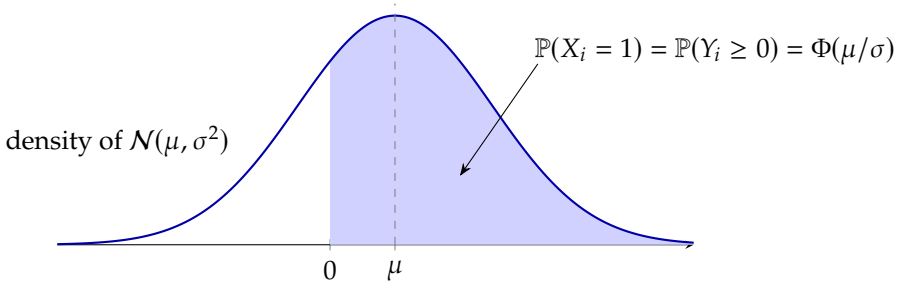


Figure 5: The shaded area under the $\mathcal{N}(\mu, \sigma^2)$ density is the Bernoulli parameter $\Phi(\mu/\sigma)$; it is unchanged when μ and σ are scaled together. Illustrated with $\mu = 1$, $\sigma = 1.5$.

3.4 Estimators and consistency

Given a sample and a model, we want to estimate θ, and we need to measure how well an estimator does compared with simply guessing.

Definition 3.4 [Statistic, estimator, consistency]. A statistic is any measurable function of the sample, for example $\bar{X}_n$, $\max_i X_i$, $X_1 + \log(1 + |X_n|)$, or the sample variance. An estimator of θ is a statistic whose expression does not depend on θ. An estimator $\hat{\theta}_n$ of θ is weakly (resp. strongly) consistent if $\hat{\theta}_n \to \theta$ in probability (resp. almost surely) under $\mathbb{P}_\theta$, as $n \to \infty$.

Rule of thumb: if a quantity can be computed exactly once the data are given, it is measurable. Problems can arise only for implicitly defined quantities, such as a supremum or infimum over a

large class of functions that need not be attained.

3.5 Bias and quadratic risk

Definition 3.5 [Bias and risk]. The bias of an estimator $\hat{\theta}_n$ of θ is $\mathbb{E}[\hat{\theta}_n] - \theta$; the estimator is unbiased if its bias is 0. Its quadratic risk is $\mathbb{E}\left[|\hat{\theta}_n - \theta|^2\right]$.

Unbiasedness means that over many repetitions of the experiment, the estimates would average to θ. It ignores fluctuations: by linearity and identical distribution,

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mathbb{E}[X_1],$$

so $\bar{X}_n$ and the single observation X_1 have the same bias as estimators of $\mathbb{E}[X]$ (in the kiss example, both are unbiased for p).

The risk does account for fluctuations. If it tends to 0, then $\hat{\theta}_n \to \theta$ in L^2, hence in probability, so $\hat{\theta}_n$ is weakly consistent. Thus $\bar{X}_n$ is consistent (by the LLN; strongly so by the strong law), while X_1 is unbiased but not consistent, since it ignores all further data.

Proposition 3.6 [Bias-variance decomposition]. *If $\Theta \subseteq \mathbb{R}$, then*

$$\mathbb{E}\left[(\hat{\theta}_n - \theta)^2\right] = \underbrace{\left(\mathbb{E}[\hat{\theta}_n] - \theta\right)^2}_{bias^2} + \underbrace{\mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2\right]}_{\mathrm{Var}(\hat{\theta}_n)}.$$

Proof. Insert $\mathbb{E}[\hat{\theta}_n]$ and expand the square:

$$\begin{aligned}\mathbb{E}\left[(\hat{\theta}_n - \theta)^2\right] &= \mathbb{E}\left[\left((\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n]) + (\mathbb{E}[\hat{\theta}_n] - \theta)\right)^2\right] \\ &= \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2\right] + 2\left(\mathbb{E}[\hat{\theta}_n] - \theta\right)\mathbb{E}\left[\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n]\right] + \left(\mathbb{E}[\hat{\theta}_n] - \theta\right)^2.\end{aligned}$$

The factor $\mathbb{E}[\hat{\theta}_n] - \theta$ is deterministic, so it comes out of the expectation, and the cross term vanishes because $\mathbb{E}\left[\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n]\right] = 0$. □

The bias is squared so that both terms have the same units. Both terms are nonnegative, so the risk tends to 0 exactly when both the bias and the variance do: the estimator must stop fluctuating *and* be on target. The constant estimator $\hat{\theta} = 4$ has variance 0, but its bias $4 - \theta$ is large when θ is far from 4.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 3, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 4: Parametric Inference (cont.) and Maximum Likelihood Estimation

Risk functions compare estimators, and confidence intervals are defined formally. To estimate a whole distribution, we then measure distances between distributions: the total variation distance is the natural one, but the Kullback–Leibler divergence is the one we can estimate.

4.1 Comparing estimators by their risk

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Ber}(\theta)$ and consider three estimators of θ.

- $\hat{\theta} = \bar{X}_n$. By linearity $\mathbb{E}[\bar{X}_n] = \theta$, so it is unbiased. Pulling the constant out of the variance (with a square) and using independence,

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^{n} \text{Var}(X_i) = \frac{\theta(1-\theta)}{n}.$$

  Its risk is therefore $\theta(1-\theta)/n$.

- $\hat{\theta} = 0.5$, a guess at no cost. Its bias is $0.5 - \theta$, its variance is 0, and its risk is $(0.5-\theta)^2$.

- $\hat{\theta} = X_1$, the first observation only. This is $\bar{X}_n$ with $n = 1$: bias 0, variance and risk $\theta(1-\theta)$.

X_1 and $\bar{X}_n$ have the same bias, and $\bar{X}_n$ has n times smaller variance, so $\bar{X}_n$ is always better than X_1. The guess 0.5 is not dominated: its variance is always smaller, and its bias is small when θ is near $1/2$ (it is exact at $\theta = 1/2$).

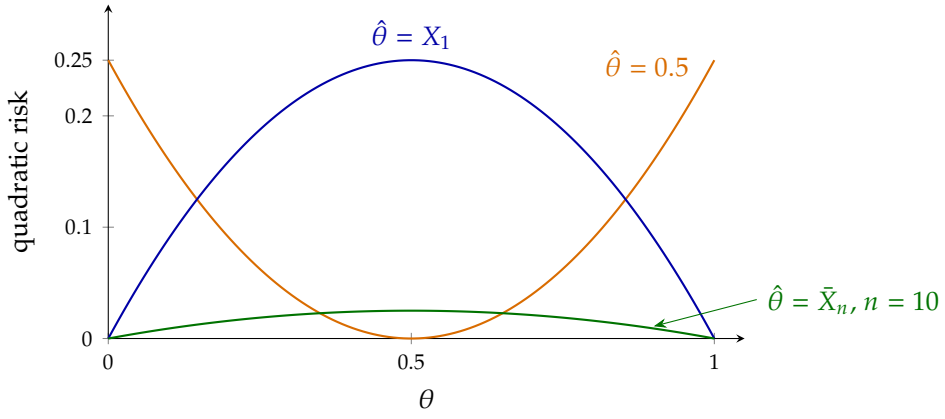


Figure 6: Quadratic risk as a function of the true θ for three estimators of a Bernoulli parameter, with $n = 10$ for $\bar{X}_n$.

The risk is a function of θ, so different estimators win for different values of θ. As n grows, the risk curve of $\bar{X}_n$ is crushed toward 0, and the interval of θ around $1/2$ where the guess wins shrinks. For an unbiased estimator the risk is the variance; in general it is the expected squared fluctuation around θ rather than around $\mathbb{E}[\hat{\theta}]$. The bias-variance trade-off is most visible with multivariate parameters: forcing the coordinates of an estimated mean vector to share information (for instance, to be equal) adds bias but averages more and lowers the variance. It is pervasive in nonparametric estimation.

4.2 Confidence intervals

Definition 4.1 [Confidence interval]. Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a statistical model based on observations $X_1, \dots, X_n$, with $\Theta \subseteq \mathbb{R}$, and let $\alpha \in (0, 1)$.

- A confidence interval of level $1 - \alpha$ for θ is any random interval $\mathcal{I}$ (depending on $X_1, \dots, X_n$) whose boundaries do not depend on θ and such that

$$\mathbb{P}_\theta[\mathcal{I} \ni \theta] \geq 1 - \alpha \qquad \text{for all } \theta \in \Theta.$$

- A confidence interval of asymptotic level $1 - \alpha$ for θ is any random interval $\mathcal{I}$ whose boundaries do not depend on θ and such that

$$\lim_{n\to\infty} \mathbb{P}_\theta[\mathcal{I} \ni \theta] \geq 1 - \alpha \qquad \text{for all } \theta \in \Theta.$$

The notation $\mathcal{I} \ni \theta$ stresses that the interval is random and θ is fixed. The requirement must hold for every θ, since the true value is unknown, and the boundaries must be computable from the data alone, just as an estimator may not depend on θ. Once the observed data are plugged in, the interval is fixed (for example $[0.42, 0.65]$) and contains θ or not: the probability is then 0 or 1, not $1 - \alpha$. Intervals built from the CLT have asymptotic level; intervals built from Hoeffding's inequality have level $1 - \alpha$ for every n.

For $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Ber}(p)$, $p \in (0, 1)$, the sample average is strongly consistent, unbiased, and has risk $p(1 - p)/n$. With $q_{\alpha/2}$ the $(1 - \alpha/2)$-quantile of $\mathcal{N}(0, 1)$, the CLT gives

$$\mathcal{I} = \left[\bar{X}_n - \frac{q_{\alpha/2}\sqrt{p(1-p)}}{\sqrt{n}},\ \bar{X}_n + \frac{q_{\alpha/2}\sqrt{p(1-p)}}{\sqrt{n}}\right], \qquad \lim_{n\to\infty} \mathbb{P}_p[\mathcal{I} \ni p] = 1 - \alpha \quad \text{for all } p \in (0, 1).$$

But $\mathcal{I}$ depends on p. Two solutions remove the dependence:

- replace $p(1 - p)$ by its upper bound $1/4$; enlarging the interval can only increase its coverage;
- replace p by $\bar{X}_n$ and use Slutsky's theorem.

The first interval is always wider, since $\bar{X}_n(1 - \bar{X}_n) \leq 1/4$ as well; it loses only through the CLT

approximation, while the plug-in interval relies on asymptotics a second time. Hoeffding's interval is wider still but needs no asymptotics. The choice depends on how critical it is that the error bars be valid. For the plug-in interval, the effect of replacing p by $\bar{X}_n$ is governed by the slope of $x \mapsto 1/\sqrt{x(1-x)}$ near the true p.

4.3 Total variation distance

Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a statistical model for an i.i.d. sample $X_1, \dots, X_n$, with true parameter θ^*, so $X_1 \sim \mathbb{P}_{\theta^*}$. The real goal is the distribution: find an estimator $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ such that $\mathbb{P}_{\hat{\theta}}$ is close to $\mathbb{P}_{\theta^*}$, meaning that $|\mathbb{P}_{\hat{\theta}}(A) - \mathbb{P}_{\theta^*}(A)|$ is small for *all* events $A \subset E$ simultaneously.

Definition 4.2 [Total variation distance]. The total variation distance between $\mathbb{P}_\theta$ and $\mathbb{P}_{\theta'}$ is

$$\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = \max_{A\subset E} \big|\mathbb{P}_\theta(A) - \mathbb{P}_{\theta'}(A)\big|.$$

If $\mathrm{TV}(\mathbb{P}_{\hat{\theta}}, \mathbb{P}_{\theta^*}) \le 0.01$, then every probability computed under $\mathbb{P}_{\hat{\theta}}$ lies within 0.01 of the corresponding true probability. The maximum over all events is not directly computable, but it reduces to the PMF or density. If E is discrete (Bernoulli, binomial, Poisson, ...), X has a PMF $p_\theta(x) = \mathbb{P}_\theta(X = x)$ with $p_\theta \ge 0$ and $\sum_{x\in E} p_\theta(x) = 1$. If E is continuous (Gaussian, exponential, ...) and X has a density, $\mathbb{P}_\theta(X \in A) = \int_A f_\theta(x)\,dx$ with $f_\theta \ge 0$ and $\int_E f_\theta = 1$.

Proposition 4.3 [TV from PMFs and densities].

$$\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = \frac{1}{2}\sum_{x\in E}\big|p_\theta(x) - p_{\theta'}(x)\big| \quad \text{or} \quad \mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = \frac{1}{2}\int_E \big|f_\theta(x) - f_{\theta'}(x)\big|\,dx.$$

Proof. We treat densities; for PMFs replace integrals by sums. Write $f = f_\theta$, $g = f_{\theta'}$, and $A^* = \{x : f(x) > g(x)\}$. Since $\int_E (f - g) = 1 - 1 = 0$, we have $\int_{(A^*)^c}(g - f) = \int_{A^*}(f - g)$, so

$$\frac{1}{2}\int_E |f - g| = \frac{1}{2}\int_{A^*}(f - g) + \frac{1}{2}\int_{(A^*)^c}(g - f) = \int_{A^*}(f - g) = \mathbb{P}_\theta(A^*) - \mathbb{P}_{\theta'}(A^*).$$

This is at most the maximum over A. Conversely, for any event A,

$$\mathbb{P}_\theta(A) - \mathbb{P}_{\theta'}(A) = \int_A (f - g) \le \int_{A\cap A^*}(f - g) \le \int_{A^*}(f - g),$$

because $f - g \le 0$ off A^* and $f - g > 0$ on A^*. The same argument with f and g exchanged bounds $\mathbb{P}_{\theta'}(A) - \mathbb{P}_\theta(A)$ by $\int_{(A^*)^c}(g - f) = \int_{A^*}(f - g)$. Hence the maximum is attained at A^* and equals $\frac{1}{2}\int_E |f - g|$. □

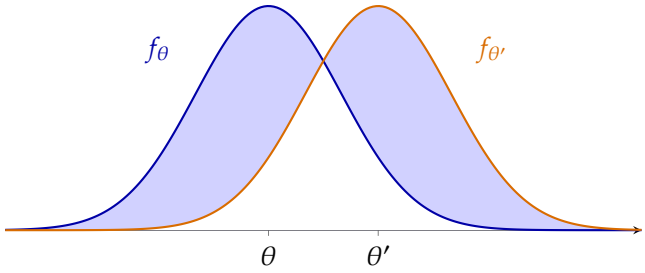


Figure 7: The shaded area between two densities is $\int |f_\theta - f_{\theta'}| = 2\,\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'})$. Illustrated with two unit-variance Gaussians.

Proposition 4.4 [TV is a distance]**.**

- $\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = \mathrm{TV}(\mathbb{P}_{\theta'}, \mathbb{P}_\theta)$ *(symmetric);*
- $\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) \geq 0$*;*
- *if* $\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = 0$*, then* $\mathbb{P}_\theta = \mathbb{P}_{\theta'}$ *(definite);*
- $\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) \leq \mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta''}) + \mathrm{TV}(\mathbb{P}_{\theta''}, \mathbb{P}_{\theta'})$ *(triangle inequality).*

Proof. Symmetry and nonnegativity are clear from either formula. If the maximum of the nonnegative numbers $|\mathbb{P}_\theta(A) - \mathbb{P}_{\theta'}(A)|$ is 0, each of them is 0, so the two measures agree on every event. The triangle inequality follows by integrating $|f_\theta - f_{\theta'}| \leq |f_\theta - f_{\theta''}| + |f_{\theta''} - f_{\theta'}|$. □

This suggests an estimation strategy: build an estimator $\widehat{\mathrm{TV}}(\mathbb{P}_\theta, \mathbb{P}_{\theta^*})$ for every $\theta \in \Theta$, and take $\hat{\theta}$ minimizing $\theta \mapsto \widehat{\mathrm{TV}}(\mathbb{P}_\theta, \mathbb{P}_{\theta^*})$. The true function is minimized, with value 0, at θ^*, so if the estimate is uniformly close, its minimizer should be close to θ^*. The problem is that it is unclear how to build $\widehat{\mathrm{TV}}$: estimators usually replace expectations by averages, and TV is not an expectation under $\mathbb{P}_{\theta^*}$ of any computable quantity.

4.4 Kullback–Leibler divergence

Among the many other ways to compare distributions, choose a more convenient one.

Definition 4.5 [Kullback–Leibler divergence]**.** The Kullback–Leibler (KL) divergence between $\mathbb{P}_\theta$ and $\mathbb{P}_{\theta'}$ is

$$\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = \begin{cases} \displaystyle\sum_{x \in E} p_\theta(x) \log \frac{p_\theta(x)}{p_{\theta'}(x)} & \text{if } E \text{ is discrete,} \\ \displaystyle\int_E f_\theta(x) \log \frac{f_\theta(x)}{f_{\theta'}(x)}\, dx & \text{if } E \text{ is continuous.} \end{cases}$$

The order of the arguments matters. KL lacks the direct meaning of TV, but it is much easier to manipulate; for example, for unit-variance Gaussians, $\mathrm{KL}(\mathcal{N}(\theta, 1), \mathcal{N}(\theta', 1)) = (\theta - \theta')^2/2$.

Proposition 4.6 [Properties of the KL divergence].

- $\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) \neq \mathrm{KL}(\mathbb{P}_{\theta'}, \mathbb{P}_\theta)$ *in general;*
- $\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) \geq 0$*;*
- *if* $\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = 0$*, then* $\mathbb{P}_\theta = \mathbb{P}_{\theta'}$ *(definite);*
- $\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) \not\leq \mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta''}) + \mathrm{KL}(\mathbb{P}_{\theta''}, \mathbb{P}_{\theta'})$ *in general.*

Proof of nonnegativity and definiteness. In the discrete case, with sums over the support of p_θ, Jensen's inequality for the concave function log gives

$$-\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P}_{\theta'}) = \sum_x p_\theta(x) \log \frac{p_{\theta'}(x)}{p_\theta(x)} \leq \log \sum_x p_\theta(x) \frac{p_{\theta'}(x)}{p_\theta(x)} = \log \sum_x p_{\theta'}(x) \leq \log 1 = 0.$$

Since log is strictly concave, equality forces the ratio $p_{\theta'}/p_\theta$ to be constant on the support of p_θ, and equality in the last step forces $p_{\theta'}$ to put all its mass there; together these give $p_{\theta'} = p_\theta$. Densities are handled the same way. □

So KL is not a distance (it is not symmetric, and symmetrizing it still does not give the triangle inequality); it is called a divergence. Nonnegativity and definiteness are all we need, and the asymmetry is the key to estimating it. Put the true distribution first:

$$\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \mathbb{E}_{\theta^*}\left[\log \frac{p_{\theta^*}(X)}{p_\theta(X)}\right] = \mathbb{E}_{\theta^*}\big[\log p_{\theta^*}(X)\big] - \mathbb{E}_{\theta^*}\big[\log p_\theta(X)\big].$$

The logarithm turns the ratio into a difference. The first term does not depend on θ (it is minus the entropy of $\mathbb{P}_{\theta^*}$), so $\theta \mapsto \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$ has the form "constant" $-\mathbb{E}_{\theta^*}[\log p_\theta(X)]$. The remaining term is an expectation under the distribution that generated the data, so by the LLN, $\mathbb{E}_{\theta^*}[h(X)]$ is estimated by $\frac{1}{n}\sum_{i=1}^n h(X_i)$:

$$\widehat{\mathrm{KL}}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \text{"constant"} - \frac{1}{n}\sum_{i=1}^n \log p_\theta(X_i).$$

The constant depends on the unknown θ^*, but not on θ. Shifting a function by a constant does not move its minimizer, so minimizing $\widehat{\mathrm{KL}}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$ over $\theta \in \Theta$ is the same as minimizing the computable function $\theta \mapsto -\frac{1}{n}\sum_{i=1}^n \log p_\theta(X_i)$ (with f_θ in place of p_θ for densities).

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 4, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 5: Maximum Likelihood Estimation (cont.)

Minimizing an estimate of the Kullback–Leibler divergence leads to the maximum likelihood principle. Maximizing the likelihood is then a calculus problem, which is easy when the log-likelihood is concave.

5.1 From the KL divergence to the likelihood

Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a statistical model for an i.i.d. sample $X_1, \dots, X_n$ with true parameter θ^*. The KL divergence is nonnegative and vanishes when its two arguments are equal, so θ^* minimizes $\theta \mapsto \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$.

Proposition 5.1 [Unique minimizer]. *If the model is identified, then θ^* is the unique minimizer of $\theta \mapsto \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$.*

Proof. By definiteness, $\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = 0$ only if $\mathbb{P}_\theta = \mathbb{P}_{\theta^*}$, so $\mathbb{P}_{\theta^*}$ is the unique minimizing distribution. By identifiability, $\theta \mapsto \mathbb{P}_\theta$ is injective, so θ^* is the only parameter producing it. □

For a discrete model with PMF p_θ, Lecture 4 wrote $\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \mathbb{E}_{\theta^*}[\log p_{\theta^*}(X)] - \mathbb{E}_{\theta^*}[\log p_\theta(X)]$ and replaced the expectation by an average:

$$\widehat{\mathrm{KL}}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \text{“constant”} - \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(X_i).$$

The constant depends on θ^* but not on θ, so it does not move the minimizer. Dropping it, removing the factor $1/n$, and using that log is increasing,

$$\min_{\theta\in\Theta} \widehat{\mathrm{KL}}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) \iff \max_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(X_i) \iff \max_{\theta\in\Theta} \sum_{i=1}^{n} \log p_\theta(X_i) \iff \max_{\theta\in\Theta} \prod_{i=1}^{n} p_\theta(X_i).$$

Here $\iff$ means the problems have the same maximizers or minimizers, not the same optimal values. This is the maximum likelihood principle. With densities, p_θ is replaced by f_θ throughout.

The product $\prod_i f_\theta(x_i)$ is the joint density of an i.i.d. sample at $(x_1, \dots, x_n)$, which explains why it measures how likely θ is to have generated the data. In the model $\{\mathcal{N}(\theta, 1) : \theta \in \mathbb{R}\}$ with a single observation x_1, the density value $f_\theta(x_1)$ is largest when the bell curve is centered at x_1, so the maximizer is $\theta = x_1$. With n observations, the product is small if any observation sits in a tail, and it becomes large when the curve covers all of them.

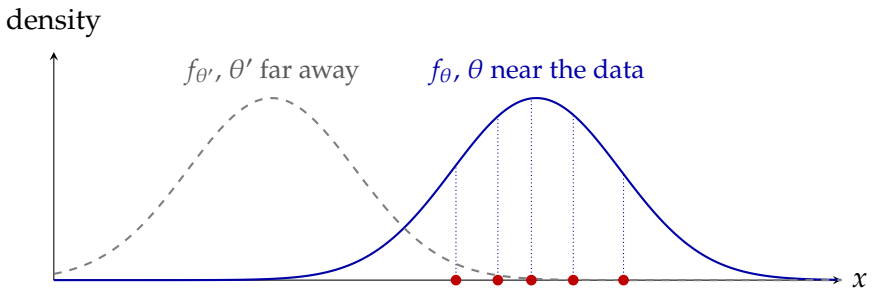


Figure 8: Likelihood in the model $\mathcal{N}(\theta, 1)$ for five illustrative observations (dots). The density centered among the data gives every observation a moderate value $f_\theta(x_i)$; the shifted density gives most observations a value near zero, so its product is far smaller.

5.2 Interlude: maximizing concave functions

Since $\min_\theta(-h(\theta))$ and $\max_\theta h(\theta)$ have the same solutions, it suffices to discuss maximization. For an arbitrary function this can be hard: setting the derivative to zero only finds critical points, which may be local maxima, local minima, or neither. For instance, the polynomial $\theta \mapsto \prod_{i=1}^n (\theta - X_i)$ can have $n - 1$ critical points, and every one must be checked. Many standard statistical models avoid this difficulty because their log-likelihoods are concave.

Definition 5.2 [Concave and convex functions]. A twice differentiable function $h : \Theta \subset \mathbb{R} \to \mathbb{R}$ is concave if $h''(\theta) \le 0$ for all $\theta \in \Theta$, and strictly concave if $h''(\theta) < 0$ for all $\theta \in \Theta$. It is (strictly) convex if $-h$ is (strictly) concave, that is, $h'' \ge 0$ ($h'' > 0$).

The inequality must hold everywhere, not just at a maximum. Examples:

- $h(\theta) = -\theta^2$ on $\mathbb{R}$: $h'' = -2 < 0$, strictly concave; this downward parabola is the canonical picture.
- $h(\theta) = \sqrt{\theta}$ on $(0, \infty)$: $h''(\theta) = -\frac{1}{4}\theta^{-3/2} < 0$, strictly concave.
- $h(\theta) = \log \theta$ on $(0, \infty)$: $h''(\theta) = -1/\theta^2 < 0$.
- $h(\theta) = \sin \theta$ on $[0, \pi]$: $h'' = -\sin\theta \le 0$, so it is concave there. On a longer interval the sine has inflection points and is not concave.
- $h(\theta) = 2\theta - 3$ on $\mathbb{R}$: $h'' = 0$, so it is both concave and convex, but neither strictly.

For $h : \Theta \subset \mathbb{R}^d \to \mathbb{R}$ with $d \ge 2$, the gradient $\nabla h(\theta) \in \mathbb{R}^d$ has entries $\partial h/\partial\theta_i$, and the Hessian $\nabla^2 h(\theta) \in \mathbb{R}^{d\times d}$ has entries $\partial^2 h/\partial\theta_i\partial\theta_j$. Then

$$h \text{ is concave} \iff x^\top \nabla^2 h(\theta)\, x \le 0 \quad \text{for all } x \in \mathbb{R}^d,\ \theta \in \Theta,$$
$$h \text{ is strictly concave} \iff x^\top \nabla^2 h(\theta)\, x < 0 \quad \text{for all } x \ne 0,\ \theta \in \Theta.$$

For $d = 1$, $x\, h''(\theta)\, x = h''(\theta)x^2$, which recovers the one-dimensional definition.

Example: *Two functions on* $\mathbb{R}^2$. For $h(\theta) = -\theta_1^2 - 2\theta_2^2$,

$$\nabla h(\theta) = \begin{pmatrix} -2\theta_1 \\ -4\theta_2 \end{pmatrix}, \qquad \nabla^2 h(\theta) = \begin{pmatrix} -2 & 0 \\ 0 & -4 \end{pmatrix}, \qquad x^\top \nabla^2 h(\theta)\, x = -2x_1^2 - 4x_2^2,$$

which is negative for $x \neq 0$, so h is strictly concave. For $h(\theta) = -(\theta_1 - \theta_2)^2$ the Hessian has rows $(-2, 2)$ and $(2, -2)$, and $x^\top \nabla^2 h\, x = -2(x_1 - x_2)^2 \leq 0$, with equality when $x_1 = x_2$: concave but not strictly. Similarly, $h(\theta) = \log(\theta_1 + \theta_2)$ is concave on $(0, \infty)^2$.

The last two functions follow a useful recipe: a concave function of a linear combination of the coordinates is concave. Indeed, if $h(\theta) = g(a^\top \theta)$ with $g'' \leq 0$, then $\nabla^2 h(\theta) = g''(a^\top \theta)\, aa^\top$ and $x^\top \nabla^2 h(\theta)\, x = g''(a^\top \theta)(a^\top x)^2 \leq 0$.

Proposition 5.3 [Maximizing a strictly concave function]. *Let* Θ *be convex and* h *strictly concave on* Θ. *Then* h *has at most one maximizer. If* θ_0 *is an interior point with* $\nabla h(\theta_0) = 0$ *(*$h'(\theta_0) = 0$ *when* $d = 1$*), then* θ_0 *is the unique global maximizer.*

Proof. For $d = 1$, $h'' < 0$ makes h' strictly decreasing on the interval Θ. If $h'(\theta_0) = 0$, then $h' > 0$ to the left of θ_0 and $h' < 0$ to its right, so h increases up to θ_0 and decreases after it. For $d \geq 2$, fix $\theta \neq \theta_0$ in Θ and apply this to $g(t) = h(\theta_0 + t(\theta - \theta_0))$ on $[0, 1]$, which lies in Θ by convexity. Its second derivative is $(\theta - \theta_0)^\top \nabla^2 h\, (\theta - \theta_0) < 0$ and $g'(0) = \nabla h(\theta_0)^\top (\theta - \theta_0) = 0$, so $h(\theta) = g(1) < g(0) = h(\theta_0)$. Uniqueness of a maximizer follows the same way: two maximizers would give a strictly concave g on $[0, 1]$ with equal values at both ends, so g would be larger in between. □

Convexity of Θ matters: pieces of downward parabolas over two disjoint intervals satisfy $h'' < 0$ yet can have two global maxima. The equation $\nabla h(\theta) = 0$ may have no closed form, but convex optimization solves it numerically and efficiently. In one dimension the sign of the derivative tells whether the maximizer lies to the right or to the left of the current point; in higher dimensions the gradient rules out half of the directions, which underlies gradient ascent. In this course, the maximizer often has a closed form.

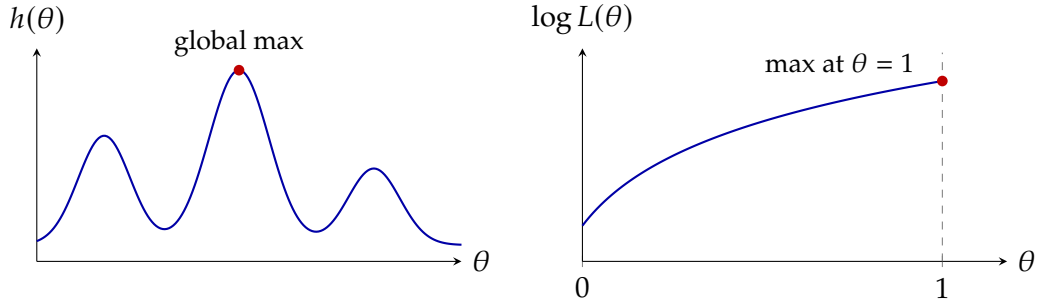


Figure 9: Left (schematic): a function with five critical points, three local maxima and two local minima; only one is the global maximum. Right (schematic): a concave log-likelihood that increases on $\Theta = [0, 1]$. Its derivative never vanishes on Θ, and the maximum is attained at the boundary.

5.3 The likelihood

Definition 5.4 [Likelihood, discrete case]. Let E be discrete (finite or countable). The likelihood of the model is the map

$$L_n : E^n \times \Theta \to \mathbb{R}, \qquad (x_1, \dots, x_n, \theta) \mapsto \mathbb{P}_\theta[X_1 = x_1, \dots, X_n = x_n].$$

The definition makes sense for dependent observations as well. For an i.i.d. sample, independence gives $L_n(x_1, \dots, x_n, \theta) = \prod_{i=1}^n \mathbb{P}_\theta[X_i = x_i]$, which is the only case considered in this course. The arguments x_i are generic points of E, not realizations of the data. Sometimes one writes $L(x_1, \dots, x_n; \theta)$ to separate the two kinds of argument.

Example: *Bernoulli trials.* Let $X_i \overset{\text{iid}}{\sim} \text{Ber}(p)$ with $E = \{0, 1\}$ and $\Theta = (0, 1)$. The PMF has the compact form $\mathbb{P}_p[X = x] = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$; plugging in $x = 1$ and $x = 0$ checks it. For all $(x_1, \dots, x_n) \in \{0, 1\}^n$,

$$L(x_1, \dots, x_n, p) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i}.$$

Example: *Poisson model.* Let $X_i \overset{\text{iid}}{\sim} \text{Poiss}(\lambda)$ with $E = \mathbb{N}$ and $\Theta = (0, \infty)$. For all $(x_1, \dots, x_n) \in \mathbb{N}^n$,

$$L(x_1, \dots, x_n, \lambda) = \prod_{i=1}^n e^{-\lambda}\frac{\lambda^{x_i}}{x_i!} = e^{-n\lambda}\frac{\lambda^{\sum_{i=1}^n x_i}}{x_1! \cdots x_n!}.$$

The denominator does not depend on λ, so it will not affect the maximizer.

For continuous distributions, the probability of observing exactly $(x_1, \dots, x_n)$ is zero, so the probability is replaced by the joint density. This is also exactly what the estimated KL divergence produces.

Definition 5.5 [Likelihood, continuous case]. If every $\mathbb{P}_\theta$ has a density f_θ, the likelihood is

$$L : E^n \times \Theta \to \mathbb{R}, \qquad (x_1, \dots, x_n, \theta) \mapsto \prod_{i=1}^n f_\theta(x_i).$$

Example: *Gaussian model.* Let $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $E = \mathbb{R}$ and $\Theta = \mathbb{R} \times (0, \infty)$. For all $(x_1, \dots, x_n) \in \mathbb{R}^n$ and $(\mu, \sigma^2) \in \Theta$, pulling the normalizing constants out of the product and turning the product of exponentials into the exponential of a sum gives

$$L(x_1, \dots, x_n, \mu, \sigma^2) = \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\Big(-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2\Big).$$

5.4 The maximum likelihood estimator

Definition 5.6 [Maximum likelihood estimator]. Let L be the likelihood of the model for an i.i.d. sample $X_1, \dots, X_n$. The maximum likelihood estimator (MLE) of θ is

$$\hat{\theta}_n^{\mathrm{MLE}} = \arg\max_{\theta \in \Theta} L(X_1, \dots, X_n, \theta),$$

provided it exists.

The likelihood is a deterministic function, but the estimator evaluates its first n arguments at the random sample, so $\hat{\theta}_n^{\mathrm{MLE}}$ is a random variable. It is kept random as long as possible, to study its fluctuations, and data are plugged in at the end. On an unbounded parameter set the likelihood may have no maximum, in which case Θ must be restricted.

Because log is increasing, the MLE also maximizes the log-likelihood,

$$\hat{\theta}_n^{\mathrm{MLE}} = \arg\max_{\theta \in \Theta} \log L(X_1, \dots, X_n, \theta), \qquad \log L(X_1, \dots, X_n, \theta) = \sum_{i=1}^{n} \log f_\theta(X_i),$$

with p_θ in place of f_θ in the discrete case. The logarithm has no intrinsic role; it turns products, powers, and exponentials into sums, which are easier to differentiate.

Example: *Bernoulli MLE.* The log-likelihood is $\left(\sum_i x_i\right) \log p + \left(n - \sum_i x_i\right) \log(1-p)$. Its second derivative, $-\sum_i x_i / p^2 - (n - \sum_i x_i)/(1-p)^2$, is negative when $0 < \sum_i x_i < n$, so the log-likelihood is strictly concave. Setting the derivative to zero,

$$\frac{\sum_i x_i}{\hat{p}} - \frac{n - \sum_i x_i}{1 - \hat{p}} = 0 \iff (1-\hat{p}) \sum_i x_i = \hat{p}\left(n - \sum_i x_i\right) \iff \hat{p}\, n = \sum_i x_i.$$

Hence $\hat{p}_n^{\mathrm{MLE}} = \bar{X}_n$.

Example: *Poisson MLE.* The log-likelihood is $-\log(x_1! \cdots x_n!) + \left(\sum_i x_i\right) \log \lambda - n\lambda$. The first term does not involve λ, and setting the derivative to zero gives $\sum_i x_i / \hat{\lambda} = n$. Hence $\hat{\lambda}_n^{\mathrm{MLE}} = \bar{X}_n$.

Example: *Gaussian MLE.* The log-likelihood is $-\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$. Setting both coordinates of the gradient to zero gives

$$\frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu) = 0, \qquad -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (x_i - \mu)^2 = 0.$$

The first equation gives $\hat{\mu} = \bar{X}_n$, and then the second gives $\hat{\sigma}^2 = \frac{1}{n} \sum_i (X_i - \bar{X}_n)^2 = \hat{S}_n$, the sample variance. So $(\hat{\mu}_n, \hat{\sigma}_n^2) = (\bar{X}_n, \hat{S}_n)$.

Setting the derivative to zero finds the maximum of a concave log-likelihood only when that maximum is attained inside Θ. If the log-likelihood is monotone on Θ, as on the right of Figure 9, the maximum is at the boundary, and the first-order equation has no solution in Θ; solving it

formally produces nonsense such as $1/\hat{\theta} = 0$. The boundary point can depend on the data; the MLE is then often $\max_i X_i$. It helps to sketch the log-likelihood before differentiating.

5.5 Fisher information

The precision of the MLE is governed by the curvature of the log-likelihood. Let

$$\ell(\theta) = \log L_1(X, \theta), \qquad \theta \in \Theta \subset \mathbb{R}^d,$$

be the log-likelihood of a single observation $X \sim \mathbb{P}_\theta$. It is a random function of θ; assume it is almost surely twice differentiable.

Definition 5.7 [Fisher information]. Under some regularity conditions, the Fisher information of the model is the $d \times d$ matrix

$$I(\theta) = \mathbb{E}\big[\nabla\ell(\theta)\nabla\ell(\theta)^\top\big] - \mathbb{E}\big[\nabla\ell(\theta)\big]\mathbb{E}\big[\nabla\ell(\theta)\big]^\top = -\mathbb{E}\big[\nabla^2\ell(\theta)\big].$$

The first expression is the covariance matrix of the random vector $\nabla\ell(\theta)$. If $\Theta \subset \mathbb{R}$,

$$\boxed{I(\theta) = \operatorname{Var}\big[\ell'(\theta)\big] = -\mathbb{E}\big[\ell''(\theta)\big]}.$$

The regularity conditions allow derivatives and integrals to be interchanged; Lecture 6 proves that the two expressions agree. The second form measures the curvature of the log-likelihood near its maximum. A sharply curved maximizer barely moves when expectations are replaced by averages, while a flat one is very sensitive to these fluctuations. Hence a large Fisher information is desirable: the asymptotic variance of $\sqrt{n}(\hat{\theta}_n^{\mathrm{MLE}} - \theta^*)$ is $1/I(\theta^*)$.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 5, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 6: Maximum Likelihood Estimation (cont.) and the Method of Moments

The Fisher information is the curvature of the expected log-likelihood, and it gives the asymptotic variance of the MLE. The method of moments is a second general estimation strategy: match population moments to empirical ones.

6.1 Fisher information

For a d-dimensional parameter the Fisher information is a $d \times d$ matrix, since it is built from $\nabla \ell \, \nabla \ell^\top$. It measures how much information about θ the model carries, and it depends on the parametrization: describing a centered Gaussian by σ, σ^2, or σ^4 gives different values.

Focus on $\Theta \subset \mathbb{R}$, and let X have density f_θ. The log-likelihood of one observation, $\ell(\theta) = \log f_\theta(X)$, is a random function of θ, and expectations and variances below are taken with $X \sim \mathbb{P}_\theta$. The Fisher information is of interest at every θ, but mostly at the true parameter θ^*.

Proposition 6.1 [Two forms of the Fisher information]. *Assume that derivatives in θ and integrals in x can be interchanged. Then*

$$I(\theta) = \operatorname{Var}\big[\ell'(\theta)\big] = -\mathbb{E}\big[\ell''(\theta)\big] = \int \frac{\big(\partial_\theta f_\theta(x)\big)^2}{f_\theta(x)}\, dx.$$

Proof. Differentiate $\int f_\theta(x)\, dx = 1$ once and twice under the integral:

$$\int \partial_\theta f_\theta(x)\, dx = 0, \qquad \int \partial_\theta^2 f_\theta(x)\, dx = 0.$$

Since $\ell' = \partial_\theta f_\theta / f_\theta$, the quotient rule gives $\ell'' = \partial_\theta^2 f_\theta / f_\theta - (\partial_\theta f_\theta)^2 / f_\theta^2$. Integrating against f_θ cancels one factor of f_θ in each term:

$$-\mathbb{E}[\ell''(\theta)] = -\int \partial_\theta^2 f_\theta \, dx + \int \frac{(\partial_\theta f_\theta)^2}{f_\theta}\, dx = \int \frac{(\partial_\theta f_\theta)^2}{f_\theta}\, dx.$$

From the other side, $\mathbb{E}[\ell'(\theta)] = \int \partial_\theta f_\theta \, dx = 0$, so $\operatorname{Var}[\ell'(\theta)] = \mathbb{E}[\ell'(\theta)^2] = \int (\partial_\theta f_\theta)^2 / f_\theta^2 \cdot f_\theta \, dx$, the same integral. □

All three expressions are useful; which one is quickest depends on the model. The fact that ℓ is the logarithm of a density is what makes them agree.

6.2 Curvature and the asymptotic normality of the MLE

The MLE minimizes an estimate of $\theta \mapsto \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$, which is minimized at θ^*. As n grows, the estimate stays uniformly close to the true function, so their minimizers are close. How close depends on the shape of the true function. If it is sharply curved at θ^*, every function within a narrow band of it has its minimizer near θ^*. If it is flat at the bottom, a band of the same width allows the minimizer to move far away.

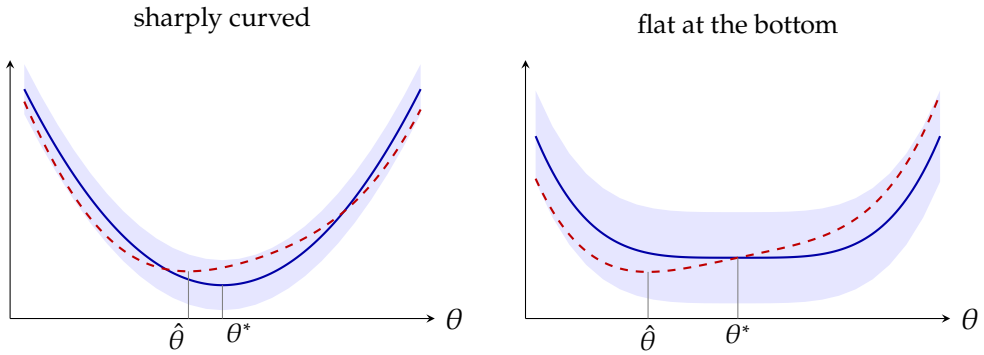


Figure 10: Schematic. The solid curve is $\theta \mapsto \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$; the dashed curve is an estimate within the same shaded band in both panels. When the curve is sharply curved (left), the estimated minimizer $\hat{\theta}$ is close to θ^*; when it is flat at the bottom (right), $\hat{\theta}$ can be far from θ^*.

Curvature at a minimum is the second derivative. Since $\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \text{const} - \mathbb{E}_{\theta^*}[\log f_\theta(X)]$,

$$\frac{d^2}{d\theta^2} \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)\Big|_{\theta=\theta^*} = -\mathbb{E}_{\theta^*}\left[\ell''(\theta^*)\right] = I(\theta^*).$$

Maximizing the likelihood flips the picture upside down, which explains the minus sign; the curvature is the same. So the Fisher information should control the precision of the MLE: a larger $I(\theta^*)$ should mean a smaller variance.

Theorem 6.2 [Asymptotic normality of the MLE]. *Let $\theta^* \in \Theta$ be the true parameter. Assume:*

1. *the model is identified;*
2. *for all $\theta \in \Theta$, the support of $\mathbb{P}_\theta$ does not depend on θ;*
3. *θ^* is not on the boundary of Θ;*
4. *$I(\theta)$ is invertible in a neighborhood of θ^*;*
5. *a few more technical conditions hold.*

Then, with respect to $\mathbb{P}_{\theta^*}$,

$$\hat{\theta}_n^{\mathrm{MLE}} \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*, \qquad \sqrt{n}\left(\hat{\theta}_n^{\mathrm{MLE}} - \theta^*\right) \xrightarrow[n\to\infty]{(d)} \mathcal{N}\left(0, I(\theta^*)^{-1}\right).$$

Each assumption has a role.

- Identifiability makes θ^*, not just $\mathbb{P}_{\theta^*}$, the unique minimizer of the KL divergence.
- If the support moved with θ, the log-ratio inside the KL divergence would involve $\log 0$.
- The argument uses derivatives and local fluctuations around θ^*. At a boundary point the maximum can occur where the derivative is not zero, and curvature plays no role.
- In one dimension, invertibility means $I(\theta^*) \neq 0$, so the quadratic term of the Taylor expansion at θ^* is present. If it vanished and the first nonzero term were of order four, the rate would be $n^{1/4}$ rather than $\sqrt{n}$; flatter optima give slower rates.
- The technical conditions include the interchange of derivatives and integrals used above.

The first conclusion is consistency. The second is a central limit theorem even though $\hat{\theta}_n^{\mathrm{MLE}}$ need not be an average, or even have a closed form. The asymptotic variance is exactly the inverse curvature, with no other constants: flat means large variance, curved means small variance. In the multivariate case, $I(\theta^*)^{-1}$ is the asymptotic covariance matrix. Heuristically, treating $\sqrt{n}(\hat{\theta}_n^{\mathrm{MLE}} - \theta^*)$ as exactly Gaussian, the bias vanishes and the quadratic risk satisfies

$$\mathbb{E}\big[(\hat{\theta}_n^{\mathrm{MLE}} - \theta^*)^2\big] \approx \frac{1}{n\, I(\theta^*)},$$

up to constants that vanish asymptotically. When the MLE is an average, the ordinary central limit theorem gives the same result, but the theorem supplies the variance without computing it directly.

6.3 Moments determine a distribution

Theorem 6.3 [Weierstrass approximation theorem]. *Let f be continuous on $[a, b]$. For every $\varepsilon > 0$ there exist d and $a_0, a_1, \ldots, a_d \in \mathbb{R}$ such that*

$$\max_{x \in [a,b]} \Big| f(x) - \sum_{k=0}^{d} a_k x^k \Big| < \varepsilon.$$

Continuous functions on an interval are uniformly approximated by polynomials; the degree d depends on ε and tends to infinity as $\varepsilon \to 0$.

Let $X_1, \ldots, X_n$ be an i.i.d. sample from an identified model $(E, \{\mathbb{P}_\theta\}_{\theta \in \Theta})$ in which every $\mathbb{P}_\theta$ has a density f_θ, and let θ^* be the true parameter. Two densities that integrate every bounded continuous function to the same value coincide almost everywhere. Hence, if

$$\int h(x) f_{\theta^*}(x)\, dx = \int h(x) f_\theta(x)\, dx \quad \text{for all bounded continuous } h,$$

then $\theta = \theta^*$. The left side is $\mathbb{E}_{\theta^*}[h(X)]$, so replacing it by an average suggests an estimator $\hat{\theta}$ with

$$\frac{1}{n}\sum_{i=1}^{n} h(X_i) = \int h(x) f_{\hat{\theta}}(x)\,dx \quad \text{for all bounded continuous } h.$$

These are infinitely many equations. By the Weierstrass theorem it is enough to take polynomials $h(x) = \sum_{k=0}^{d} a_k x^k$, which is still an infinite family; by linearity it is then enough to take the monomials $h(x) = x^k$:

$$\frac{1}{n}\sum_{i=1}^{n} X_i^k = \int x^k f_{\hat{\theta}}(x)\,dx, \qquad k = 1, \dots, d.$$

With $k = 0$, which holds automatically, this is $d + 1$ equations.

Definition 6.4 [Moments]. The kth moment of $\mathbb{P}_\theta$ is $m_k(\theta) = \int x^k f_\theta(x)\,dx = \mathbb{E}_\theta[X^k]$.

The mean is the first moment, and the variance is $m_2 - m_1^2$. The Weierstrass theorem has limitations as a basis for a method: it requires continuous functions (not a real problem), it works only on a bounded interval $[a, b]$, and it does not say how many moments d are needed.

When E is finite, the number of moments is exact. Let $E = \{x_1, \dots, x_r\}$ with distinct values and PMF p. The PMF has $r - 1$ free parameters, since $p(x_r) = 1 - \sum_{j<r} p(x_j)$.

Proposition 6.5 [Moments determine a finite distribution]. *Let $E = \{x_1, \dots, x_r\}$ consist of distinct values. A PMF on E is uniquely determined by its first $r - 1$ moments $m_1, \dots, m_{r-1}$.*

Proof. For $k = 1, \dots, r-1$, $m_k = \mathbb{E}[X^k] = \sum_{j=1}^{r} p(x_j) x_j^k$, and $\sum_j p(x_j) = 1$. These r equations are linear in the r unknowns $p(x_1), \dots, p(x_r)$:

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_r \\ x_1^2 & x_2^2 & \cdots & x_r^2 \\ \vdots & & \ddots & \vdots \\ x_1^{r-1} & x_2^{r-1} & \cdots & x_r^{r-1} \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} p(x_1) \\ p(x_2) \\ \vdots \\ p(x_{r-1}) \\ p(x_r) \end{pmatrix} = \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{r-1} \\ 1 \end{pmatrix}.$$

Up to moving the row of ones to the top, the matrix is a Vandermonde matrix, with determinant $\pm \prod_{1 \le j < k \le r} (x_k - x_j)$. The x_j are distinct, so the determinant is nonzero, and the unique solution is the inverse matrix applied to $(m_1, \dots, m_{r-1}, 1)^\top$. □

So the PMF values and the first $r - 1$ moments carry the same information, for any distribution on r points; this is the idea behind Gaussian quadrature. Moments therefore contain the information needed to recover a PDF or PMF, and accurate estimates of them may recover the distribution. In a parametric model, where knowing $\mathbb{P}_\theta$ amounts to knowing θ, fewer moments usually suffice, case by case. Rule of thumb: for $\theta \in \Theta \subset \mathbb{R}^d$, use d moments.

6.4 The method of moments

Let $X_1, \dots, X_n$ be an i.i.d. sample from $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ with $\Theta \subseteq \mathbb{R}^d$. Define the POPULATION MOMENTS $m_k(\theta) = \mathbb{E}_\theta[X_1^k]$, the EMPIRICAL MOMENTS $\hat{m}_k = \overline{X_n^k} = \frac{1}{n}\sum_{i=1}^n X_i^k$, for $1 \le k \le d$, and

$$\psi : \Theta \subset \mathbb{R}^d \to \mathbb{R}^d, \qquad \theta \mapsto \big(m_1(\theta), \dots, m_d(\theta)\big).$$

Assume ψ is one to one, so that $\theta = \psi^{-1}(m_1(\theta), \dots, m_d(\theta))$.

Definition 6.6 [MOMENTS ESTIMATOR]. The MOMENTS ESTIMATOR of θ is

$$\hat{\theta}_n^{\mathrm{MM}} = \psi^{-1}(\hat{m}_1, \dots, \hat{m}_d),$$

provided it exists.

Its quality depends on how well behaved ψ^{-1} is. Empirical moments are at distance of order $1/\sqrt{n}$ from the population moments, by the central limit theorem. If ψ^{-1} is steep, equivalently if ψ is flat, these small fluctuations become large errors in $\hat{\theta}$. For the linear map in the Vandermonde case, the amplification is governed by the condition number of the matrix, the ratio of its largest to its smallest eigenvalue in absolute value; numerical stability becomes statistical stability. So the derivative of ψ^{-1} should appear in the asymptotic variance.

For the analysis, let $M(\theta) = (m_1(\theta), \dots, m_d(\theta))$, $\hat{M} = (\hat{m}_1, \dots, \hat{m}_d)$, and let $\Sigma(\theta) = \mathbb{V}_\theta(X, X^2, \dots, X^d)$ be the covariance matrix of the random vector $(X, X^2, \dots, X^d)$ with $X \sim \mathbb{P}_\theta$. Its entries are

$$\Sigma_{jk}(\theta) = \mathbb{E}_\theta[X^j X^k] - \mathbb{E}_\theta[X^j]\mathbb{E}_\theta[X^k] = m_{j+k}(\theta) - m_j(\theta)m_k(\theta).$$

Assume ψ^{-1} is continuously differentiable at $M(\theta)$, and write $\nabla\psi^{-1}\big|_{M(\theta)}$ for its $d \times d$ gradient matrix there.

Consistency. Each $\hat{m}_k$ is an average, so by the (weak or strong) law of large numbers $\hat{m}_k \to m_k(\theta)$ (in probability or almost surely). Since ψ^{-1} is continuous, the continuous mapping theorem gives $\hat{\theta}_n^{\mathrm{MM}} = \psi^{-1}(\hat{M}) \to \psi^{-1}(M(\theta)) = \theta$. Injectivity of ψ is the key assumption here.

Central limit theorem. The univariate central limit theorem gives the limit of each $\sqrt{n}(\hat{m}_k - m_k(\theta))$ separately, which describes only the diagonal of the limiting covariance matrix. The multivariate central limit theorem describes the joint behavior:

$$\sqrt{n}\big(\hat{M} - M(\theta)\big) \xrightarrow[n\to\infty]{(d)} \mathcal{N}\big(0, \Sigma(\theta)\big) \qquad \text{(w.r.t. } \mathbb{P}_\theta\text{)}.$$

To pass from moments to parameters, recall the delta method: if $\sqrt{n}(T_n - \theta) \to \mathcal{N}(0, \sigma^2)$ and g is differentiable, then $\sqrt{n}(g(T_n) - g(\theta)) \to \mathcal{N}(0, g'(\theta)\sigma^2 g'(\theta))$. Writing the variance in this symmetric form suggests the multivariate version, with gradients in place of derivatives.

Theorem 6.7 [Multivariate delta method]. *Let $(T_n)_{n\geq 1}$ be random vectors in $\mathbb{R}^p$ with $\sqrt{n}(T_n - \theta) \xrightarrow{(d)} \mathcal{N}(0, \Sigma)$ for some $\theta \in \mathbb{R}^p$ and symmetric positive semidefinite $\Sigma \in \mathbb{R}^{p\times p}$. Let $g : \mathbb{R}^p \to \mathbb{R}^k$ be continuously differentiable at θ, with gradient matrix $\nabla g(\theta) = \big(\partial g_j/\partial\theta_i\big)_{1\leq i\leq p,\, 1\leq j\leq k} \in \mathbb{R}^{p\times k}$. Then*

$$\sqrt{n}\big(g(T_n) - g(\theta)\big) \xrightarrow[n\to\infty]{(d)} \mathcal{N}\big(0, \nabla g(\theta)^\top \Sigma\, \nabla g(\theta)\big).$$

Applying it with $T_n = \hat{M}$ and $g = \psi^{-1}$ gives the behavior of the moments estimator.

Theorem 6.8 [Asymptotic normality of the moments estimator]. *Under the assumptions above, with respect to $\mathbb{P}_\theta$,*

$$\sqrt{n}\big(\hat{\theta}_n^{\mathrm{MM}} - \theta\big) \xrightarrow[n\to\infty]{(d)} \mathcal{N}\big(0, \Gamma(\theta)\big), \qquad \Gamma(\theta) = \Big[\nabla\psi^{-1}\big|_{M(\theta)}\Big]^\top \Sigma(\theta) \Big[\nabla\psi^{-1}\big|_{M(\theta)}\Big].$$

A steep ψ^{-1} has a large gradient and produces a large asymptotic variance, as anticipated.

6.5 MLE versus the method of moments

- In general, the MLE is more accurate: it has smaller quadratic risk.
- The MLE is sometimes intractable, while moments are easy to compute.
- If the likelihood is concave, optimization algorithms such as interior point methods or gradient ascent find the MLE.
- If the likelihood is not concave, only local heuristics are available, such as expectation-maximization, and they can get stuck at local maxima. A common combination starts a local likelihood maximization at the moments estimator, which is likely to lie near the right peak.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 6, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 7: Parametric Hypothesis Testing

A hypothesis test answers a yes/no question about the parameter. Unlike estimation, where the central limit theorem mostly served to analyze an estimator, a test uses the size of random fluctuations in its design.

7.1 Two motivating problems

Example: *Cherry Blossom Run.* This 10-mile race takes place every year in Washington, D.C. In 2009 it had 14974 participants with an average running time of 103.5 minutes. Were runners faster in 2012? Select n runners from the 2012 race at random and let $X_1, \dots, X_n$ be their running times.

The histogram of the 2009 times is bell shaped, and a Gaussian density with mean 103.5 and variance 373 matches it well. The question is whether $X_1 \sim \mathcal{N}(103.5, 373)$. This can fail in many ways: $\mathbb{E}[X_1] \neq 103.5$, $\mathrm{Var}[X_1] \neq 373$, or X_1 may not be Gaussian. The specific question is whether $\mathbb{E}[X_1] < 103.5$. So make the modeling assumptions

1. $\mathrm{Var}[X_1] = 373$, the same as in 2009;
2. X_1 is Gaussian.

The only unknown is $\mu = \mathbb{E}[X_1]$, and the question becomes: is $\mu = 103.5$, or is $\mu < 103.5$? The hypothesis $X_1 \sim \mathcal{N}(103.5, 373)$ can now be rejected only in favor of $\mathcal{N}(\mu, 373)$ with $\mu < 103.5$. Values $\mu > 103.5$ are not considered.

Such assumptions matter because a richer set of alternatives makes it easier to find one that fits the data by chance alone. For instance, if all observations are distinct, the distribution that puts mass $1/n$ on each observed value explains the sample better than any Gaussian.

The simple heuristic "if $\bar{X}_n < 103.5$, then $\mu < 103.5$" can go wrong, for instance if the sample happens to contain only fast runners: $\bar{X}_n$ is close to μ but is not equal to it, and its fluctuations are of order $1/\sqrt{n}$ by the central limit theorem. A better heuristic is

$$\text{"if } \bar{X}_n < 103.5 - (\text{something} \xrightarrow[n\to\infty]{} 0), \text{ then } \mu < 103.5\text{"},$$

where the buffer accounts for the fluctuations of $\bar{X}_n$. Here an expected value is compared with a fixed reference number. Such a reference is rarely available; the usual situation compares two unknown quantities.

Example: *Clinical trial.* A cough syrup is given to a test group, and a placebo to a control group. Let μ_{drug} and μ_{control} be the expected numbers of expectorations per hour under the syrup and under the placebo. The placebo has an effect of unknown size, so there is no reference number:

the question is whether $\mu_{\text{drug}} < \mu_{\text{control}}$. Model the data as $X_1, \ldots, X_{n_{\text{drug}}} \overset{\text{iid}}{\sim} \text{Poiss}(\mu_{\text{drug}})$ and $Y_1, \ldots, Y_{n_{\text{control}}} \overset{\text{iid}}{\sim} \text{Poiss}(\mu_{\text{control}})$, independent of each other. The group sizes need not be equal. The heuristic is now "if $\bar{X}_{\text{drug}} < \bar{X}_{\text{control}} - (\text{something})$, then $\mu_{\text{drug}} < \mu_{\text{control}}$", where the something tends to zero as both n_{drug} and n_{control} grow, at rates $1/\sqrt{n_{\text{drug}}}$ and $1/\sqrt{n_{\text{control}}}$. The buffer is deliberate. By the central limit theorem, $\bar{X}_{\text{drug}} - \bar{X}_{\text{control}}$ is approximately Gaussian with mean $\mu_{\text{drug}} - \mu_{\text{control}}$. If the syrup does nothing, this distribution is centered at 0, and without a buffer half of all useless drugs would be approved. The burden of proof is put on the claim of an effect.

7.2 Calibrating with the central limit theorem

Test whether a coin is fair: $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Ber}(p)$, and the question is $p = 1/2$ versus $p \neq 1/2$.

Example 1: A coin is tossed 80 times and shows 54 heads, so $\bar{X}_n = 54/80 = 0.675$. If $p = 0.5$, the central limit theorem gives

$$\sqrt{n}\,\frac{\bar{X}_n - 0.5}{\sqrt{0.5(1-0.5)}} \approx \mathcal{N}(0,1), \qquad \text{and here} \qquad \sqrt{80}\,\frac{0.675 - 0.5}{0.5} \approx 3.13.$$

Under the hypothesis $p = 0.5$, the value of p is known, so it should be plugged in everywhere, including the variance. One may instead estimate the variance by $\bar{X}_n(1 - \bar{X}_n)$, by Slutsky's theorem; here this gives 3.34. If in fact $p = 0.6$, the statistic equals an approximately standard Gaussian term plus $\sqrt{n}(0.6 - 0.5)/\sqrt{0.25}$, a shift that grows like $\sqrt{n}$. So large values are evidence against $p = 0.5$.

A standard Gaussian lies in $[-1, 1]$ with probability about 0.68, in $[-2, 2]$ with probability about 0.95, and in $[-3, 3]$ with probability about 0.997. A value of 3.13 is far in the tail, so it seems quite reasonable to reject $p = 0.5$. Such an outcome is still possible for a fair coin, just as 80 heads in 80 tosses has probability $2^{-80} > 0$: the uncertainty cannot be removed, only made small. Since the probability of any single value of a continuous variable is zero, what is controlled is the probability of a value at least this extreme.

Example 2: A coin is tossed 30 times and shows 13 heads, so $\bar{X}_n = 13/30 \approx 0.433$ and $\sqrt{30}\,(\bar{X}_n - 0.5)/0.5 \approx -0.73$. This is a plausible realization of $Z \sim \mathcal{N}(0,1)$, so the data do not suggest that the coin is unfair. This does not show that the coin is fair; there is just not enough evidence against it.

Where to draw the line? On the scale of raw counts the answer changes with every problem (13 out of 30, 54 out of 80, and so on). Standardizing maps every such problem onto the scale of a standard Gaussian, where one only needs to know which values are unusually large. A second step maps a threshold to a probability of error, which is easier to choose. The usual threshold 1.96 is exceeded in absolute value with probability 5%: a test using it, repeated many times on fair coins, wrongly declares the coin unfair 5% of the time. For a 1% error, the threshold is 2.58.

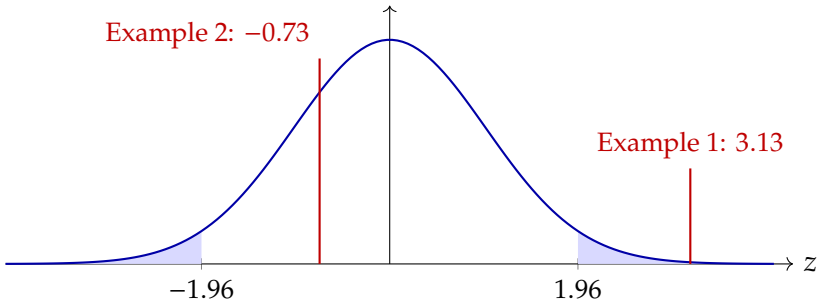


Figure 11: The two standardized coin statistics on the scale of the standard Gaussian density. The shaded tails beyond ±1.96 have total probability 5%. Example 1 (3.13) lies far in the right tail; Example 2 (−0.73) is a typical value.

In all three problems, the conclusion that would be a discovery (faster runners, an effective drug, an unfair coin) is made deliberately hard to reach. Data are used to look for evidence against a status quo.

7.3 Statistical formulation

Definition 7.1 [Null and alternative hypotheses]. Let $X_1, \ldots, X_n$ be i.i.d. with statistical model $(E, (\mathbb{P}_\theta)_{\theta \in \Theta})$, and let Θ_0 and Θ_1 be disjoint subsets of Θ. Consider

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \in \Theta_1.$$

H_0 is the null hypothesis and H_1 the alternative hypothesis. If the true θ is believed to lie in $\Theta_0 \cup \Theta_1$, we test H_0 against H_1, deciding whether to reject H_0.

Parameters outside $\Theta_0 \cup \Theta_1$ are excluded beforehand. In the running example, $\Theta_0 = \{103.5\}$, $\Theta_1 = (-\infty, 103.5)$, and $\mu > 103.5$ is not considered. In the clinical trial, $H_0 : \mu_{\text{drug}} = \mu_{\text{control}}$ and $H_1 : \mu_{\text{drug}} < \mu_{\text{control}}$. For the coin, $H_0 : p = 1/2$ and $H_1 : p \neq 1/2$.

H_0 and H_1 play asymmetric roles. H_0 is the status quo and H_1 the discovery; the data are only used to try to disprove H_0. In a close call one sides with H_0, and a lack of evidence does not mean that H_0 is true: one says "we fail to reject H_0", never "we accept H_0". This is the principle of a court, where a defendant is innocent until proven guilty and an acquittal does not establish innocence. A regulator leans toward H_0; a scientist hoping for a discovery leans toward H_1.

Definition 7.2 [Test]. A test is a statistic $\psi \in \{0, 1\}$, that is, a measurable function of the data that does not depend on unknown parameters, such that H_0 is not rejected if $\psi = 0$ and is rejected if $\psi = 1$.

A $\{0, 1\}$-valued function is an indicator: $\mathbf{1}\{A\}$ equals 1 if the statement A is true and 0 otherwise.

For the coin,

$$\psi = \mathbf{1}\left\{\left|\sqrt{n}\,\frac{\bar{X}_n - 0.5}{\sqrt{0.5(1-0.5)}}\right| > C\right\} \quad \text{for some } C > 0.$$

As a function of $\bar{X}_n$, this test rejects when $\bar{X}_n$ is either too small or too large. With the variance estimate $\bar{X}_n(1-\bar{X}_n)$ in the denominator, the standardized statistic is a nonlinear function of $\bar{X}_n$, and the set where it exceeds C in absolute value can be a more complicated union of intervals. The test is more than the average: it includes the threshold C, which encodes what counts as a large value for a Gaussian.

Definition 7.3 [Rejection region, errors, power]. Let ψ be a test.

- Its rejection region is $R_\psi = \{x \in E^n : \psi(x) = 1\}$, so $\psi = \mathbf{1}_{R_\psi}$.
- Its type 1 error (rejecting H_0 when it is true) is $\alpha_\psi : \Theta_0 \to \mathbb{R}$, $\theta \mapsto \mathbb{P}_\theta[\psi = 1]$.
- Its type 2 error (not rejecting H_0 although H_1 is true) is $\beta_\psi : \Theta_1 \to \mathbb{R}$, $\theta \mapsto \mathbb{P}_\theta[\psi = 0]$.
- Its power is the number $\pi_\psi = \inf_{\theta\in\Theta_1}\big(1 - \beta_\psi(\theta)\big) = \inf_{\theta\in\Theta_1} \mathbb{P}_\theta[\psi = 1]$.

The errors are functions because θ can range over a whole set. In the running example Θ_0 is a single point, so α_ψ is one number. Were Θ_0 the half-line $[103.5, \infty)$, the largest type 1 error would occur at the point closest to the alternative. The power is the probability of a correct rejection in the worst case over the alternative. Errors occur only off the diagonal of the following table.

	H_0 true	H_1 true
fail to reject H_0 ($\psi = 0$)	correct	type 2 error
reject H_0 ($\psi = 1$)	type 1 error	correct

In a US court, H_0 is "the defendant is innocent". A type 1 error convicts an innocent person; a type 2 error lets a guilty person go free. Acquitting everyone avoids all type 1 errors but makes the type 2 error as large as possible; convicting everyone does the reverse. Both errors cannot be made small at once, and ad hoc combinations, such as minimizing their sum or a weighted sum, are no more justified than one another. The approach of hypothesis testing is to constrain the type 1 error to an acceptable level, such as 5%, and to make the type 2 error as small as possible under this constraint.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 7, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 8: Parametric Hypothesis Testing (cont.)

A test is calibrated by fixing its level and choosing the threshold from the limiting distribution of its test statistic. The p-value summarizes the evidence on a single scale. The chi-squared distribution is a first new limiting distribution beyond the Gaussian.

8.1 The power function

For a test ψ of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, consider the single function $\theta \mapsto \mathbb{P}_\theta[\psi = 1]$ on $\Theta_0 \cup \Theta_1$, the POWER FUNCTION of ψ. On Θ_0 it is the type 1 error α_ψ. On Θ_1 it is $1 - \beta_\psi$, the probability of a correct rejection, and the power π_ψ is its infimum there.

Example: *A one-sided coin test.* Let $X_i \overset{\text{iid}}{\sim} \text{Ber}(p)$ and test $H_0 : p \le 1/2$ against $H_1 : p > 1/2$ with

$$\psi = \mathbf{1}\Big\{\sqrt{n}\,\frac{\bar{X}_n - 0.5}{\sqrt{\bar{X}_n(1-\bar{X}_n)}} > c\Big\},$$

with the statistic set to 0 when $\bar{X}_n \in \{0, 1\}$. Near $p = 0$ almost every toss is a tail and ψ essentially never rejects; the probability of rejection increases with p. With a type 1 error budget of 5%, c is chosen so that the power function equals 5% at the boundary $p = 1/2$, and it stays below 5% on Θ_0. Because the power function is continuous and increasing, it is close to 5% just to the right of $1/2$, so $\pi_\psi = 5\%$: the power equals the level.

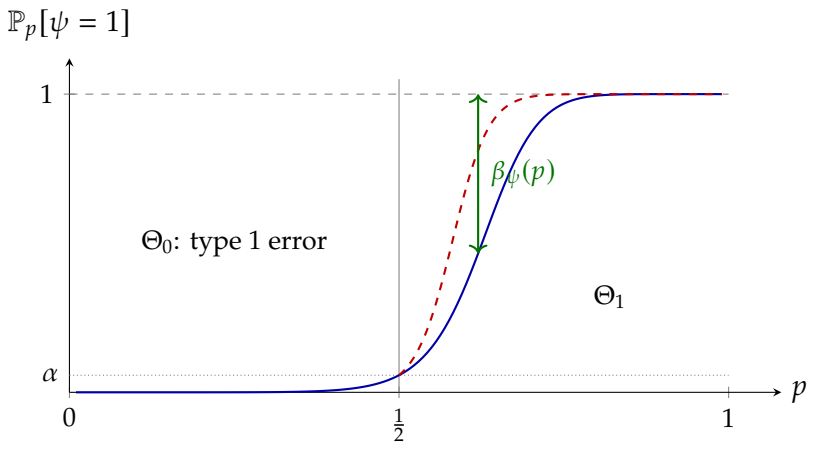


Figure 12: Schematic power function $p \mapsto \mathbb{P}_p[\psi = 1]$ of the one-sided test. On $\Theta_0 = [0, 1/2]$ it is the type 1 error and stays below the level α; on $\Theta_1 = (1/2, 1]$ the gap to 1 is the type 2 error $\beta_\psi(p)$. A uniformly more powerful test of the same level has a higher curve on Θ_1 (dashed).

Two tests can have the same power, since the infimum is attained at the boundary, while one of

them has a higher power function at every point of Θ_1; it is then UNIFORMLY MORE POWERFUL. Its behavior on Θ_0 is irrelevant as long as it respects the budget there. A uniformly most powerful test need not exist: two power functions can cross, and the choice then depends on where in Θ_1 the parameter is expected to be.

8.2 Level and test statistics

Definition 8.1 [LEVEL AND ASYMPTOTIC LEVEL]. A test ψ has LEVEL α if $\alpha_\psi(\theta) \le \alpha$ for all $\theta \in \Theta_0$, and ASYMPTOTIC LEVEL α if $\lim_{n\to\infty} \alpha_\psi(\theta) \le \alpha$ for all $\theta \in \Theta_0$.

The level is an upper bound: a test of level 5% also has level 10%. Since lowering the type 1 error raises the type 2 error, one uses the whole budget, choosing the threshold so that the largest type 1 error is exactly α. Exact type 1 errors are often hard to compute, for instance for an awkward binomial, but their limits follow from the central limit theorem, hence the asymptotic level.

In general a test has the form $\psi = \mathbf{1}\{T_n > c\}$ for a statistic T_n, the TEST STATISTIC, and a threshold $c \in \mathbb{R}$; the rejection region is $R_\psi = \{T_n > c\}$. This covers two-sided regions $\{|Q_n| > c\}$ with $T_n = |Q_n|$, and left-sided regions $\{Q_n < -c\}$ with $T_n = -Q_n$.

A test statistic must be computable from the data. The standardized mean $\sqrt{n}(\bar{X}_n - p)/\sqrt{p(1-p)}$ depends on the unknown p, but under $H_0 : p = 1/2$ the value $p = 1/2$ can be plugged in. When Θ_0 is a larger set, the worst type 1 error occurs at the point of Θ_0 closest to Θ_1, and one calibrates there. What is needed is a statistic whose limiting distribution under H_0 is pivotal (Lecture 2), that is, does not depend on unknown quantities, and whose quantiles are tabulated or can be computed.

8.3 The two-sided coin test

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Ber}(p)$ with $p \in (0,1)$ unknown, and test $H_0 : p = 1/2$ against $H_1 : p \neq 1/2$ at asymptotic level $\alpha \in (0,1)$. For $\beta \in (0,1)$ let q_β denote the $(1-\beta)$-quantile of $\mathcal{N}(0,1)$, so that $\mathbb{P}[Z > q_\beta] = \beta$ for $Z \sim \mathcal{N}(0,1)$. Let

$$T_n = \sqrt{n}\,\frac{|\hat{p}_n - 0.5|}{\sqrt{0.5(1-0.5)}},$$

where $\hat{p}_n = \bar{X}_n$ is the MLE. The direction of a deviation is not known in advance, so the test rejects for large $|\hat{p}_n - 0.5|$.

If H_0 is true, the central limit theorem gives $\mathbb{P}_{1/2}[T_n > c] \to \mathbb{P}[|Z| > c]$. This holds only for $p = 1/2$, the single point of Θ_0; for other p the statistic is not asymptotically standard Gaussian. By symmetry, $\mathbb{P}[|Z| > c] = \alpha$ exactly when each tail has probability $\alpha/2$, that is, $c = q_{\alpha/2}$. So

$$\boxed{\psi_\alpha = \mathbf{1}\{T_n > q_{\alpha/2}\}}$$

has asymptotic level α. For $\alpha = 5\%$, $q_{\alpha/2} = 1.96$. In Example 1 of Lecture 7 (54 heads in 80 tosses), $T_n \approx 3.13 > 1.96$ and H_0 is rejected at asymptotic level 5%. In Example 2 (13 heads in 30 tosses), $T_n \approx 0.73 < 1.96$ and H_0 is not rejected.

8.4 p-values

Lowering α moves $q_{\alpha/2}$ to the right. In Example 1 the test rejects as long as $q_{\alpha/2} < 3.13$; the tipping point is the level α' with $q_{\alpha'/2} = 3.13$, that is, $\alpha' = \mathbb{P}[|Z| > 3.13]$. Levels above α' reject and levels below it do not.

Definition 8.2 [P-VALUE]. The (asymptotic) P-VALUE of a test ψ_α is the smallest (asymptotic) level α at which ψ_α rejects H_0. It is random, since it depends on the sample.

Proposition 8.3 [GOLDEN RULE]. *p-value* $\leq \alpha \iff$ *H_0 is rejected by ψ_α at the (asymptotic) level α.*

The smaller the p-value, the more confidently one can reject H_0. For $\psi_\alpha = \mathbf{1}\{T_n > q_{\alpha/2}\}$, the test rejects iff $q_{\alpha/2} < T_n$, iff $\alpha > \mathbb{P}[|Z| > T_n]$, so

$$\text{p-value} = \mathbb{P}\big[|Z| > T_n\big] = 2\big(1 - \Phi(T_n)\big),$$

where Φ is the standard Gaussian CDF and the probability is over Z only, with T_n held at its observed value. Pictorially, the p-value is the tail area obtained by replacing $q_{\alpha/2}$ with T_n. Example 1: $\mathbb{P}[|Z| > 3.13] \approx 0.002 \ll 0.01$. Example 2: $\mathbb{P}[|Z| > 0.73] \approx 0.47$.

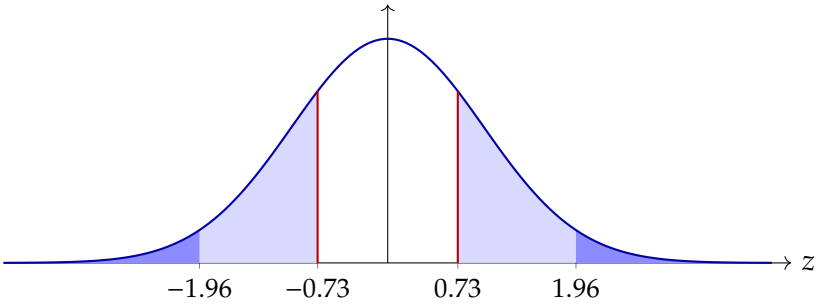


Figure 13: Example 2: the p-value is the total shaded area beyond $\pm T_n = \pm 0.73$, about 0.47. The test of level $\alpha = 5\%$ rejects only beyond $\pm q_{\alpha/2} = \pm 1.96$ (dark tails), so it does not reject; the p-value exceeds α.

A p-value puts every test on the same scale $[0, 1]$, whatever the limiting distribution: the CDF of that distribution is built into its computation. Small values mean strong evidence against H_0, large values weak evidence. A standard test takes the data and a level and returns reject or not; a p-value takes only the data and lets anyone apply their own level. Reporting the p-value conveys how much evidence there was, which a bare decision at 5% does not. The p-value is not the probability of making an error, although it is often read that way.

The hypotheses must be fixed before the data are seen. After observing 13 heads in 30 tosses,

switching the alternative to $p < 1/2$ halves the p-value. All the calculations above assume that Θ_0 and Θ_1 are deterministic and that the only randomness is in the data. Choosing hypotheses from the data, or running many tests and reporting the favorable one, invalidates the level; this is p-hacking.

8.5 The Neyman–Pearson paradigm

Type 1 and type 2 errors cannot be minimized together. The Neyman–Pearson paradigm fixes the type 1 error: among all tests of level (or asymptotic level) α, find one with maximal power. The trivial test $\psi = 0$, which never rejects H_0, has perfect level $\alpha = 0$ but power $\pi_\psi = 0$; in court, it acquits everyone. Neyman–Pearson theory provides most powerful tests of a given level in several cases. The paradigm suits testing, with its status quo. Other binary decisions, such as classifying images, often weight the two errors equally and minimize their sum instead.

8.6 The chi-squared distribution

Definition 8.4 [Chi-squared distribution]. For a positive integer d, the χ^2 distribution with d degrees of freedom, χ^2_d, is the law of $Z_1^2 + Z_2^2 + \cdots + Z_d^2$, where $Z_1, \ldots, Z_d \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.

If $Z \sim \mathcal{N}_d(0, I_d)$, then $\|Z\|_2^2 \sim \chi^2_d$: by Pythagoras, a squared Euclidean norm is a sum of squared coordinates. For instance, a dart aimed at a target with independent standard Gaussian errors in the horizontal and vertical directions lands at a squared distance from the center distributed as χ^2_2. The density of χ^2_d is

$$f_d(x) = \frac{x^{d/2-1} e^{-x/2}}{2^{d/2}\,\Gamma(d/2)}, \qquad x > 0.$$

Since $\mathbb{E}[Z_i^2] = 1$ and $\mathrm{Var}[Z_i^2] = \mathbb{E}[Z_i^4] - 1 = 2$,

$$\mathbb{E}[\chi^2_d] = d, \qquad \mathrm{Var}[\chi^2_d] = 2d.$$

As d grows, the mass moves to the right and spreads out (Figure 14). For $d = 2$, $\Gamma(1) = 1$ gives $f_2(x) = \frac{1}{2}e^{-x/2}$, so $\chi^2_2 = \mathrm{Exp}(1/2)$, which is not obvious from the definition.

The chi-squared distribution appears through the sample variance

$$S_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - (\bar{X}_n)^2.$$

Theorem 8.5 [Cochran]. *If $X_1, \ldots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ and S_n is the sample variance, then*

$$\frac{nS_n}{\sigma^2} \sim \chi^2_{n-1}.$$

The scaling is transparent. Subtracting μ from every X_i does not change S_n, so take $\mu = 0$. Without centering at $\bar{X}_n$, $\frac{1}{\sigma^2}\sum_i X_i^2$ is a sum of n squared standard Gaussians, that is, χ^2_n; scaling the data by a constant scales S_n by its square. The decomposition $\sum_i X_i^2 = nS_n + n(\bar{X}_n)^2$ splits this into two independent pieces, and centering with the sample mean removes exactly one degree of freedom. This reflects orthogonality to the span of the constant vector. The statement is exact, not asymptotic, because the data are assumed Gaussian, and nS_n/σ^2 can serve as a pivot for tests about the variance.

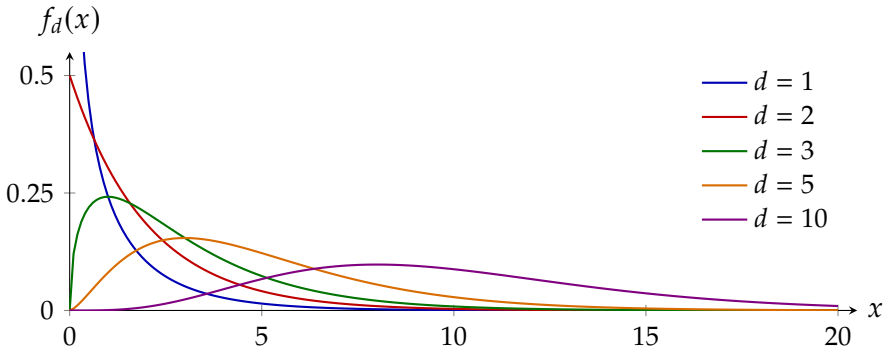


Figure 14: Densities of χ^2_d for $d = 1, 2, 3, 5, 10$. For $d = 1$ the density is infinite at 0, for $d = 2$ it is the exponential density $\frac{1}{2}e^{-x/2}$, and from $d = 3$ on it starts at 0 and its bulk moves to the right as d increases.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 8, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 9: Parametric Hypothesis Testing (cont.)

Throughout, q_α denotes the $(1-\alpha)$-quantile of the relevant pivotal distribution, so that the right tail beyond q_α has probability α. A test rejects when a pivotal statistic exceeds such a quantile.

9.1 Student's t distribution

Definition 9.1 [Student's t distribution]. For a positive integer d, Student's t distribution with d degrees of freedom, written t_d, is the law of

$$\frac{Z}{\sqrt{V/d}}, \qquad Z \sim \mathcal{N}(0,1), \quad V \sim \chi^2_d, \quad Z \perp\!\!\!\perp V.$$

This is the distribution that appears when a Gaussian is standardized by its *empirical* rather than its true variance. Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with sample variance $S_n = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2$. With σ known, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \sim \mathcal{N}(0,1)$ exactly. When σ is unknown, replace it by $\sqrt{S_n}$ and multiply and divide by σ:

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sqrt{S_n}} = \underbrace{\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma}}_{Z} \cdot \frac{\sqrt{n}}{\sqrt{nS_n/\sigma^2}} = \frac{Z}{\sqrt{V/n}}, \qquad V = \frac{nS_n}{\sigma^2}.$$

By Cochran's theorem (Lecture 8), $V \sim \chi^2_{n-1}$ and $\bar{X}_n \perp\!\!\!\perp S_n$, so $Z \perp\!\!\!\perp V$. The only mismatch with the definition is that V has $n-1$ degrees of freedom but is divided by n. Rescaling fixes it:

$$\boxed{\sqrt{n-1}\,\frac{\bar{X}_n - \mu}{\sqrt{S_n}} = \frac{Z}{\sqrt{V/(n-1)}} \sim t_{n-1}}.$$

Equivalently, with the unbiased estimator $\tilde{S}_n = \frac{1}{n-1}\sum_i (X_i - \bar{X}_n)^2 = \frac{n}{n-1} S_n$ (indeed $\mathbb{E}[S_n] = \frac{n-1}{n}\sigma^2$), the statistic $\sqrt{n}(\bar{X}_n - \mu)/\sqrt{\tilde{S}_n}$ is t_{n-1}.

The statement is exact for every n, but it requires the data to be Gaussian; no central limit theorem is involved. Its distribution depends only on the known sample size, so its quantiles can be tabulated, one row per number of degrees of freedom. The extra randomness of the denominator gives t_d heavier tails than $\mathcal{N}(0,1)$, so its quantiles are larger and the resulting confidence intervals wider. By the law of large numbers $S_n \to \sigma^2$, so $t_d \to \mathcal{N}(0,1)$ as $d \to \infty$; for about 50 or more observations the two are practically indistinguishable, and the difference matters only for small samples.

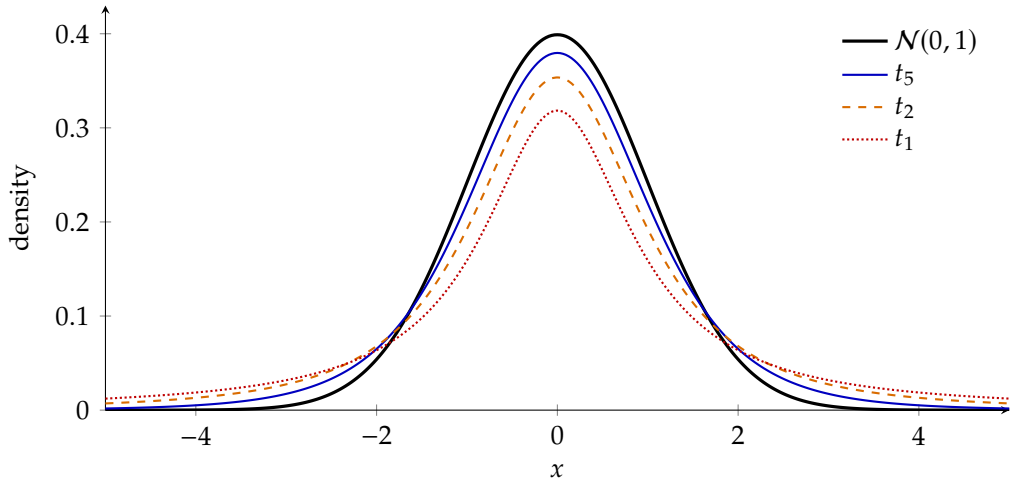


Figure 15: Densities of t_1, t_2, t_5 and $\mathcal{N}(0,1)$, plotted from their exact formulas. Fewer degrees of freedom move mass from the center into the tails.

9.2 Wald's test

Let $X_1, \dots, X_n$ be an i.i.d. sample from a model $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ with $\Theta \subseteq \mathbb{R}^d$, and let $\theta_0 \in \Theta$ be fixed. Test

$$H_0 : \theta = \theta_0 \qquad \text{vs.} \qquad H_1 : \theta \neq \theta_0.$$

Assume the technical conditions under which the MLE is asymptotically normal. Under H_0 the true parameter is θ_0, so

$$\sqrt{n}\big(\hat\theta_n^{\mathrm{MLE}} - \theta_0\big) \xrightarrow[n\to\infty]{(d)} \mathcal{N}_d\big(0, I(\theta_0)^{-1}\big) \quad \text{w.r.t. } \mathbb{P}_{\theta_0}.$$

To standardize a Gaussian vector $Y \sim \mathcal{N}_d(0,\Sigma)$ with Σ symmetric positive definite, use the matrix square root (take square roots of the eigenvalues): $\Sigma^{-1/2}Y \sim \mathcal{N}_d(0, \Sigma^{-1/2}\Sigma\Sigma^{-1/2}) = \mathcal{N}_d(0, I_d)$. With $\Sigma = I(\theta_0)^{-1}$, $\Sigma^{-1/2} = I(\theta_0)^{1/2}$. By Slutsky's theorem and continuity of I, the information may be evaluated at the MLE instead:

$$\sqrt{n}\, I\big(\hat\theta_n^{\mathrm{MLE}}\big)^{1/2}\big(\hat\theta_n^{\mathrm{MLE}} - \theta_0\big) \xrightarrow[n\to\infty]{(d)} \mathcal{N}_d(0, I_d) \quad \text{w.r.t. } \mathbb{P}_{\theta_0}.$$

Using $I(\theta_0)$ or $I(\hat\theta_n^{\mathrm{MLE}})$ is a choice, as for the variance in the earlier Bernoulli test of $p = 1/2$; only the second extends to nulls that do not specify a single parameter.

The left-hand side is asymptotically pivotal and computable. A vector is far from zero when its norm is large, and if $Z \sim \mathcal{N}_d(0, I_d)$ then $\|Z\|_2^2 = \sum_j Z_j^2 \sim \chi^2_d$, since the coordinates are independent standard Gaussians. Hence

$$T_n = n\big(\hat\theta_n^{\mathrm{MLE}} - \theta_0\big)^{\top} I\big(\hat\theta_n^{\mathrm{MLE}}\big)\big(\hat\theta_n^{\mathrm{MLE}} - \theta_0\big) \xrightarrow[n\to\infty]{(d)} \chi^2_d \quad \text{w.r.t. } \mathbb{P}_{\theta_0}.$$

Definition 9.2 [Wald's test]. Wald's test with asymptotic level $\alpha \in (0,1)$ is $\psi = \mathbf{1}\{T_n > q_\alpha\}$, where q_α is the $(1-\alpha)$-quantile of χ^2_d.

The threshold comes from the type I error: we need $\mathbb{P}_{\theta_0}[T_n > c] \to \alpha$, and the limit law of T_n under H_0 is χ^2_d, which has no unknown parameters. The test is also valid for alternatives such as $\theta > \theta_0$, $\theta < \theta_0$, or $\theta = \theta_1$ (in one-sided cases, by using the boundary value θ_0 closest to the alternative).

Example: *Why weight by the Fisher information.* Suppose $d = 3$, $\theta_0 = (1,1,2)$, and $\hat\theta_n^{\mathrm{MLE}} = (1.2, 0.9, 2.1)$. A naive distance is the unweighted sum $0.2^2 + 0.1^2 + 0.1^2 = 0.06$. Wald's statistic is instead the quadratic form $n\sum_{j,k} x_j x_k I_{jk}$ with $x = \hat\theta_n^{\mathrm{MLE}} - \theta_0$. If I is diagonal, this weights each coordinate by its information (the inverse of its variance); otherwise cross-terms appear. Directions of small variance count more. The Fisher information thus replaces Euclidean balls by ellipsoids whose axes need not be aligned with the coordinates.

9.3 The likelihood ratio test

For two simple hypotheses $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$, the natural rule asks which parameter is more likely: reject when $L_n(\theta_1)/L_n(\theta_0) > c$. Taking $c = 1$ gives no control of the type I error, so c is calibrated so that the level is α; $c \to \infty$ never rejects. The Neyman–Pearson lemma states that this test has the smallest type II error among tests of level α. Two simple hypotheses rarely arise in practice, so we use the same principle for composite nulls.

Let $\Theta \subseteq \mathbb{R}^d$ and, for fixed numbers $\theta^{(0)}_{r+1}, \dots, \theta^{(0)}_d$, let

$$H_0 : (\theta_{r+1}, \dots, \theta_d) = \big(\theta^{(0)}_{r+1}, \dots, \theta^{(0)}_d\big),$$

with $\theta_1, \dots, \theta_r$ unrestricted. Write Θ_0 for the parameters satisfying H_0; it is a whole family, not a point. For example, with $d = 3$, $H_0 : \theta_2 = 1,\ \theta_3 = 2$ allows every vector $(\theta_1, 1, 2)$. Let

$$\hat\theta_n = \operatorname*{arg\,max}_{\theta\in\Theta} \ell_n(\theta) \quad \text{(MLE)}, \qquad \hat\theta_n^c = \operatorname*{arg\,max}_{\theta\in\Theta_0} \ell_n(\theta) \quad \text{(constrained MLE)}.$$

A maximum over a larger set is at least as large, so $\ell_n(\hat\theta_n) \geq \ell_n(\hat\theta_n^c)$. If H_0 holds, the true parameter lies in Θ_0, the unconstrained MLE is close to it, and the two maxima should be nearly equal. If H_0 fails, the unconstrained maximum should be much larger.

Theorem 9.3 [Likelihood ratio test]. *Let*

$$T_n = 2\big(\ell_n(\hat\theta_n) - \ell_n(\hat\theta_n^c)\big) \geq 0.$$

If H_0 is true and the MLE technical conditions hold, then $T_n \xrightarrow[n\to\infty]{(d)} \chi^2_{d-r}$ w.r.t. $\mathbb{P}_\theta$, for every $\theta \in \Theta_0$. The likelihood ratio test *with asymptotic level α is $\psi = \mathbf{1}\{T_n > q_\alpha\}$, where q_α is the $(1-\alpha)$-quantile of χ^2_{d-r}.*

Since the limit is the same for every $\theta \in \Theta_0$, the maximal type I error over the composite null is asymptotically α. The degrees of freedom count the coordinates pinned down by H_0: the null is an affine subspace of dimension r in $\mathbb{R}^d$, defined by $d - r$ constraints. Only the dimension matters, so the same holds for a null of the form $(U\theta)_{r+1:d} = (\tau_{r+1}, \dots, \tau_d)$ with U invertible. In the example above, $T_n \to \chi^2_2$.

Wald's statistic and the likelihood ratio statistic generally differ, just as the MLE and the method of moments do. For the mean of a Gaussian with known covariance they coincide: there $2(\ell_n(\hat{\theta}_n) - \ell_n(\theta_0)) = n(\bar{X}_n - \theta_0)^\top \Sigma^{-1}(\bar{X}_n - \theta_0)$.

9.4 Testing implicit hypotheses

Sometimes the hypothesis concerns a function of the parameter, such as $\theta_1 - \theta_2$ (comparing two skills of the same subject) or a combination of two measurements. Let $X_1, \dots, X_n$ be i.i.d., let $\theta \in \mathbb{R}^d$ be a parameter of the distribution of X_1 (a moment, a model parameter, ...), and let $g : \mathbb{R}^d \to \mathbb{R}^k$ be continuously differentiable with $k < d$. Test

$$H_0 : g(\theta) = 0 \qquad \text{vs.} \qquad H_1 : g(\theta) \neq 0.$$

For example $g(\theta) = (\theta_1, \theta_2)$ ($k = 2$), or $g(\theta) = \theta_1 - \theta_2$ ($k = 1$). The case $g(\theta) = \theta - 0.5$ for a Bernoulli parameter is the earlier Bernoulli test of $p = 1/2$.

Suppose an asymptotically normal estimator is available, not necessarily the MLE: $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{(d)} \mathcal{N}_d(0, \Sigma(\theta))$. The multivariate delta method gives

$$\sqrt{n}\big(g(\hat{\theta}_n) - g(\theta)\big) \xrightarrow[n\to\infty]{(d)} \mathcal{N}_k\big(0, \Gamma(\theta)\big), \qquad \Gamma(\theta) = \nabla g(\theta)^\top \Sigma(\theta) \nabla g(\theta) \in \mathbb{R}^{k\times k},$$

where $\nabla g(\theta) \in \mathbb{R}^{d\times k}$ has the gradients of the k coordinates of g as columns. Assume $\Sigma(\theta)$ is invertible and $\nabla g(\theta)$ has rank k; then $\Gamma(\theta)$ is invertible and $\sqrt{n}\,\Gamma(\theta)^{-1/2}(g(\hat{\theta}_n) - g(\theta)) \to \mathcal{N}_k(0, I_k)$. If Γ is continuous, Slutsky's theorem lets us replace $\Gamma(\theta)$ by $\Gamma(\hat{\theta}_n)$. Here the plug-in is unavoidable: H_0 fixes $g(\theta)$, not θ, so there is no θ_0 to substitute unless g can be inverted. Under H_0, $g(\theta) = 0$, and taking squared norms,

$$T_n = n\, g(\hat{\theta}_n)^\top \Gamma(\hat{\theta}_n)^{-1} g(\hat{\theta}_n) \xrightarrow[n\to\infty]{(d)} \chi^2_k.$$

The test with asymptotic level α is $\psi = \mathbf{1}\{T_n > q_\alpha\}$, with q_α the $(1-\alpha)$-quantile of χ^2_k. The invertibility assumptions must be checked in each model; they fail for the multinomial model of Lecture 10.

Note. For $k = 1$ nothing new happens. Then $T_n = W_n^2$ with $W_n = \sqrt{n}\, g(\hat{\theta}_n)/\sqrt{\Gamma(\hat{\theta}_n)} \to \mathcal{N}(0,1)$, and the $(1-\alpha)$-quantile of χ_1^2 is the square of the $(1-\alpha/2)$-quantile $q_{\alpha/2}$ of $\mathcal{N}(0,1)$. So $\{T_n > q_\alpha(\chi_1^2)\} = \{|W_n| > q_{\alpha/2}\}$: the chi-squared test is exactly the two-sided Gaussian test used so far. Geometry, and with it the chi-squared distribution with several degrees of freedom, enters only in higher dimensions.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 9, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 10: Parametric Hypothesis Testing (cont.)

The chi-squared test checks whether a distribution on a finite set equals a given one. It is Wald's idea applied to a model whose asymptotic covariance matrix is singular.

10.1 The multinomial model

Let $E = \{a_1, \dots, a_K\}$ be a finite space, and let

$$\Delta_K = \left\{ \mathbf{p} = (p_1, \dots, p_K) \in (0,1)^K : \sum_{j=1}^{K} p_j = 1 \right\}$$

be the probability simplex. For $\mathbf{p} \in \Delta_K$ and $X \sim \mathbb{P}_{\mathbf{p}}$, set $\mathbb{P}_{\mathbf{p}}[X = a_j] = p_j$ for $j = 1, \dots, K$. The family $(\mathbb{P}_{\mathbf{p}})_{\mathbf{p} \in \Delta_K}$ contains every distribution on E with positive probabilities.

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathbb{P}_{\mathbf{p}}$ for an unknown $\mathbf{p} \in \Delta_K$, and fix $\mathbf{p}^0 \in \Delta_K$. We test, with asymptotic level $\alpha \in (0,1)$,

$$H_0 : \mathbf{p} = \mathbf{p}^0 \qquad \text{vs.} \qquad H_1 : \mathbf{p} \neq \mathbf{p}^0.$$

For $\mathbf{p}^0 = (1/K, \dots, 1/K)$ this tests whether $\mathbb{P}_{\mathbf{p}}$ is the uniform distribution on E.

10.2 The maximum likelihood estimator

With counts $N_j = \#\{i = 1, \dots, n : X_i = a_j\}$, the likelihood is

$$L_n(X_1, \dots, X_n, \mathbf{p}) = p_1^{N_1} p_2^{N_2} \cdots p_K^{N_K}.$$

Proposition 10.1 [Multinomial MLE]. *The MLE is $\hat{\mathbf{p}}_j = N_j/n$ for $j = 1, \dots, K$.*

Proof. The maximization must respect the constraint $\sum_j p_j = 1$: without it, $\log L_n = \sum_j N_j \log p_j$ increases in every p_j and has no maximizer. With a Lagrange multiplier λ, stationarity of $\sum_j N_j \log p_j - \lambda(\sum_j p_j - 1)$ gives $N_j/p_j = \lambda$, so $p_j = N_j/\lambda$. Summing over j gives $\lambda = \sum_j N_j = n$. The objective is concave, so this stationary point is the maximizer. □

10.3 The chi-squared test

Theorem 10.2 [Chi-squared statistic]. *If H_0 is true, then $\sqrt{n}(\hat{\mathbf{p}} - \mathbf{p}^0)$ is asymptotically normal, and*

$$T_n = n \sum_{j=1}^{K} \frac{(\hat{\mathbf{p}}_j - \mathbf{p}_j^0)^2}{\mathbf{p}_j^0} \xrightarrow[n\to\infty]{(d)} \chi^2_{K-1}.$$

Proof. Write $e_{X_i} \in \mathbb{R}^K$ for the indicator vector with a 1 in position j when $X_i = a_j$. Then $\hat{\mathbf{p}}$ is the average of the i.i.d. vectors e_{X_i}, with mean $\mathbf{p}^0$ under H_0 and covariance $\mathbb{E}[e_X e_X^\top] - \mathbf{p}^0 \mathbf{p}^{0\top}$. Since $e_X e_X^\top$ is diagonal with $\mathbb{E}$ equal to $D = \mathrm{diag}(\mathbf{p}^0)$, the multivariate CLT gives

$$\sqrt{n}(\hat{\mathbf{p}} - \mathbf{p}^0) \xrightarrow{(d)} \mathcal{N}_K(0, \Sigma), \qquad \Sigma = D - \mathbf{p}^0 \mathbf{p}^{0\top}.$$

This Σ is singular: $\Sigma \mathbf{1} = \mathbf{p}^0 - \mathbf{p}^0 = 0$, reflecting that the coordinates of $\hat{\mathbf{p}} - \mathbf{p}^0$ always sum to zero. So it is not the inverse of a Fisher information matrix, and Wald's test cannot be applied as stated. Standardize by $D^{-1/2}$ instead. With $\mathbf{s} = (\sqrt{\mathbf{p}_1^0}, \dots, \sqrt{\mathbf{p}_K^0})$,

$$Z_n = \sqrt{n}\, D^{-1/2}(\hat{\mathbf{p}} - \mathbf{p}^0) \xrightarrow{(d)} \mathcal{N}_K\big(0, D^{-1/2}\Sigma D^{-1/2}\big) = \mathcal{N}_K(0, I_K - \mathbf{s}\mathbf{s}^\top).$$

Since $\|\mathbf{s}\|^2 = \sum_j \mathbf{p}_j^0 = 1$, the matrix $P = I_K - \mathbf{s}\mathbf{s}^\top$ is the orthogonal projection onto the $(K-1)$-dimensional complement of $\mathbf{s}$. In an orthonormal basis whose last vector is $\mathbf{s}$, a vector $Y \sim \mathcal{N}_K(0, P)$ has $K-1$ independent standard Gaussian coordinates and a last coordinate equal to zero, so $\|Y\|^2 \sim \chi^2_{K-1}$. Finally $T_n = \|Z_n\|^2$, and the continuous mapping theorem gives the claim. □

Definition 10.3 [Chi-squared test]. The χ^2 test with asymptotic level α is $\psi_\alpha = \mathbf{1}\{T_n > q_\alpha\}$, where q_α is the $(1-\alpha)$-quantile of χ^2_{K-1}. Its asymptotic p-value is $\mathbb{P}[Z > T_n \mid T_n]$, where $Z \sim \chi^2_{K-1}$ and $Z \perp\!\!\!\perp T_n$.

In terms of counts, with $n\mathbf{p}_j^0$ the expected count of cell j under H_0,

$$T_n = \sum_{j=1}^{K} \frac{(N_j - n\mathbf{p}_j^0)^2}{n\mathbf{p}_j^0},$$

the familiar sum of squared differences between observed and expected counts, divided by the expected counts.

Note. The weights $1/\mathbf{p}_j^0$ are not arbitrary. The matrix D^{-1} is a generalized inverse of Σ: using $\mathbf{p}^{0\top} D^{-1} \mathbf{p}^0 = \sum_j \mathbf{p}_j^0 = 1$,

$$\Sigma D^{-1} \Sigma = D - 2\mathbf{p}^0 \mathbf{p}^{0\top} + \mathbf{p}^0 (\mathbf{p}^{0\top} D^{-1} \mathbf{p}^0) \mathbf{p}^{0\top} = \Sigma.$$

So $T_n = n(\hat{\mathbf{p}} - \mathbf{p}^0)^\top D^{-1} (\hat{\mathbf{p}} - \mathbf{p}^0)$ is Wald's statistic with this generalized inverse in place of the Fisher information, and the constraint $\sum_j p_j = 1$ removes one degree of freedom.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 10, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 11: Parametric Hypothesis Testing (cont.) and Testing Goodness of Fit

As before, q_α denotes the $(1-\alpha)$-quantile of the relevant pivotal distribution.

11.1 The chi-squared test in practice

Two data sets show the kind of question the χ^2 test of Lecture 10 answers.

- *Zodiac signs of CEOs.* The signs of 256 heads of Fortune 500 companies were recorded. Under uniformity each of the 12 signs has expected count $256/12 \approx 21.33$; one sign (Pisces) has 29. The test of $\mathbf{p}^0 = (1/12, \dots, 1/12)$, the center of the simplex, decides whether such deviations are evidence against uniformity.
- *Composition of juries.* For 275 jurors classified by racial group, the benchmark is the population proportions $\mathbf{p}^0 = (0.72, 0.07, 0.12, 0.09)$ (white, black, Hispanic, other). Here $\mathbf{p}^0$ is not uniform.

The weights $1/\mathbf{p}_j^0$ in the statistic come from standardizing by the asymptotic covariance; they are not a matter of taste. The Fisher information does not exist in the parameterization by $(p_1, \dots, p_K)$ because of the constraint $\sum_j p_j = 1$. Writing the model in terms of $(p_1, \dots, p_{K-1})$, with $p_K = 1 - \sum_{j<K} p_j$, removes the redundant direction and makes it exist.

11.2 Student's test

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ unknown, and fix $\mu_0 \in \mathbb{R}$. Test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.

If σ^2 is known, $T_n = \sqrt{n}(\bar{X}_n - \mu_0)/\sigma \sim \mathcal{N}(0,1)$ exactly under H_0, so $\psi_\alpha = \mathbf{1}\{|T_n| > q_{\alpha/2}\}$ has (non-asymptotic) level α. If σ^2 is unknown, let S_n be the sample variance and

$$\widetilde{T}_n = \sqrt{n-1}\,\frac{\bar{X}_n - \mu_0}{\sqrt{S_n}}.$$

By Cochran's theorem, $\bar{X}_n \perp\!\!\!\perp S_n$ and $nS_n/\sigma^2 \sim \chi^2_{n-1}$, so under H_0, $\widetilde{T}_n \sim t_{n-1}$ (Lecture 9).

Definition 11.1 [Student's test]. Student's test with level $\alpha \in (0,1)$ is $\psi_\alpha = \mathbf{1}\{|\widetilde{T}_n| > q_{\alpha/2}\}$, with $q_{\alpha/2}$ the $(1-\alpha/2)$-quantile of t_{n-1}. For $H_1 : \mu > \mu_0$ it is $\psi'_\alpha = \mathbf{1}\{\widetilde{T}_n > q_\alpha\}$, with q_α the $(1-\alpha)$-quantile of t_{n-1}.

The t_{n-1} quantiles exceed the Gaussian ones, so more evidence is needed to reject: the test accounts

for the fluctuations of S_n. Its advantages are that it is non-asymptotic and can be run on small samples (say 15 observations); for large n it agrees with the Gaussian test, which is why software reports t tests by default. Its drawback is that it relies on the sample being Gaussian. With a small sample this is an assumption to be checked, not a consequence of a limit theorem. Checking it is a goodness-of-fit problem with null hypothesis "the data are Gaussian". Unusually, we then hope to keep H_0, and since there are so many ways of not being Gaussian, such tests have low power.

11.3 Two-sample tests

Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be independent random variables with $\mathbb{E}[X_i] = \mu_X$ and $\mathbb{E}[Y_i] = \mu_Y$ for all i. Test $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$.

Large samples, known variances. Assume without loss of generality that all variances equal 1. By the CLT and independence of the two samples, $\mathrm{Var}(\bar{X}_n - \bar{Y}_m) = 1/n + 1/m$, and as $n, m \to \infty$ with $n/m \to \gamma$,

$$\sqrt{n}\big(\bar{X}_n - \bar{Y}_m - (\mu_X - \mu_Y)\big) \xrightarrow{(d)} \mathcal{N}(0, 1 + \gamma).$$

Under H_0, $\sqrt{n}(\bar{X}_n - \bar{Y}_m)/\sqrt{1 + n/m} = (\bar{X}_n - \bar{Y}_m)/\sqrt{1/n + 1/m} \to \mathcal{N}(0,1)$, giving the test

$$\psi_\alpha = \mathbf{1}\Big\{\frac{|\bar{X}_n - \bar{Y}_m|}{\sqrt{1/n + 1/m}} > q_{\alpha/2}\Big\}.$$

Gaussian samples, unknown variances. If $X_i \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y_i \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, then $\bar{X}_n - \bar{Y}_m \sim \mathcal{N}(\mu_X - \mu_Y, \sigma_X^2/n + \sigma_Y^2/m)$, so under H_0 the ratio $(\bar{X}_n - \bar{Y}_m)/\sqrt{\sigma_X^2/n + \sigma_Y^2/m}$ is exactly $\mathcal{N}(0,1)$. With the variances replaced by the unbiased sample variances S_X^2, S_Y^2, the TWO-SAMPLE T TEST uses

$$\frac{\bar{X}_n - \bar{Y}_m}{\sqrt{S_X^2/n + S_Y^2/m}} \approx t_N, \qquad N = \frac{\big(S_X^2/n + S_Y^2/m\big)^2}{\dfrac{S_X^4}{n^2(n-1)} + \dfrac{S_Y^4}{m^2(m-1)}},$$

an approximation (Welch–Satterthwaite) in which N need not be an integer.

11.4 Goodness-of-fit tests

Given i.i.d. copies of a random variable X, a GOODNESS-OF-FIT TEST asks whether a hypothesized distribution fits the data: does X have distribution $\mathcal{N}(0,1)$, as required by Student's test? Is it $\mathcal{U}([0,1])$? Does it have PMF $p_1 = 0.3$, $p_2 = 0.5$, $p_3 = 0.2$? The χ^2 test is an example. The key characteristic is that there is no parametric model: the question is about the whole distribution, not one parameter.

Example: *p-values are uniform under the null.* For $H_0 : \mu = 0$ with known σ, the two-sided p-value at an observed mean $\bar{x}_n$ is $\mathbb{P}[|Z| > \sqrt{n}|\bar{x}_n|/\sigma] = 2\Phi(-\sqrt{n}|\bar{x}_n|/\sigma)$, with $Z \sim \mathcal{N}(0,1)$ and Φ its cdf. Plugging in the random $\bar{X}_n$ makes the p-value a statistic $2\Phi(-|Z|)$, with $Z = \sqrt{n}\bar{X}_n/\sigma \sim \mathcal{N}(0,1)$ under H_0. Since $-|Z| \le 0$, it takes values in $[0,1]$, and for $t \in [0,1]$,

$$\mathbb{P}\big[2\Phi(-|Z|) \le t\big] = \mathbb{P}\big[|Z| \ge -\Phi^{-1}(t/2)\big] = 2\Phi\big(\Phi^{-1}(t/2)\big) = t.$$

So the p-value is $\mathcal{U}([0,1])$ under H_0, and the same holds for any test whose statistic has a continuous null distribution. To validate a new test, one can simulate many data sets under H_0, compute their p-values, and test whether these are uniform: a goodness-of-fit problem.

11.5 The empirical cdf

Let $X_1, \dots, X_n$ be i.i.d. real random variables. The cdf $F(t) = \mathbb{P}[X_1 \le t]$, $t \in \mathbb{R}$, completely characterizes the distribution of X_1. Writing $F(t) = \mathbb{E}[\mathbf{1}\{X_1 \le t\}]$ and replacing the expectation by an average gives an estimator.

Definition 11.2 [Empirical cdf]. The empirical cdf of the sample is

$$F_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \le t\} = \frac{\#\{i = 1, \dots, n : X_i \le t\}}{n}, \qquad t \in \mathbb{R}.$$

For each t, the indicators are i.i.d. Bernoulli with parameter $F(t)$, so the strong law of large numbers gives $F_n(t) \to F(t)$ almost surely. A uniform statement is much stronger.

Theorem 11.3 [Glivenko–Cantelli, fundamental theorem of statistics].

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \xrightarrow[n\to\infty]{a.s.} 0.$$

Pointwise convergence lets the sample size needed for a given accuracy, $n(t, \varepsilon, \delta)$ with $\mathbb{P}[|F_n(t) - F(t)| > \delta] < \varepsilon$, depend on t; if it grew without bound in t, no single n would serve all t and the supremum would not converge. Glivenko–Cantelli rules this out.

The Bernoulli variance is $F(t)(1 - F(t))$, so the CLT gives, for each t,

$$\sqrt{n}\big(F_n(t) - F(t)\big) \xrightarrow{(d)} \mathcal{N}\big(0, F(t)(1 - F(t))\big).$$

The supremum of these Gaussians is not Gaussian. Indexed by t, they form a stochastic process, and its limit is a Brownian bridge $\mathbb{B}$ on $[0,1]$: a Gaussian process like Brownian motion but pinned to $\mathbb{B}(0) = \mathbb{B}(1) = 0$, matching the variance $F(t)(1 - F(t))$, which vanishes as $t \to \pm\infty$. Its marginal variance at u is $u(1-u)$, the pointwise variance above with $u = F(t)$.

Theorem 11.4 [Donsker]. *If F is continuous, then*

$$\sqrt{n}\sup_{t\in\mathbb{R}}|F_n(t)-F(t)| \xrightarrow[n\to\infty]{(d)} \sup_{0\le u\le 1}|\mathbb{B}(u)|.$$

The time scales differ: $\sqrt{n}(F_n(t)-F(t))$ behaves like $\mathbb{B}(F(t))$, and as t ranges over $\mathbb{R}$, $F(t)$ ranges over $(0,1)$. The distribution of $\sup_u|\mathbb{B}(u)|$ is known explicitly and tabulated.

11.6 The Kolmogorov–Smirnov test

Let $X_1,\dots,X_n$ be i.i.d. real random variables with unknown cdf F, let F^0 be a *continuous* cdf, and test

$$H_0: F = F^0 \qquad \text{vs.} \qquad H_1: F \neq F^0.$$

Goodness-of-fit hypotheses are formulated through cdfs because the empirical cdf gives a handle on them. Under H_0, $F_n(t)\approx F^0(t)$ for all t.

Definition 11.5 [Kolmogorov–Smirnov test]. Let $T_n = \sup_{t\in\mathbb{R}}\sqrt{n}\,|F_n(t)-F^0(t)|$. By Donsker's theorem, under H_0, $T_n \xrightarrow{(d)} Z = \sup_{0\le u\le 1}|\mathbb{B}(u)|$. The Kolmogorov–Smirnov (KS) test with asymptotic level α is $\delta_\alpha^{KS} = \mathbf{1}\{T_n > q_\alpha\}$, where q_α is the $(1-\alpha)$-quantile of Z. Its p-value is $\mathbb{P}[Z > T_n \mid T_n]$.

The statistic is nonnegative and Z is not symmetric, so the test is one-sided: the threshold is q_α, not $q_{\alpha/2}$.

Computing T_n. The supremum over all t reduces to $2n$ values. Let $X_{(1)}\le X_{(2)}\le\cdots\le X_{(n)}$ be the order statistics, the sample sorted increasingly.

Proposition 11.6 [Practical formula].

$$T_n = \sqrt{n}\max_{i=1,\dots,n}\max\left\{\left|\frac{i-1}{n}-F^0(X_{(i)})\right|,\ \left|\frac{i}{n}-F^0(X_{(i)})\right|\right\}.$$

Proof. F_n is piecewise constant, jumping by $1/n$ at each observation, with $F_n = (i-1)/n$ just before $X_{(i)}$ and $F_n(X_{(i)}) = i/n$; it is 0 before $X_{(1)}$ and 1 from $X_{(n)}$ on. On an interval where F_n equals a constant c, the function F^0 is nondecreasing and continuous, so $|c-F^0(t)|$ is largest at an endpoint. At a left endpoint $X_{(i)}$ this is $|i/n - F^0(X_{(i)})|$ (with $c = i/n$); toward a right endpoint $X_{(i+1)}$ it approaches $|i/n - F^0(X_{(i+1)})|$, which is the "just before the jump" term for index $i+1$. The two unbounded end intervals contribute $F^0(X_{(1)})$ and $1-F^0(X_{(n)})$, which are also of these forms. Hence the supremum is a maximum over these $2n$ numbers. □

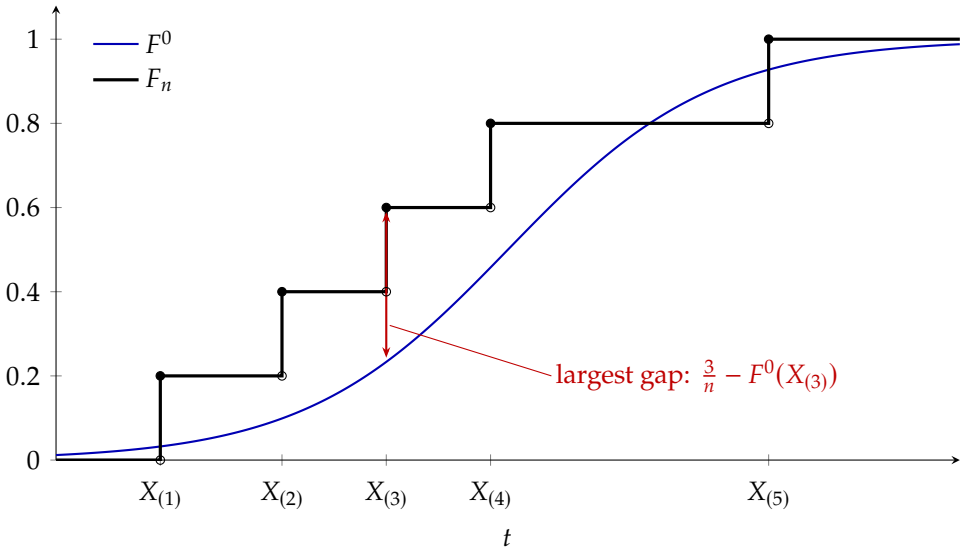


Figure 16: Schematic: the empirical cdf of five illustrative observations (filled dots are attained values, open dots the values just before each jump) against a continuous cdf F^0. The largest vertical gap, which defines $T_n/\sqrt{n}$, occurs at an observation.

Pivotality. T_n is a PIVOTAL STATISTIC: under H_0 its distribution does not depend on F^0, for every n, so it is easy to simulate. Let $U_i = F^0(X_i)$ and let G_n be the empirical cdf of $U_1, \dots, U_n$. Under H_0, $U_1, \dots, U_n \overset{\text{iid}}{\sim} \mathcal{U}([0,1])$, because F^0 is continuous. Moreover $X_i \le t$ implies $F^0(X_i) \le F^0(t)$, and the converse fails only when X_i falls in an interval where F^0 is flat, which has probability zero. So $F_n(t) = G_n(F^0(t))$ almost surely, and since $F^0(t)$ sweeps $(0,1)$,

$$T_n = \sup_{0 \le x \le 1} \sqrt{n}\,|G_n(x) - x|.$$

To tabulate the quantile $q_\alpha^{(n)}$ of T_n for a given n, simulate M i.i.d. copies $T_n^1, \dots, T_n^M$ of this statistic for some large M, and take their sample $(1-\alpha)$-quantile $\hat{q}_\alpha^{(n,M)}$. The test $\delta_\alpha = \mathbf{1}\{T_n > \hat{q}_\alpha^{(n,M)}\}$ has approximate level α, and its p-value is approximately $\#\{j = 1, \dots, M : T_n^j > T_n\}/M$. Such quantiles are precomputed in tables; for instance, for $n = 10$ and $\alpha = 5\%$, the threshold for $\sup_t |F_n(t) - F^0(t)|$ is about 0.41.

Continuity of F^0 is needed for this limit and for pivotality. Note also that F^0 must be fully specified, such as $\mathcal{N}(1,5)$; the question "is X Gaussian?" with estimated mean and variance requires a different table.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 11, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 12: Testing Goodness of Fit (cont.)

As before, q_α denotes the $(1-\alpha)$-quantile of the relevant limiting or pivotal distribution, F_n is the empirical cdf of the i.i.d. sample $X_1, \dots, X_n$, and $X_{(1)} \le \cdots \le X_{(n)}$ are its order statistics.

12.1 Other distances between cdfs

The Kolmogorov–Smirnov test measures the distance between F_n and F in the sup norm. Other distances lead to other tests:

$$\begin{aligned}
&\text{Kolmogorov–Smirnov:} && d(F_n, F) = \sup_{t\in\mathbb{R}} |F_n(t) - F(t)|,\\
&\text{Cramér–von Mises:} && d^2(F_n, F) = \int_{\mathbb{R}} [F_n(t) - F(t)]^2 \, dt,\\
&\text{Anderson–Darling:} && d^2(F_n, F) = \int_{\mathbb{R}} \frac{[F_n(t) - F(t)]^2}{F(t)(1-F(t))} \, dt.
\end{aligned}$$

Cramér–von Mises uses a squared L^2 distance; Anderson–Darling weights it by the inverse pointwise variance $F(t)(1-F(t))$, which emphasizes the tails. For each, the suitably scaled statistic has a pivotal asymptotic distribution with tabulated quantiles. Each test is more powerful against different alternatives, and since a goodness-of-fit test is usually run hoping to keep H_0, there is rarely a reason to prefer one. Kolmogorov–Smirnov is the most widely used.

12.2 Composite goodness of fit: the Kolmogorov–Lilliefors test

The usual question is not whether $X \sim \mathcal{N}(1, 5)$ but whether X is Gaussian, that is, $X \sim \mathcal{N}(\mu, \sigma^2)$ for some unknown μ, σ^2. The natural Gaussian to compare with has the empirical mean and variance, $\hat\mu = \bar X_n$ and $\hat\sigma^2 = S_n$. The plug-in statistic is

$$\sup_{t\in\mathbb{R}} \left| F_n(t) - \Phi_{\hat\mu,\hat\sigma^2}(t) \right|,$$

where $\Phi_{\hat\mu,\hat\sigma^2}$ is the cdf of $\mathcal{N}(\hat\mu, \hat\sigma^2)$. Donsker's theorem no longer applies to it, and comparing it with Kolmogorov–Smirnov quantiles is a common and serious mistake. Its own quantiles can be computed, because it is still pivotal under H_0.

Proposition 12.1 [Pivotality of the plug-in statistic]. *If $X_1, \dots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$, the distribution of $\sup_t |F_n(t) - \Phi_{\hat\mu,\hat\sigma^2}(t)|$ does not depend on μ or σ^2.*

Proof. If $Y = \hat{\sigma}Z + \hat{\mu}$ with $Z \sim \mathcal{N}(0,1)$, then $\Phi_{\hat{\mu},\hat{\sigma}^2}(t) = \mathbb{P}[Z \le (t-\hat{\mu})/\hat{\sigma}] = \Phi((t-\hat{\mu})/\hat{\sigma})$, with Φ the standard Gaussian cdf. Substitute $u = (t-\hat{\mu})/\hat{\sigma}$, which ranges over $\mathbb{R}$ as t does:

$$\sup_{t\in\mathbb{R}} \left| F_n(t) - \Phi_{\hat{\mu},\hat{\sigma}^2}(t) \right| = \sup_{u\in\mathbb{R}} \left| \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\left\{ \frac{X_i - \hat{\mu}}{\hat{\sigma}} \le u \right\} - \Phi(u) \right|.$$

Write $X_i = \sigma Z_i + \mu$ with $Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ and divide numerator and denominator by σ:

$$\frac{X_i - \hat{\mu}}{\hat{\sigma}} = \frac{Z_i - (\hat{\mu}-\mu)/\sigma}{\hat{\sigma}/\sigma}.$$

Here $(\hat{\mu}-\mu)/\sigma = \bar{Z}_n$ and $\hat{\sigma}^2/\sigma^2 = \frac{1}{n}\sum_i (Z_i - \bar{Z}_n)^2$ are functions of $Z_1, \dots, Z_n$ alone. (Their laws are $\mathcal{N}(0, 1/n)$ and χ^2_{n-1}/n, and they are independent by Cochran's theorem, so neither their marginals nor their dependence involve μ, σ.) The whole statistic is therefore a function of $Z_1, \dots, Z_n$, whose law is fixed. □

Using these quantiles is the Kolmogorov–Lilliefors test; they were first tabulated by simulation. They are considerably smaller than the Kolmogorov–Smirnov quantiles: for $n = 10$ and $\alpha = 5\%$, the threshold for the sup distance is about 0.26, against 0.41 for a fully specified F^0. The reason is that $\hat{\mu}$ and $\hat{\sigma}^2$ are fitted to the same data, so the fitted cdf follows F_n more closely than the true cdf would. The two tests use the same statistic, so a failure to reject with Kolmogorov–Lilliefors implies a failure to reject with Kolmogorov–Smirnov, and on the whole range between the two thresholds they disagree. Using the Kolmogorov–Smirnov table after plugging in estimates rejects normality far too rarely.

12.3 Quantile-quantile plots

Numerical tests give a verdict but not a diagnosis. When the p-value is borderline, or to see *how* the data depart from a distribution, a visual check helps. What matters about a Gaussian is mostly its tails, which decay like $e^{-x^2/2}$; bumps near the center matter less. Plotting F_n against F is uninformative, because both are close to 0 or 1 exactly where the tails live. Instead, compare the inverse functions, which map $(0,1)$ to the whole real line.

F_n is not invertible, but it takes the value i/n on $[X_{(i)}, X_{(i+1)})$, so define $F_n^{-1}(i/n) = X_{(i)}$, the ith smallest observation.

Definition 12.2 [QQ plot]. The quantile-quantile (QQ) plot of the sample against a cdf F is the scatter plot of the points

$$\left(F^{-1}\left(\tfrac{1}{n}\right), F_n^{-1}\left(\tfrac{1}{n}\right)\right),\ \left(F^{-1}\left(\tfrac{2}{n}\right), F_n^{-1}\left(\tfrac{2}{n}\right)\right),\ \dots,\ \left(F^{-1}\left(\tfrac{n-1}{n}\right), F_n^{-1}\left(\tfrac{n-1}{n}\right)\right).$$

The x-coordinates are the theoretical quantiles and the y-coordinates, the order statistics, are the empirical quantiles.

If the data come from F, then $F_n \approx F$ and the points lie near the line $y = x$. It is not a formal test but a quick and easy check that a distribution is plausible. In a NORMAL QQ PLOT, F is Gaussian. Since $\Phi_{\mu,\sigma^2}(t) = \Phi((t-\mu)/\sigma)$, plotting against the standard Gaussian quantiles turns $y = x$ into a line with slope $\hat{\sigma}$ and intercept $\hat{\mu}$; software either draws this reference line or rescales the axes so that the reference is again the 45° line. Any straight line therefore indicates a good fit.

Even for Gaussian data the plot is never perfectly straight. With 10,000 standard Gaussian observations it is essentially the line; with 50 or 10 the extreme points scatter, because the tails consist of rare values. Conversely, a sample from t_{15}, whose tails are heavier, is hard to tell from Gaussian already at 50 observations: t_{15} is close to a Gaussian with a slightly larger variance.

Reading the tails. The right end of the plot concerns large values. If the points there lie below the reference line, the empirical quantiles are smaller than the theoretical ones at the same level i/n, so the right tail is *lighter* than Gaussian; above the line means a heavier right tail. At the left end the comparison flips sign, since the quantiles are negative: points above the line mean a lighter left tail, points below a heavier one. Hence

- light tails on both sides (e.g. uniform) give an S shape, flat at both ends;
- heavy tails on both sides (e.g. Cauchy) give the reverse S, steep at both ends;
- the exponential has no left tail (light) and a right tail decaying like e^{-x} (heavy), giving a convex U shape.

Lighter tails are better-behaved noise: large deviations are then harder to explain by randomness alone.

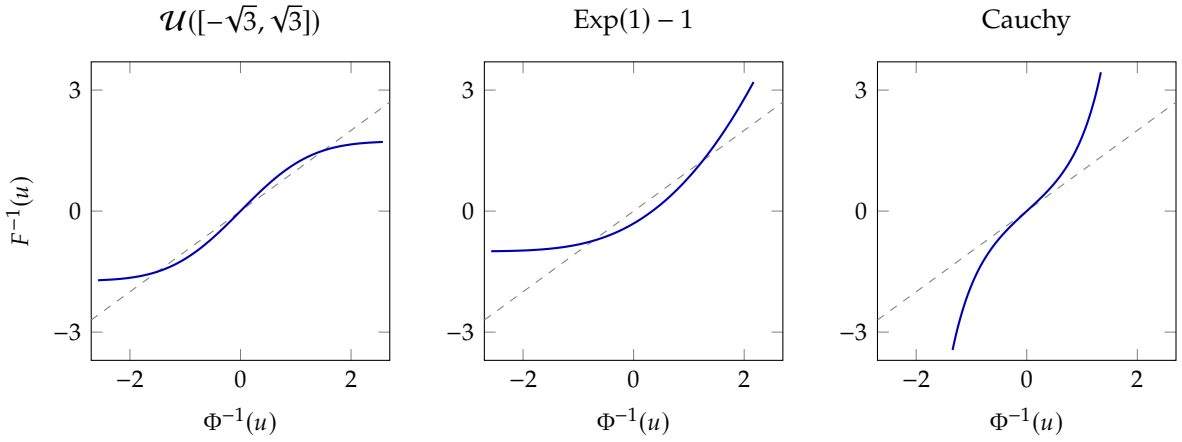


Figure 17: Population QQ curves $u \mapsto (\Phi^{-1}(u), F^{-1}(u))$ against the standard Gaussian, computed from the exact quantile functions (the uniform and exponential are standardized to mean 0 and variance 1; the Cauchy curve is cut off at $|y| = 3.6$). The dashed line is $y = x$. A sample QQ plot scatters around such a curve.

12.4 The chi-squared goodness-of-fit test: finite case

The χ^2 test of Lecture 10 compares the data with a probability vector $\mathbf{p}^0$ given in advance. Usually these probabilities come from a model whose parameter must be estimated. Let $X_1, \dots, X_n$ be i.i.d. on a finite space $E = \{a_1, \dots, a_K\}$ with distribution $\mathbb{P}$, and let $(\mathbb{P}_\theta)_{\theta\in\Theta}$ with $\Theta \subseteq \mathbb{R}^d$ be a parametric family on E. Set

$$p_j(\theta) = \mathbb{P}_\theta[Y = a_j] \ \ (Y \sim \mathbb{P}_\theta), \qquad p_j = \mathbb{P}[X_1 = a_j].$$

We test whether the statistical model $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ fits the data:

$$H_0 : \mathbb{P} \in (\mathbb{P}_\theta)_{\theta\in\Theta} \qquad \text{vs.} \qquad H_1 : \mathbb{P} \notin (\mathbb{P}_\theta)_{\theta\in\Theta}.$$

Equivalently, H_0 states that $p_j = p_j(\theta)$ for all j, for some $\theta \in \Theta$.

Let $\hat\theta$ be the MLE of θ assuming H_0, and $\hat p_j = \frac{1}{n}\sum_i \mathbf{1}\{X_i = a_j\} = \#\{i : X_i = a_j\}/n$. If H_0 is true, both $\hat p_j$ and $p_j(\hat\theta)$ estimate p_j, so $\hat p_j \approx p_j(\hat\theta)$ for all j. Define

$$T_n = n \sum_{j=1}^{K} \frac{\big(\hat p_j - p_j(\hat\theta)\big)^2}{p_j(\hat\theta)}.$$

Theorem 12.3 [Chi-squared goodness of fit]. *Under some technical assumptions, if H_0 is true, then $T_n \xrightarrow[n\to\infty]{(d)} \chi^2_{K-d-1}$, where $d < K-1$ is the dimension of θ. The test with asymptotic level α is $\delta_\alpha = \mathbf{1}\{T_n > q_\alpha\}$, with q_α the $(1-\alpha)$-quantile of χ^2_{K-d-1}, and its p-value is $\mathbb{P}[Z > T_n \mid T_n]$ with $Z \sim \chi^2_{K-d-1}$, $Z \perp\!\!\!\perp T_n$.*

Degrees of freedom count the directions in which the truth can differ from the null. Without a model, a point of the $(K-1)$-dimensional simplex can move away from $\mathbf{p}^0$ in $K-1$ directions. A d-dimensional family $\{\mathbf{p}(\theta)\}$ is a d-dimensional surface in the simplex, and moving along it does not violate H_0, so only $K-1-d$ directions remain. The larger the family, the fewer the degrees of freedom; if the family were the whole simplex, there would be none. Unlike the Kolmogorov–Lilliefors statistic, T_n is not pivotal for finite n; only its limit is free of the parameter.

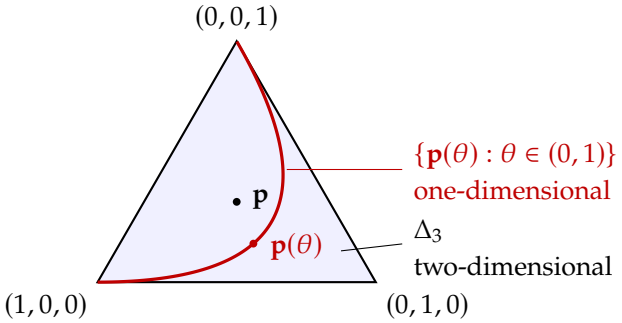


Figure 18: For $K = 3$, the simplex Δ_3 is a triangle. The binomial family $\mathbf{p}(\theta) = ((1-\theta)^2, 2\theta(1-\theta), \theta^2)$ on $\{0, 1, 2\}$ is a curve inside it; a distribution $\mathbf{p}$ off the curve violates H_0. Testing membership in the family leaves $K - 1 - d = 1$ degree of freedom.

Example: *Binomial data.* Suppose each observation is a count in $\{0, 1, \dots, N\}$ and H_0 says it is $\mathrm{Bin}(N, \theta)$ for some $\theta \in (0, 1)$. Then $K = N + 1$ cells, $p_j(\theta) = \binom{N}{j}\theta^j(1-\theta)^{N-j}$ for $j = 0, \dots, N$, $\hat{\theta}$ is the MLE, and $d = 1$, so $T_n \to \chi^2_{K-2}$. The binomial serves only as a recipe for the test probabilities, which replace the given vector of Lecture 10.

12.5 The chi-squared goodness-of-fit test: infinite case

If E is infinite (e.g. $E = \mathbb{N}$ or $E = \mathbb{R}$), partition it into K disjoint BINS, $E = A_1 \cup \cdots \cup A_K$, and define, for $\theta \in \Theta$ and $j = 1, \dots, K$,

$$p_j(\theta) = \mathbb{P}_\theta[Y \in A_j], \qquad p_j = \mathbb{P}[X_1 \in A_j], \qquad \hat{p}_j = \frac{\#\{i : X_i \in A_j\}}{n},$$

with $\hat{\theta}$ as in the finite case. With T_n defined by the same formula, Theorem 12.3 holds unchanged: under technical assumptions and H_0, $T_n \to \chi^2_{K-d-1}$ and $\delta_\alpha = \mathbf{1}\{T_n > q_\alpha\}$. Binning makes the χ^2 test usable for continuous families too: to test normality, bin the line and use $\hat{\mu}, \hat{\sigma}^2$ for θ ($d = 2$). The guarantee is only asymptotic, whereas the Kolmogorov–Smirnov and Kolmogorov–Lilliefors quantiles are valid for each n.

Practical issues remain.

- *Choice of K and of the bins.* The rules resemble those for histograms: more than one bin, and bins not so small that they are nearly empty (at least one observation each). The conclusion only says whether the binned distribution fits, so the bins must be reasonable: with the two bins $(-\infty, 0)$ and $[0, \infty)$, a Gaussian cannot be told apart from a variable taking values ± 1.
- *Computation of $p_j(\theta)$.* For an interval bin this is a difference of cdf values, typically computed numerically; for less standard families it is an integral.

Example: *Poisson counts.* Let $E = \mathbb{N}$ and $H_0 : \mathbb{P} \in (\mathrm{Poiss}(\lambda))_{\lambda > 0}$, as for photon arrivals or insurance claim counts. If one expects $\lambda \le \lambda_{\max}$, choose $A_1 = \{0\}, A_2 = \{1\}, \dots, A_{K-1} = \{K-2\}$ and $A_K = \{K-1, K, K+1, \dots\}$, with K large enough that $p_K(\lambda_{\max}) \approx 0$. The last bin collects the whole tail.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 12, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 13: Regression

Regression tries to predict one variable from another. The explained (or response, or dependent) variable Y is modeled as a function of the explanatory variable X, plus noise. The data are pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$, independent copies of a generic pair (X, Y). Statements about "the distribution" refer to the joint law of (X, Y).

13.1 Linear models and examples

The simplest nontrivial model is a line,

$$Y_i \approx a + bX_i, \qquad i = 1, \ldots, n,$$

for unknown $a, b \in \mathbb{R}$. The points never lie exactly on a line; the noise ε_i in $Y_i = a + bX_i + \varepsilon_i$ accounts for everything the line does not explain, not only for randomness. The goal is to make the model rigorous (what is assumed about ε?), construct estimators $\hat{a}, \hat{b}$, and then do inference on a and b. When $X \in \mathbb{R}$ this is univariate regression; when $X \in \mathbb{R}^p$ with $p \geq 2$ it is multivariate regression, with

$$Y_i \approx a + X_i^\top b = a + b_1 X_i^{(1)} + \cdots + b_p X_i^{(p)}, \qquad a \in \mathbb{R},\ b \in \mathbb{R}^p.$$

The model is still linear: each coordinate of X gets its own coefficient.

Example: *Demand and price.* $D_i \approx a + bp_i$; typically $b < 0$, since demand falls as price rises.

Example: *Ideal gas law.* From $PV = nRT$, taking logarithms gives $\log P = \log(nR) - \log V + \log T$. With nR fixed, this is

$$\log P_i \approx a + b \log V_i + c \log T_i, \qquad a = \log(nR),\ b = -1,\ c = 1,$$

with measurement error playing the role of noise.

Relationships that are not linear can often be made linear by transforming variables (logarithms, exponentials, squares, products of variables). Choosing the transformation is domain-specific.

13.2 Theoretical linear regression of Y on X

First suppose the joint distribution of (X, Y) is known, as it would be with infinitely many observations. Among all lines, which one fits best? Measure the misfit of the line $y = a + bx$ at the random point (X, Y) by the *vertical* distance $Y - (a + bX)$, square it, and take the expectation. (One could also use the perpendicular or horizontal distance, or another power; the squared vertical distance is the mathematically convenient choice, and no distributional model is assumed at this

point.)

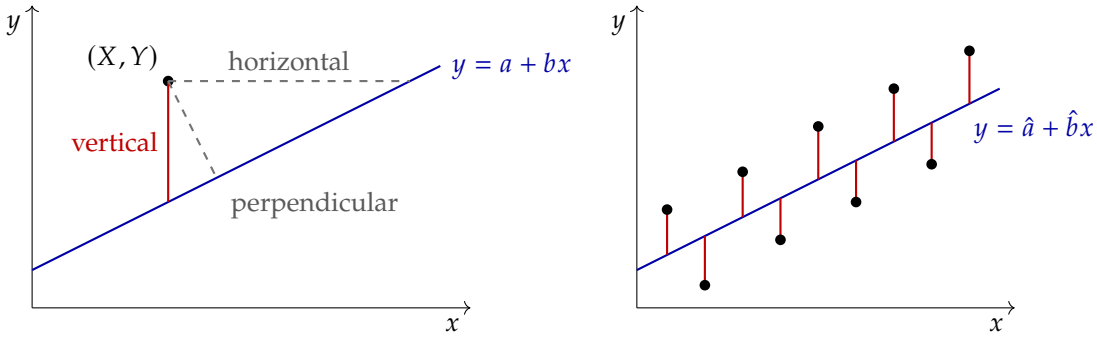


Figure 19: Left: three ways to measure the distance from a point to the line $y = a + bx$; least squares uses the vertical one. Right: the least squares line minimizes the sum of the squared lengths of the vertical segments. Both panels are schematic; the points are illustrative.

Definition 13.1 [Theoretical linear regression]. Let X, Y be real random variables, not necessarily independent, with finite second moments and $\mathrm{Var}(X) \neq 0$. The theoretical linear regression of Y on X is the best approximation of Y in quadratic mean by a linear function of X: the random variable $a + bX$, where (a, b) minimizes $\mathbb{E}\big[(Y - a - bX)^2\big]$.

The condition $\mathrm{Var}(X) \neq 0$ excludes a degenerate cloud in which every point has the same x-coordinate, where no slope can be determined.

Proposition 13.2 [Population least squares]. *The minimizer is*

$$\boxed{b = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}}, \qquad \boxed{a = \mathbb{E}[Y] - b\,\mathbb{E}[X] = \mathbb{E}[Y] - \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}\mathbb{E}[X]}.$$

Proof. The objective is a convex quadratic in (a, b): its Hessian is $2\begin{pmatrix} 1 & \mathbb{E}X \\ \mathbb{E}X & \mathbb{E}X^2 \end{pmatrix}$, with determinant $4\,\mathrm{Var}(X) > 0$. So the minimizer is the unique zero of the gradient. Differentiating under the expectation,

$$\begin{aligned} \partial_a: \quad & \mathbb{E}[Y - a - bX] = 0 \quad && \Longleftrightarrow \quad a + b\,\mathbb{E}[X] = \mathbb{E}[Y], \\ \partial_b: \quad & \mathbb{E}[X(Y - a - bX)] = 0 \quad && \Longleftrightarrow \quad a\,\mathbb{E}[X] + b\,\mathbb{E}[X^2] = \mathbb{E}[XY]. \end{aligned}$$

Multiply the first equation by $\mathbb{E}[X]$ and subtract it from the second: $b\big(\mathbb{E}[X^2] - (\mathbb{E}X)^2\big) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$, that is, $b\,\mathrm{Var}(X) = \mathrm{Cov}(X, Y)$. Since $\mathrm{Var}(X) \neq 0$, this gives b, and the first equation gives a. □

Only the first two moments matter. The slope is the covariance measured in units of $\mathrm{Var}(X)$; the intercept just places the line so that it passes through $(\mathbb{E}X, \mathbb{E}Y)$.

13.3 What the noise must satisfy

Define $\varepsilon = Y - (a + bX)$ for the optimal (a, b), so that

$$Y = a + bX + \varepsilon.$$

The two first-order conditions say exactly that

$$\mathbb{E}[\varepsilon] = 0 \qquad \text{and} \qquad \operatorname{Cov}(X, \varepsilon) = \mathbb{E}[X\varepsilon] = 0.$$

Conversely, textbooks usually start from the model.

Proposition 13.3 [Model form of the regression]. *Suppose $Y = a + bX + \varepsilon$ for some $a, b \in \mathbb{R}$ and a centered random variable ε with $\operatorname{Cov}(X, \varepsilon) = 0$. Then $a + bX$ is the theoretical linear regression of Y on X. Both conditions hold, for instance, if $X \perp\!\!\!\perp \varepsilon$ with $\mathbb{E}\varepsilon = 0$, or if $\mathbb{E}[\varepsilon \mid X] = 0$.*

Proof. By bilinearity, $\operatorname{Cov}(X, Y) = b \operatorname{Var}(X) + \operatorname{Cov}(X, \varepsilon) = b \operatorname{Var}(X)$, and $\mathbb{E}[Y] = a + b\mathbb{E}[X]$. These are the formulas of Proposition 13.2. If $\mathbb{E}[\varepsilon \mid X] = 0$, then $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[X\varepsilon] = \mathbb{E}\big[X\,\mathbb{E}[\varepsilon \mid X]\big] = 0$. □

So under these weak assumptions on ε, the coefficients that generated the data are the ones found by minimizing the expected *square*, not the fourth or any other power. Other criteria arise as maximum likelihood estimators under other noise densities. Inference (for example t-tests) will need stronger assumptions, such as Gaussian noise.

13.4 The least squares estimator

In the statistical problem the expectations are unknown. We observe i.i.d. pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ with distribution of (X, Y) and

$$Y_i = a + bX_i + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i] = 0, \quad \operatorname{Cov}(X_i, \varepsilon_i) = 0,$$

with the same a and b for every i. The true line is never seen; an estimated line $y = \hat{a} + \hat{b}x$ is built from the data, and the questions are how to compute $\hat{a}, \hat{b}$ and how far they are from a, b.

Definition 13.4 [Least squares estimator]. The least squared error (LSE) estimator of (a, b) is the minimizer of the sum of squared errors

$$\sum_{i=1}^{n} (Y_i - a - bX_i)^2.$$

Proposition 13.5 [Univariate LSE]. *Write $\overline{XY} = \frac{1}{n}\sum_i X_i Y_i$, $\overline{X^2} = \frac{1}{n}\sum_i X_i^2$, and let $\bar{X}, \bar{Y}$ be the sample*

means. If the X_i are not all equal,

$$\hat{b} = \frac{\overline{XY} - \bar{X}\,\bar{Y}}{\overline{X^2} - \bar{X}^2}, \qquad \hat{a} = \bar{Y} - \hat{b}\,\bar{X}.$$

Proof. Minimizing the sum is the same as minimizing the average, which is $\mathbb{E}[(Y - a - bX)^2]$ under the empirical distribution of the pairs (X_i, Y_i). Proposition 13.2 applied to that distribution replaces each expectation by the corresponding average; its condition $\mathrm{Var}(X) \neq 0$ becomes $\overline{X^2} - \bar{X}^2 \neq 0$, which holds when the X_i are not all equal. □

Thus "replace expectations by averages, then minimize" and "minimize, then replace expectations by averages" agree here. The first form is the one that generalizes: in the multivariate problem there is no simple closed form to plug into, but the empirical criterion can still be minimized. Geometrically, the LSE line minimizes the sum of squared vertical segments in Figure 19.

Other criteria give other estimators. The LEAST ABSOLUTE DEVIATION estimator minimizes $\sum_i |Y_i - a - bX_i|$; it has no closed form but can be computed numerically, and it is used far less than least squares.

13.5 Multivariate regression

Usually several measurements are combined to predict Y. For example, body density is hard to measure directly (it requires immersing a person in water), but it can be predicted well from easy measurements such as wrist circumference, height, and so on.

Definition 13.6 [MULTIVARIATE LINEAR MODEL]. For $i = 1, \dots, n$,

$$Y_i = \mathbf{X}_i^\top \beta + \varepsilon_i,$$

where $\mathbf{X}_i \in \mathbb{R}^p$ is the vector of EXPLANATORY VARIABLES or COVARIATES, $Y_i \in \mathbb{R}$ is the dependent variable, $\beta \in \mathbb{R}^p$ is the unknown parameter, and the noise satisfies $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Cov}(\mathbf{X}_i, \varepsilon_i) = \mathbf{0}$ (coordinatewise).

Without loss of generality the first coordinate of every $\mathbf{X}_i$ is 1; then $\beta = (a, b^\top)^\top$ and $\beta_1 = a$ is the INTERCEPT. In a data table this means adding a column of ones to the measured covariates, so $p - 1$ covariates are actually measured. With two measured covariates the points live in $\mathbb{R}^3$, and the model says they lie close to a plane.

Definition 13.7 [MULTIVARIATE LSE]. The LEAST SQUARES ESTIMATOR of β is

$$\hat{\beta} = \arg\min_{t \in \mathbb{R}^p} \sum_{i=1}^{n} \left(Y_i - \mathbf{X}_i^\top t\right)^2,$$

the empirical version of minimizing $\mathbb{E}\big[(Y - \mathbf{X}^\top t)^2\big]$.

Setting each partial derivative to zero gives a linear system of p equations in p unknowns, involving the covariances of every $X^{(j)}$ with Y and every cross covariance of $X^{(j)}$ with $X^{(k)}$. Matrix notation lets us write and solve this system at once.

13.6 Matrix form

Stack the responses into $\mathbf{Y} = (Y_1, \dots, Y_n)^\top \in \mathbb{R}^n$ and the noise into $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\top \in \mathbb{R}^n$ (unobserved). Let $\mathbf{X}$ be the $n \times p$ matrix whose rows are $\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top$, called the DESIGN: one row per observation and one column per covariate, the first column being all ones. The n scalar equations become

$$\boxed{\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}}.$$

Since a squared Euclidean norm is the sum of squared coordinates,

$$\sum_{i=1}^{n} \big(Y_i - \mathbf{X}_i^\top t\big)^2 = \|\mathbf{Y} - \mathbf{X}t\|_2^2, \qquad \text{so} \qquad \hat{\beta} = \operatorname*{arg\,min}_{t \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}t\|_2^2.$$

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 13, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 14: Regression (cont.)

Continue with the linear model $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$ from Lecture 13, with $\mathbf{Y}, \boldsymbol{\varepsilon} \in \mathbb{R}^n$, design $\mathbf{X} \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$. The least squares estimator minimizes $\|\mathbf{Y} - \mathbf{X}t\|_2^2$ over $t \in \mathbb{R}^p$.

14.1 The normal equations

Expand the criterion, using $\|v\|_2^2 = v^\top v$ and $(AB)^\top = B^\top A^\top$:

$$\|\mathbf{Y} - \mathbf{X}t\|_2^2 = \|\mathbf{Y}\|_2^2 - 2\,\mathbf{Y}^\top \mathbf{X}t + t^\top \mathbf{X}^\top \mathbf{X}t.$$

The rules of differentiation are as in one dimension, as long as the order of matrix products is respected. The first term is constant in t, the second is linear (like ax), and the third is quadratic (like ax^2). Writing gradients as column vectors,

$$\nabla_t \|\mathbf{Y} - \mathbf{X}t\|_2^2 = 2\,\mathbf{X}^\top \mathbf{X}t - 2\,\mathbf{X}^\top \mathbf{Y}.$$

The Hessian $2\mathbf{X}^\top \mathbf{X}$ is positive semidefinite, so the criterion is convex and its minimizers are exactly the solutions of the normal equations (or score equations)

$$\mathbf{X}^\top \mathbf{X}\,\hat{\beta} = \mathbf{X}^\top \mathbf{Y}. \tag{14.1}$$

Proposition 14.1 [Closed form of the LSE]. *If* $\operatorname{rank}(\mathbf{X}) = p$*, then the* $p \times p$ *matrix* $\mathbf{X}^\top \mathbf{X}$ *is invertible and*

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}}.$$

The two conditions are the same, since $\operatorname{rank}(\mathbf{X}^\top \mathbf{X}) = \operatorname{rank}(\mathbf{X})$. If $\mathbf{X}^\top \mathbf{X}$ is not invertible, there is $v \neq 0$ with $\mathbf{X}^\top \mathbf{X} v = 0$, and then for every $\lambda \in \mathbb{R}$,

$$\mathbf{X}^\top \mathbf{X}(\hat{\beta} + \lambda v) = \mathbf{X}^\top \mathbf{X}\hat{\beta} + \lambda\, \mathbf{X}^\top \mathbf{X} v = \mathbf{X}^\top \mathbf{Y}.$$

So $\hat{\beta} + \lambda v$ also solves (14.1), and the least squares estimator is not well defined. Since $\operatorname{rank}(\mathbf{X}) \leq \min(n, p)$, this always happens when $p > n$: there are more unknowns than equations, one equation per observation. In genomics, for instance, one may have $n \approx 100$ patients and $p \approx 100{,}000$ genes. Here we stay in the regime $p < n$, typically p of order 10 and n of order 100.

14.2 Geometry: least squares as a projection

Even when β itself is not of interest, $\mathbf{X}\hat{\beta}$ is: it estimates the noiseless signal $\mathbf{X}\beta$, moving each observation onto the fitted line or plane. In imaging, for example, $\mathbf{Y}$ is an image stacked into a vector, the columns of $\mathbf{X}$ are a fixed library of patterns (wavelets, Gabor frames), and one wants the denoised image $\mathbf{X}\beta$ rather than its coefficients.

The set $\{\mathbf{X}t : t \in \mathbb{R}^p\}$ is the column span of $\mathbf{X}$, a p-dimensional subspace of $\mathbb{R}^n$. The observation $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$ is pushed off this subspace by the noise, and least squares picks the point of the subspace closest to $\mathbf{Y}$: its orthogonal projection.

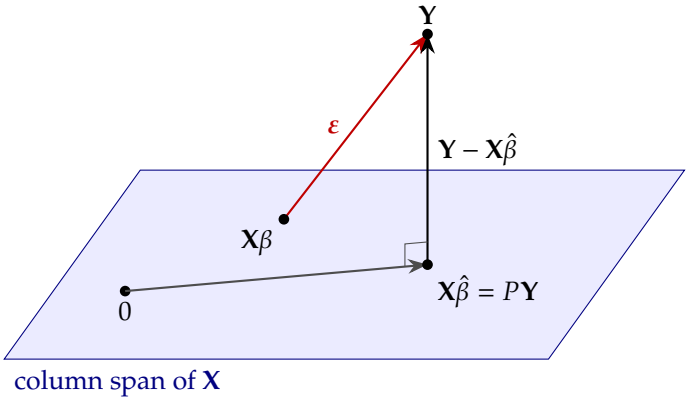


Figure 20: Least squares as orthogonal projection (schematic, $n = 3$, $p = 2$). The noise $\boldsymbol{\varepsilon}$ pushes $\mathbf{Y}$ off the column span of $\mathbf{X}$; $\mathbf{X}\hat{\beta} = P\mathbf{Y}$ is the closest point of the span, and the residual $\mathbf{Y} - \mathbf{X}\hat{\beta}$ is orthogonal to it.

Proposition 14.2 [Projection]. *If* $\operatorname{rank}(\mathbf{X}) = p$, *then*

$$\mathbf{X}\hat{\beta} = P\mathbf{Y}, \qquad P = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top,$$

and P is the orthogonal projection onto the column span of $\mathbf{X}$. It satisfies $P^2 = P$ and $P^\top = P$; its eigenvalues are 0 or 1, and $\operatorname{tr} P = p$. Moreover

$$\|\mathbf{Y}\|_2^2 = \|\mathbf{X}\hat{\beta}\|_2^2 + \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|_2^2.$$

Proof. Direct computation gives $P^2 = P$, $P^\top = P$, and $P\mathbf{X} = \mathbf{X}$, so P fixes the column span and maps into it. By the normal equations, $\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\hat{\beta}) = 0$: the residual is orthogonal to every column, which characterizes the orthogonal projection. A symmetric idempotent matrix has eigenvalues λ with $\lambda^2 = \lambda$, eigenvalue 1 on the span and 0 on its orthogonal complement, so its trace is the dimension p of the span. The identity is Pythagoras' theorem for the orthogonal vectors $\mathbf{X}\hat{\beta}$ and $\mathbf{Y} - \mathbf{X}\hat{\beta}$. □

The projection $\mathbf{X}\hat{\beta}$ is unique even when the coordinates $\hat{\beta}$ are not: if the columns are redundant, a point of their span has many coordinate vectors.

14.3 Gaussian linear model with deterministic design

So far nothing was assumed about $\boldsymbol{\varepsilon}$. Projecting back onto the span is sensible only if the noise is centered and has no privileged direction. Inference uses the following model.

Definition 14.3 [Gaussian linear model]. The Gaussian linear model with deterministic design assumes:

1. The design matrix $\mathbf{X}$ is deterministic and $\operatorname{rank}(\mathbf{X}) = p$.
2. The model is homoscedastic: $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d.
3. The noise vector is Gaussian, $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$, for some known or unknown $\sigma^2 > 0$.

"Homoscedastic" means all ε_i share the same scale σ^2. A deterministic design is an assumption about the values observed (for example that $\mathbf{X}^\top\mathbf{X}$ is invertible), which could not be guaranteed for a random design; a random design is handled by conditioning on $\mathbf{X}$. A library of image patterns is genuinely deterministic. Gaussianity is needed for the exact distributions below; for consistency alone, finite variance would suffice.

The key tool is the behavior of Gaussian vectors under linear maps: if $Z \sim \mathcal{N}(\mu, \Sigma)$ and B is a deterministic matrix with compatible dimensions, then

$$BZ \sim \mathcal{N}\big(B\mu,\ B\Sigma B^\top\big).$$

Theorem 14.4 [Properties of the LSE]. *In the model of Definition 14.3:*

1. *LSE = MLE, and* $\hat\beta \sim \mathcal{N}_p\big(\beta, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\big)$.
2. *Quadratic risk:* $\mathbb{E}\big[\|\hat\beta - \beta\|_2^2\big] = \sigma^2 \operatorname{tr}\big((\mathbf{X}^\top\mathbf{X})^{-1}\big)$.
3. *Prediction error:* $\mathbb{E}\big[\|\mathbf{Y} - \mathbf{X}\hat\beta\|_2^2\big] = \sigma^2(n-p)$.
4. $\hat\sigma^2 = \dfrac{1}{n-p}\|\mathbf{Y} - \mathbf{X}\hat\beta\|_2^2$ *is an unbiased estimator of* σ^2, *and* $(n-p)\dfrac{\hat\sigma^2}{\sigma^2} \sim \chi^2_{n-p}$.
5. $\hat\beta \perp\!\!\!\perp \hat\sigma^2$.

Proof. Distribution. Substitute $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$:

$$\hat\beta = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\beta + \boldsymbol{\varepsilon}) = \beta + B\boldsymbol{\varepsilon}, \qquad B = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top.$$

B is $p \times n$, so $B\boldsymbol{\varepsilon}$ is a p-dimensional centered Gaussian vector with covariance $\sigma^2 BB^\top = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$, using the symmetry of $(\mathbf{X}^\top\mathbf{X})^{-1}$.
MLE. The model for $\mathbf{Y}$ is $\{\mathcal{N}_n(\mathbf{X}\beta, \sigma^2 I_n) : \beta \in \mathbb{R}^p\}$; we have one observation of the vector $\mathbf{Y}$. Its log-likelihood is

$$\ell(\beta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{Y} - \mathbf{X}\beta\|_2^2,$$

and maximizing it over β is the same as minimizing $\|\mathbf{Y} - \mathbf{X}\beta\|_2^2$. This is special to Gaussian noise:

i.i.d. double exponential (Laplace) noise would instead lead to the least absolute deviation estimator.

Risk. Since $\mathbb{E}\hat{\beta}_j = \beta_j$, $\mathbb{E}\|\hat{\beta} - \beta\|_2^2 = \sum_j \operatorname{Var}(\hat{\beta}_j)$, the sum of the diagonal entries of the covariance matrix, which is $\sigma^2 \operatorname{tr}\big((\mathbf{X}^\top \mathbf{X})^{-1}\big)$.

Residuals. By Proposition 14.2, $\mathbf{Y} - \mathbf{X}\hat{\beta} = (I_n - P)\mathbf{Y}$. The matrix $I_n - P$ is the orthogonal projection onto the orthogonal complement of the column span, which has dimension $n - p$. It kills $\mathbf{X}\beta$, so

$$\mathbf{Y} - \mathbf{X}\hat{\beta} = (I_n - P)\boldsymbol{\varepsilon} \sim \mathcal{N}_n\big(0, \sigma^2 (I_n - P)(I_n - P)^\top\big) = \mathcal{N}_n\big(0, \sigma^2 (I_n - P)\big).$$

For a centered Gaussian vector, $\mathbb{E}\|Z\|^2 = \operatorname{tr}\operatorname{Cov}(Z)$, so the prediction error is $\sigma^2 \operatorname{tr}(I_n - P) = \sigma^2(n - p)$, and dividing by $n - p$ gives the unbiased $\hat{\sigma}^2$.

Chi-squared. Diagonalize $I_n - P = UDU^\top$ with U orthogonal and D diagonal with $n - p$ ones and p zeros. The norm is invariant under rotation, and $\boldsymbol{\varepsilon}' = U^\top \boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$ because a Gaussian with covariance $\sigma^2 I_n$ treats all directions alike. Hence

$$\|\mathbf{Y} - \mathbf{X}\hat{\beta}\|_2^2 = \|UDU^\top \boldsymbol{\varepsilon}\|_2^2 = \|D\boldsymbol{\varepsilon}'\|_2^2 = \sum_{j:\, D_{jj}=1} (\varepsilon'_j)^2 \sim \sigma^2 \chi^2_{n-p}.$$

Independence (Cochran's theorem). Since $\mathbf{X}^\top P = \mathbf{X}^\top$, $\hat{\beta} - \beta = B\boldsymbol{\varepsilon} = BP\boldsymbol{\varepsilon}$ is a function of $P\boldsymbol{\varepsilon}$, while $\hat{\sigma}^2$ is a function of $(I_n - P)\boldsymbol{\varepsilon}$. These two vectors are jointly Gaussian with cross covariance $\sigma^2 P(I_n - P)^\top = \sigma^2(P - P^2) = 0$, hence independent. □

The residuals $\hat{\varepsilon}_i = Y_i - \mathbf{X}_i^\top \hat{\beta}$ estimate the noise ε_i, which explains why their normalized sum of squares estimates σ^2. The normalization is $n - p$ rather than n because the residual vector lives in an $(n - p)$-dimensional space. Geometrically, $\hat{\beta}$ comes from the projection onto the column span and $\hat{\sigma}^2$ from the projection onto its orthogonal complement; for Gaussian vectors, orthogonal implies independent.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 14, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 15: Regression (cont.)

Work in the Gaussian linear model of Lecture 14: $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$ with deterministic design of rank p and $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$. Inference rests on two facts from that lecture:

$$\hat{\beta} = \beta + \mathcal{N}_p\big(0, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\big), \qquad (n-p)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p}, \quad \hat{\beta} \perp\!\!\!\perp \hat{\sigma}^2.$$

The second is what is needed when σ^2 is unknown (it may be known, for example from the specifications of a measuring device).

15.1 Testing a single coefficient

Write the model coordinatewise as $Y_i = \beta_1 + \beta_2 X_i^{(2)} + \cdots + \beta_p X_i^{(p)} + \varepsilon_i$. If $\beta_j = 0$, the jth covariate does not appear at all and can be removed. Testing whether the jth explanatory variable is SIGNIFICANT means testing

$$H_0 : \beta_j = 0 \qquad \text{vs.} \qquad H_1 : \beta_j \neq 0, \qquad 1 \le j \le p.$$

The two-sided alternative is the interesting one: it is where a discovery lies (does shoe size help predict a client's late fees; does a given gene affect a trait?).

To read off one coordinate, write $v_j = e_j^\top v$, where e_j is the jth vector of the canonical basis of $\mathbb{R}^p$. A Gaussian vector $Z \sim \mathcal{N}(\mu, \Sigma)$ satisfies $e_j^\top Z \sim \mathcal{N}(e_j^\top\mu,\ e_j^\top\Sigma e_j)$, and $e_j^\top\Sigma e_j = \Sigma_{jj}$. Hence

$$\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2\gamma_j), \qquad \gamma_j = \big((\mathbf{X}^\top\mathbf{X})^{-1}\big)_{jj} > 0,$$

positive because $(\mathbf{X}^\top\mathbf{X})^{-1}$ is positive definite. This is the one-parameter situation $\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2/n)$ with γ_j in the role of $1/n$.

If σ is known, under H_0 we have $\hat{\beta}_j/(\sigma\sqrt{\gamma_j}) \sim \mathcal{N}(0,1)$, and we reject when this exceeds the $(1-\alpha/2)$-quantile of $\mathcal{N}(0,1)$ in absolute value (the $\alpha/2$ accounts for the two-sided alternative). If σ is unknown, replace it by $\hat{\sigma}$ and write the result as a ratio of known distributions:

$$\frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2\gamma_j}} = \frac{\hat{\beta}_j/(\sigma\sqrt{\gamma_j})}{\sqrt{\dfrac{(n-p)\hat{\sigma}^2/\sigma^2}{n-p}}}.$$

Under H_0 the numerator is $\mathcal{N}(0,1)$, the denominator is $\sqrt{\chi^2_{n-p}/(n-p)}$, and they are independent because $\hat{\sigma}^2$ is independent of $\hat{\beta}$ and hence of each of its coordinates.

Theorem 15.1 [t-test for one coefficient]. *For each j,*

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 \gamma_j}} \sim t_{n-p}.$$

Let $T_n^{(j)} = \hat{\beta}_j / \sqrt{\hat{\sigma}^2 \gamma_j}$. The test

$$\delta_\alpha^{(j)} = \mathbf{1}\big\{|T_n^{(j)}| > q_{\alpha/2}(t_{n-p})\big\},$$

where $q_{\alpha/2}(t_{n-p})$ is the $(1-\alpha/2)$-quantile of t_{n-p}, has non-asymptotic level $\alpha \in (0,1)$ for $H_0 : \beta_j = 0$.

The only changes from the known-variance test are $\hat{\sigma}$ for σ and t_{n-p} quantiles for Gaussian ones. The t distribution has heavier tails, and here its degrees of freedom are $n-p$, not n: the more parameters are estimated, the harder it is to estimate the variance. Regression software reports, for each coefficient, $\hat{\beta}_j$, $T_n^{(j)}$, the corresponding p-value, and significance stars.

15.2 Multiple testing and the Bonferroni correction

Reading that table row by row performs p tests, each with a 5% chance of a false rejection; repeated often enough, such errors start to happen. To test a group of variables, let $S \subseteq \{1, \dots, p\}$ with $|S| = k$ and test

$$H_0 : \beta_j = 0 \ \ \forall j \in S \qquad \text{vs.} \qquad H_1 : \exists j \in S,\ \beta_j \neq 0.$$

The probability of rejecting at least one true null, here $\mathbb{P}_{H_0}\big(\bigcup_{j \in S} \{\psi_j = 1\}\big)$ for individual tests ψ_j, is the family-wise error rate. If each ψ_j has level α and the rejection events are disjoint, this probability is exactly $k\alpha$.

Proposition 15.2 [Bonferroni's test]. *The test $\delta_\alpha^B = \max_{j \in S} \delta_{\alpha/k}^{(j)}$, which rejects if some $|T_n^{(j)}|$, $j \in S$, exceeds $q_{\alpha/(2k)}(t_{n-p})$, has non-asymptotic level at most α.*

Proof. By the union bound, $\mathbb{P}_{H_0}\big(\delta_\alpha^B = 1\big) \leq \sum_{j \in S} \mathbb{P}_{H_0}\big(\delta_{\alpha/k}^{(j)} = 1\big) = k \cdot \frac{\alpha}{k} = \alpha$. □

The correction is conservative: the bound is attained only if the rejection events are disjoint, which is never realistic, and the tests are strongly correlated, so independence cannot be used either. Inclusion–exclusion would require controlling all intersections. The correction is also stringent: with 5 variables at overall level 5%, each test runs at 1%. Dividing by $k = |S|$ rather than p avoids making it harder than necessary.

15.3 General linear hypotheses and the F-test

Let G be a $k \times p$ matrix with $\operatorname{rank}(G) = k \le p$, and $\lambda \in \mathbb{R}^k$. Consider

$$H_0 : G\beta = \lambda \qquad \text{vs.} \qquad H_1 : G\beta \neq \lambda.$$

The single-coefficient test is the case $G = e_j^\top$, $\lambda = 0$. Others: $\beta_2 + \beta_3 = 0$ (does the effect of one gene cancel that of another?), $\beta_5 - 2\beta_6 = 23$, several such constraints at once, or $G = I_p$, $\lambda = 0$, which tests whether $\beta = 0$, that is, whether the whole model is just noise. Software reports this last test on the final line of a regression summary.

As before, $G\hat\beta = G\beta + \mathcal{N}_k\big(0, \sigma^2 G(\mathbf{X}^\top\mathbf{X})^{-1}G^\top\big)$, so under H_0

$$G\hat\beta - \lambda \sim \mathcal{N}_k\big(0, \sigma^2 G(\mathbf{X}^\top\mathbf{X})^{-1}G^\top\big).$$

The $k \times k$ matrix $G(\mathbf{X}^\top\mathbf{X})^{-1}G^\top$ is invertible, since $(\mathbf{X}^\top\mathbf{X})^{-1}$ is positive definite and $G^\top$ has rank k. A matrix covariance cannot be removed by division; instead standardize as in Wald's test: if $Z \sim \mathcal{N}_k(0, \Sigma)$ with Σ invertible, then $\Sigma^{-1/2}Z \sim \mathcal{N}_k(0, I_k)$, so $Z^\top\Sigma^{-1}Z \sim \chi^2_k$. Therefore, under H_0,

$$\sigma^{-2}(G\hat\beta - \lambda)^\top\big(G(\mathbf{X}^\top\mathbf{X})^{-1}G^\top\big)^{-1}(G\hat\beta - \lambda) \sim \chi^2_k,$$

a pivotal distribution.

Definition 15.3 [Fisher distribution]. The Fisher distribution with p and q degrees of freedom, $F_{p,q}$, is the distribution of $\dfrac{U/p}{V/q}$, where $U \sim \chi^2_p$, $V \sim \chi^2_q$, and $U \perp\!\!\!\perp V$.

Theorem 15.4 [F-test]. *Let*

$$S_n = \frac{1}{\hat\sigma^2}\,\frac{(G\hat\beta - \lambda)^\top\big(G(\mathbf{X}^\top\mathbf{X})^{-1}G^\top\big)^{-1}(G\hat\beta - \lambda)}{k}.$$

If H_0 is true, $S_n \sim F_{k,n-p}$, and the test

$$\delta_\alpha = \mathbf{1}\{S_n > q_\alpha(F_{k,n-p})\},$$

with $q_\alpha(F_{k,n-p})$ the $(1-\alpha)$-quantile of $F_{k,n-p}$, has non-asymptotic level $\alpha \in (0,1)$.

Proof. Divide numerator and denominator by σ^2: S_n is the χ^2_k variable above divided by k, over $(n-p)\hat\sigma^2/\sigma^2 \sim \chi^2_{n-p}$ divided by $n-p$. The numerator is a function of $\hat\beta$, hence independent of $\hat\sigma^2$. □

No absolute value appears because $S_n \ge 0$; only large values are evidence against H_0. The F-test generalizes the t-test. If $T \sim t_q$, then $T = Z/\sqrt{V/q}$ with $Z \sim \mathcal{N}(0,1)$ independent of $V \sim \chi^2_q$, and

$T^2 = (Z^2/1)/(V/q) \sim F_{1,q}$ since $Z^2 \sim \chi^2_1$. Consequently

$$\mathbf{1}\{|T| > q_{\alpha/2}(t_q)\} = \mathbf{1}\{T^2 > q_{\alpha/2}(t_q)^2\} = \mathbf{1}\{T^2 > q_\alpha(F_{1,q})\},$$

so the $F_{1,q}$ quantile at level α is the square of the t_q quantile at level $\alpha/2$. With $G = e_j^\top$ and $\lambda = 0$, $S_n = (T_n^{(j)})^2$ and the two tests coincide.

15.4 The role of the design

The variance of $\hat{\beta}_j$ is $\sigma^2\gamma_j$: the noise level times a factor determined entirely by $\mathbf{X}$. If $\mathbf{X}^\top\mathbf{X} = nI_p$, the columns are orthogonal and $\gamma_j = 1/n$, the basic one-parameter case. As columns become closer to each other, $\mathbf{X}^\top\mathbf{X}$ acquires nonzero off-diagonal entries and the diagonal entries of its inverse grow. Indeed, for a positive definite A, Cauchy–Schwarz gives $1 = \left(e_j^\top A^{1/2} A^{-1/2} e_j\right)^2 \le A_{jj}\,(A^{-1})_{jj}$, so $\gamma_j \ge 1/(\mathbf{X}^\top\mathbf{X})_{jj}$, with equality exactly when column j is orthogonal to the others. A large γ_j means wide confidence intervals and low power. If the design can be chosen, its columns should be as orthogonal as possible.

This is also a reason to treat the design as deterministic: it plays the role of a parameter of the experiment, setting the geometry of the problem and the variances. If $\mathbf{X}$ is random, everything above holds conditionally on $\mathbf{X}$, provided the noise is Gaussian conditionally on $\mathbf{X}$; unconditionally, statements such as the χ^2 distribution of the residuals would fail.

15.5 Correlation, causation, and normality

Linear regression exhibits correlations, not causality. Designating one variable as Y and others as X does not mean X causes Y. In a crime data set, the coefficient of police expenditure on the crime rate was positive. This does not mean that funding police raises crime; places with more crime spend more on police. Causal direction must come from outside the statistics, as with genes and physical traits.

Every distributional result above relies on Gaussian noise; without it, we would have the estimate $\hat{\beta}$ but no tests or confidence regions. Normality can be checked on the residuals $\hat{\varepsilon}_i = Y_i - \mathbf{X}_i^\top\hat{\beta}$, visually with a normal Q–Q plot or with a goodness-of-fit test. When the design contains the intercept column, the residuals have mean exactly zero (the first normal equation is $\mathbf{1}^\top(\mathbf{Y} - \mathbf{X}\hat{\beta}) = 0$), but the variance is unknown, so the appropriate test is Kolmogorov–Lilliefors rather than Kolmogorov–Smirnov.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 15, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 16: Regression (cont.)

This unit leaves the full-rank Gaussian linear model of Lectures 14 and 15: first the design may have rank smaller than p, then p may exceed n, and finally the regression function need not be linear.

16.1 Lack of identifiability

Consider again $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$ with $\mathbf{Y} \in \mathbb{R}^n$, deterministic design $\mathbf{X} \in \mathbb{R}^{n\times p}$, unknown $\beta \in \mathbb{R}^p$, and $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$. Previously $\operatorname{rank}(\mathbf{X}) = p$, so $\mathbf{X}^\top\mathbf{X}$ could be inverted. If $\operatorname{rank}(\mathbf{X}) < p$, for example when $p > n$, then β is no longer identified: there is $v \neq 0$ with $\mathbf{X}v = 0$, so β and $\beta + v$ give the same distribution of $\mathbf{Y}$. Estimating β is then vain unless more structure is assumed.

Prediction is still possible, because $\mathbf{X}\beta$ is identified. Let $\hat{\mathbf{Y}}$ be the orthogonal projection of $\mathbf{Y}$ onto the column span of $\mathbf{X}$:

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^\dagger\mathbf{X}^\top\mathbf{Y},$$

where $A^\dagger$ is the Moore–Penrose pseudo-inverse of A. Here $\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^\dagger\mathbf{X}^\top\mathbf{Y}$ is one solution of the normal equations, and $P = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^\dagger\mathbf{X}^\top$ is the orthogonal projection onto the column span; when $\operatorname{rank}(\mathbf{X}) = p$ it reduces to the earlier formula.

Proposition 16.1 [Rank-deficient design]. *Let $k = \operatorname{rank}(\mathbf{X})$. Then*

$$\frac{\|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2}{\sigma^2} \sim \chi^2_{n-k}, \qquad \|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2 \perp\!\!\!\perp \hat{\mathbf{Y}}.$$

In particular $\mathbb{E}\big[\|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2\big] = (n-k)\sigma^2$, and $\hat{\sigma}^2 = \frac{1}{n-k}\|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2$ is an unbiased estimator of σ^2.

Proof. The argument of Lecture 14 goes through with a k-dimensional column span. Since $P\mathbf{X}\beta = \mathbf{X}\beta$, $\mathbf{Y} - \hat{\mathbf{Y}} = (I_n - P)\boldsymbol{\varepsilon}$, where $I_n - P$ projects onto a space of dimension $n-k$; diagonalizing it gives a sum of $n-k$ squared independent $\mathcal{N}(0,\sigma^2)$ variables. And $\hat{\mathbf{Y}} = \mathbf{X}\beta + P\boldsymbol{\varepsilon}$ depends only on $P\boldsymbol{\varepsilon}$, which is Gaussian and uncorrelated with $(I_n - P)\boldsymbol{\varepsilon}$, hence independent of it. □

16.2 High-dimensional regression and sparsity

Keep the same model, where $\mathbf{X}_i \in \mathbb{R}^p$ (the ith row) is the vector of covariates of the ith individual. If p is too large ($p > n$), there are too many parameters to estimate and the model overfits, although some covariates may be irrelevant. The remedy is to reduce the dimension.

The idea is sparsity: assume only a few coordinates of β are nonzero, without knowing which. Then

select a subset of covariates from the sample and estimate the corresponding coordinates of β. For $S \subseteq \{1, \dots, p\}$ let $\mathbf{X}_S$ be the submatrix of $\mathbf{X}$ keeping only the columns indexed by S, and

$$\hat{\beta}_S \in \operatorname*{arg\,min}_{t \in \mathbb{R}^S} \|\mathbf{Y} - \mathbf{X}_S t\|_2^2.$$

Larger S always fit better, so select S by minimizing the prediction error penalized by the size of the model,

$$\|\mathbf{Y} - \mathbf{X}_S \hat{\beta}_S\|_2^2 + \lambda |S|,$$

where $\lambda > 0$ is a tuning parameter.

- $\lambda = 2\hat{\sigma}^2$ gives Mallows' C_p, or the AIC criterion.
- $\lambda = \hat{\sigma}^2 \log n$ gives the BIC criterion.

Each criterion is equivalent to finding $b \in \mathbb{R}^p$ that minimizes

$$\|\mathbf{Y} - \mathbf{X}b\|_2^2 + \lambda \|b\|_0, \qquad \|b\|_0 = \sum_{j=1}^{p} \mathbf{1}\{b_j \neq 0\},$$

the number of nonzero coefficients of b. Indeed, a vector b with support S has objective at least $\|\mathbf{Y} - \mathbf{X}_S \hat{\beta}_S\|_2^2 + \lambda |S|$; conversely, $\hat{\beta}_S$ extended by zeros has support inside S, so its objective is at most that value. Both problems therefore have the same minimum value, attained by $\hat{\beta}_S$ for a minimizing S. This problem is computationally hard: it is nonconvex and requires computing 2^p estimators, one $\hat{\beta}_S$ for every $S \subseteq \{1, \dots, p\}$.

Definition 16.2 [Lasso]. Replacing $\|b\|_0$ by $\|b\|_1 = \sum_{j=1}^{p} |b_j|$ makes the problem convex. The Lasso estimator is

$$\hat{\beta}^L \in \operatorname*{arg\,min}_{b \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}b\|_2^2 + \lambda \|b\|_1,$$

where $\lambda > 0$ is a tuning parameter.

Choosing λ is difficult. A good choice yields an estimator $\hat{\beta}$ very close to β that recovers, with high probability, the support $S^* = \{j \in \{1, \dots, p\} : \beta_j \neq 0\}$. Plotting each coordinate of $\hat{\beta}^L$ against λ (the regularization path) shows the coefficients shrinking as λ grows, reaching exactly zero one after another, until all of them vanish for λ large enough.

16.3 Nonparametric regression

The linear model $Y_i = \mathbf{X}_i^\top \beta + \varepsilon_i$ with deterministic $\mathbf{X}_i$ is understood conditionally on the design. It assumes that $\mathbb{E}[Y_i \mid \mathbf{X}_i]$ is a linear function of $\mathbf{X}_i$, which is not true in general. The object of interest is the regression function

$$f(x) = \mathbb{E}[Y_i \mid \mathbf{X}_i = x], \qquad x \in \mathbb{R}^p.$$

From now on $p = 1$. One option is a parametric assumption, such as $f(x) = a+bx$, $f(x) = a+bx+cx^2$, or $f(x) = e^{a+bx}$. The problem then reduces to estimating finitely many parameters, and least squares, maximum likelihood, and the theory for the linear case can be adapted.

Without a parametric assumption, assume only that f is smooth enough to be well approximated by a piecewise constant function. Then $f(t) \approx f(x)$ for t close to x, so $Y_i \approx f(x) + \varepsilon_i$ for every i with X_i close to x. Estimate $f(x)$ by LOCAL AVERAGES: the average of the Y_i whose X_i are close to x.

Definition 16.3 [LOCAL AVERAGE ESTIMATOR]. Let $h > 0$ be the window size, or BANDWIDTH, and $I_x = \{i \in \{1, \dots, n\} : |X_i - x| < h\}$. The estimator is

$$\hat{f}_{n,h}(x) = \begin{cases} \dfrac{1}{|I_x|} \displaystyle\sum_{i \in I_x} Y_i & \text{if } I_x \neq \emptyset, \\ 0 & \text{otherwise.} \end{cases}$$

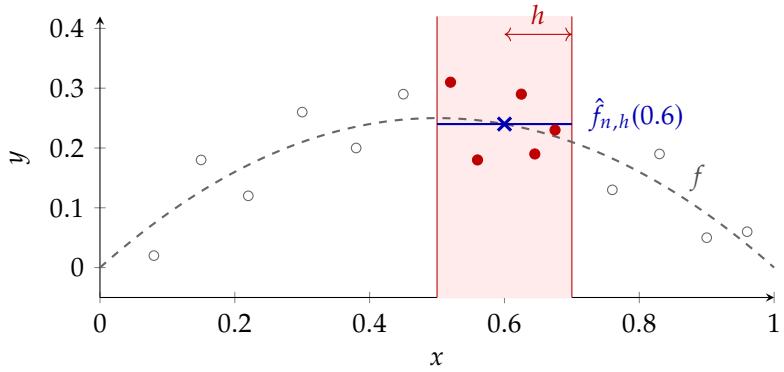


Figure 21: The local average at $x = 0.6$ with bandwidth $h = 0.1$ (schematic). Only the points with $|X_i - x| < h$ (filled) enter $\hat{f}_{n,h}(x)$, their average; the dashed curve is $f(x) = x(1-x)$. The points are illustrative, not data.

The bandwidth controls the fit. As $h \to 0$ the estimator overfits the data, averaging very few points; as $h \to \infty$ it underfits, since every point enters every window and $\hat{f}_{n,h}(x) = \bar{Y}_n$ for all x.

Example: $n = 100$, $f(x) = x(1-x)$. With $h = 0.005$ the estimate is extremely jagged, following individual points. With $h = 1$, and the X_i in $[0, 1]$, it is essentially the constant $\bar{Y}_n$. With $h = 0.2$ it is a smooth hump rising toward the middle of the interval and falling off at both ends, close in shape to f.

If the smoothness of f is known, meaning the quality of its local approximation by piecewise constant functions, there is a good choice of h depending on that smoothness. If it is unknown, other techniques are used, such as CROSS VALIDATION.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 16, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 17: Bayesian Statistics

17.1 Frequentist and Bayesian approaches

Up to now the approach has been FREQUENTIST. We observe data that were generated randomly, make assumptions on the generating process (i.i.d., Gaussian, a linear regression function, and so on), and tie the process to an object of interest, such as a parameter. That object is *unknown but fixed*, and we estimate it, build confidence intervals for it, or test hypotheses about it.

The BAYESIAN APPROACH keeps the random data and the model but changes the status of the parameter. Before seeing data we hold a PRIOR BELIEF about the parameter, describing both where we think it lies and how strongly we think so. The data are used to update it into a POSTERIOR BELIEF. The two approaches often lead to the same methods, and the prior matters less and less as the sample grows.

Example: *Proportion of women.* Let $p \in (0, 1)$ be the proportion of women in a population. Sample n people at random with replacement and let $X_i = 1$ if person i is a woman and $X_i = 0$ otherwise. The frequentist estimates p by the MLE $\bar{X}_n$, builds a confidence interval, or tests $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$. Before looking at the data, however, we believe that p is close to $1/2$: say we are 90% sure that $p \in [0.4, 0.6]$ and 95% sure that $p \in [0.3, 0.8]$. Statements of this kind, made for every range, describe a probability distribution for p. The Bayesian approach

1. includes such a prior belief mathematically in the statistical procedure, and
2. updates it using the data.

Discarding the data and declaring $p = 1/2$ would be one extreme; ignoring the belief would be the other. For many priors, the prior acts roughly like one extra observation: it matters among 5 observations and is negligible among a million. A prior that is a point mass at $1/2$, on the other hand, overrides the data completely.

The true parameter is not random. The Bayesian approach models our belief *as if* it were, through a two-layer generating process: nature first draws the parameter from the prior, then draws the data conditionally on it.

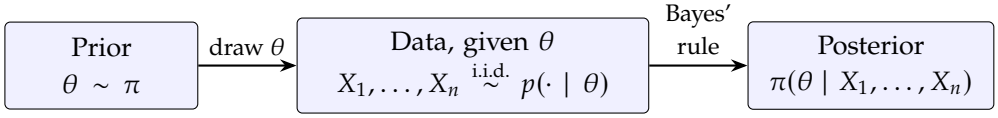


Figure 22: The Bayesian thought experiment. The parameter is drawn from the prior, the data are drawn given the parameter, and Bayes' rule turns the prior into the posterior.

17.2 Beta priors

A prior is a distribution on the parameter space Θ: the real line for a Gaussian mean, $(0, \infty)$ for an exponential rate, $(0, 1)$ for a Bernoulli parameter. A Gaussian prior is unsuitable on $(0, 1)$ because it spills outside the interval. The standard family on $(0, 1)$ is the Beta family.

Definition 17.1 [Beta distribution]. For $a, b > 0$, the Beta distribution $\mathcal{B}(a, b)$ has density

$$f(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}, \qquad x \in (0,1),$$

and $f(x) = 0$ otherwise, where $B(a, b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,dt = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function.

We write $f(x) \propto x^{a-1}(1-x)^{b-1}$: proportional to means equal up to a constant that does not depend on x. The two parameters produce a wide range of shapes, and for $a, b > 1$ the density is unimodal, which suits a population we believe homogeneous. (A multimodal prior would express several competing phenomena, which is better handled by mixture models.) With $a = b$, the density $\propto [x(1-x)]^{a-1}$ is symmetric about $1/2$, and a larger a concentrates it more sharply there.

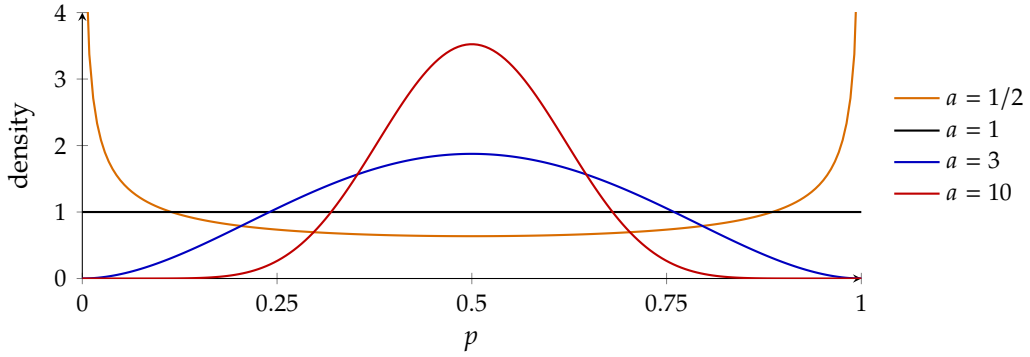


Figure 23: Exact densities of the symmetric priors $\mathcal{B}(a, a)$. As a grows the prior concentrates around $1/2$; $a = 1$ is the uniform prior and $a = 1/2$ is Jeffreys prior for the Bernoulli model (the density is unbounded at 0 and 1 and is cut off at the top of the plot).

In the example, take the prior $p \sim \mathcal{B}(a, a)$ for some $a > 0$, and given p, let $X_1, \ldots, X_n$ be i.i.d. Ber(p). Below we show that the posterior distribution, the distribution of p given the data, is

$$\mathcal{B}\Big(a + \sum_{i=1}^{n} X_i,\ a + n - \sum_{i=1}^{n} X_i\Big):$$

the first parameter gains the number of ones and the second gains the number of zeros.

17.3 Bayes' rule and the posterior distribution

Definition 17.2 [Prior and posterior]. Let $\pi(\cdot)$ be a density on the parameter space Θ, the prior distribution. Let $p_n(\cdot \mid \theta)$ be the joint density of $X_1, \dots, X_n$ conditionally on θ, where $\theta \sim \pi$; usually $X_1, \dots, X_n$ are i.i.d. conditionally on θ. The conditional distribution of θ given $X_1, \dots, X_n$ is the posterior distribution, with density $\pi(\cdot \mid X_1, \dots, X_n)$. PMFs may replace densities throughout.

Proposition 17.3 [Bayes' formula]. *For every $\theta \in \Theta$,*

$$\pi(\theta \mid X_1, \dots, X_n) = \frac{\pi(\theta)\, p_n(X_1, \dots, X_n \mid \theta)}{\int_\Theta p_n(X_1, \dots, X_n \mid t)\, \pi(t)\, dt},$$

so that $\pi(\theta \mid X_1, \dots, X_n) \propto \pi(\theta)\, p_n(X_1, \dots, X_n \mid \theta)$.

Proof. This is $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A)\mathbb{P}(A)/\mathbb{P}(B)$ for densities. The joint density of $(\theta, X_1, \dots, X_n)$ is $\pi(\theta) p_n(X_1, \dots, X_n \mid \theta)$. Dividing by the marginal density of the data, obtained by integrating θ out, gives the conditional density of θ. The denominator does not depend on θ, which gives the proportional form. □

The factor $p_n(X_1, \dots, X_n \mid \theta)$, viewed as a function of θ, is the likelihood. So the posterior is *the likelihood weighted by the prior*, then normalized to integrate to one in θ. This is why Bayesian and maximum likelihood estimation are close relatives. Following the maximum likelihood principle on the posterior gives the maximum a posteriori (MAP) estimator $\hat{\theta} = \arg\max_\theta \pi(\theta \mid X_1, \dots, X_n)$. But the posterior is a whole distribution: its mean, median, or spread can be reported too, and two posteriors with the same mode and mean can express very different confidence.

The normalizing constant is often hard to compute and usually not needed. Markov chain Monte Carlo (MCMC) methods sample from the posterior using only ratios $\pi(\theta)p_n(\cdot \mid \theta)/\big(\pi(\theta')p_n(\cdot \mid \theta')\big)$, in which the constant cancels.

Note. The denominator is often written $\int_\Theta p_n(X_1, \dots, X_n \mid t)\, d\pi(t)$, integration against the prior viewed as a measure. This covers priors with neither a density nor a PMF. For example, if $p = 1/2$ with probability $1/2$ and p is otherwise uniform on $(0, 1)$, then $\int f\, d\pi = \frac{1}{2} f(\frac{1}{2}) + \frac{1}{2}\int_0^1 f(t)\, dt$. When π has a density, $d\pi(t) = \pi(t)\, dt$.

Example: *Beta prior, Bernoulli data.* With $\pi(p) \propto p^{a-1}(1-p)^{a-1}$ on $(0, 1)$ and likelihood $p_n(X_1, \dots, X_n \mid p) = p^{\sum_i X_i}(1-p)^{n-\sum_i X_i}$,

$$\pi(p \mid X_1, \dots, X_n) \propto p^{a-1+\sum_{i=1}^n X_i}(1-p)^{a-1+n-\sum_{i=1}^n X_i}.$$

This is, up to a constant, the $\mathcal{B}(a + \sum_i X_i,\ a + n - \sum_i X_i)$ density, so it *is* that distribution, and the constant never has to be computed. With a $\mathcal{B}(a, b)$ prior the second parameter becomes $b + n - \sum_i X_i$.

Definition 17.4 [Conjugate prior]. A family of priors is conjugate to a model if the posterior belongs to the same family whenever the prior does.

Beta priors are conjugate for Bernoulli data, and Gaussian priors are conjugate for Gaussian data. With a conjugate prior one only tracks how the parameters update. Nobody believes that p is truly Beta distributed; the family is popular because it is flexible and mathematically convenient. Choosing a and b still requires care, since they encode both the location of the belief and its strength (the mean and the variance of the prior).

17.4 Non-informative and improper priors

Without prior information we may want a prior that prefers no value of θ. Since the prior only reweights the likelihood, the natural choice is to replace it by a constant: $\pi(\theta) \propto 1$. If Θ is bounded, this is the uniform prior on Θ. If Θ is unbounded, no constant density exists.

Definition 17.5 [Improper prior]. An improper prior on Θ is a measurable, nonnegative function $\pi(\cdot)$ on Θ that is not integrable.

Bayes' formula still defines a posterior from an improper prior whenever $\int_\Theta p_n(X_1, \dots, X_n \mid t)\pi(t)\, dt < \infty$. With $\pi \equiv 1$ the posterior is just the likelihood normalized to be a density in θ. Its mode is the MLE, but we may also report its mean, median, or variance.

Example: *Uniform prior, Bernoulli data.* If $p \sim \mathcal{U}(0,1)$ and given p, $X_1, \dots, X_n$ are i.i.d. Ber(p), then

$$\pi(p \mid X_1, \dots, X_n) \propto p^{\sum_i X_i}(1-p)^{n-\sum_i X_i}, \quad \text{that is,} \quad p \mid X_1, \dots, X_n \sim \mathcal{B}\Big(1 + \sum_{i=1}^n X_i,\ 1 + n - \sum_{i=1}^n X_i\Big).$$

This is the case $a = 1$ above. Its mode is $\bar{X}_n$, but its mean is $(1 + \sum_i X_i)/(n+2)$: one is added to the number of ones and one to the number of zeros. This matters mostly when all observations agree. With $n = 3$ and data $0, 0, 0$, the MLE is $\hat{p} = 0$, while the posterior mean is positive. The uniform prior gives probability zero to $p = 0$ and $p = 1$, and the estimator reflects it.

Example: *Flat prior, Gaussian data.* Let $\pi(\theta) = 1$ for all $\theta \in \mathbb{R}$ (improper) and, given θ, let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\theta, 1)$. Expanding the square and absorbing every factor free of θ into the constant,

$$\pi(\theta \mid X_1, \dots, X_n) \propto \exp\Big(-\frac{1}{2}\sum_{i=1}^n (X_i - \theta)^2\Big) \propto \exp\Big(\theta \sum_{i=1}^n X_i - \frac{n}{2}\theta^2\Big) \propto \exp\Big(-\frac{n}{2}(\theta - \bar{X}_n)^2\Big),$$

where the last step completes the square. Hence the posterior is $\mathcal{N}(\bar{X}_n, 1/n)$: centered at the sample mean and more concentrated as n grows. This is a statement about the posterior given the data, not the frequentist statement $\mathrm{Var}(\bar{X}_n) = 1/n$.

With the proper prior $\theta \sim \mathcal{N}(\mu, 1)$ instead, the extra factor $\exp(-\frac{1}{2}(\theta - \mu)^2)$ gives, after completing

the square again,

$$\theta \mid X_1, \dots, X_n \sim \mathcal{N}\Big(\frac{\sum_{i=1}^n X_i + \mu}{n+1}, \frac{1}{n+1}\Big).$$

The Gaussian prior is conjugate, and it acts exactly like one extra observation equal to μ.

17.5 Jeffreys prior

Definition 17.6 [Jeffreys prior]. Jeffreys prior is

$$\pi_J(\theta) \propto \sqrt{\det I(\theta)},$$

where $I(\theta)$ is the Fisher information matrix of the statistical model of $X_1, \dots, X_n$ in the frequentist approach (provided it exists).

The determinant measures the size of the matrix; in one dimension it is just $I(\theta)$, the inverse of the asymptotic variance of the MLE. So $\sqrt{I(\theta)}$ is inversely proportional to the asymptotic standard deviation of the MLE, and Jeffreys prior puts more weight on values of θ where the data carry more information.

Example: *The two examples.* For $\text{Ber}(p)$, $I(p) = 1/(p(1-p))$, so

$$\pi_J(p) \propto \frac{1}{\sqrt{p(1-p)}}, \qquad p \in (0,1),$$

which is the proper prior $\mathcal{B}(1/2, 1/2)$. It favors p near 0 and 1, where observations are most informative, and so pushes away from 1/2. For $\mathcal{N}(\theta, 1)$, the Fisher information is 1 (the identity in the multivariate case), so $\pi_J(\theta) \propto 1$ on $\mathbb{R}$, the improper flat prior. No location of a Gaussian is more informative than another.

Like the flat prior, Jeffreys prior is generic, and it does not depend on how the model is parametrized.

Theorem 17.7 [Reparametrization invariance]. *Let $\eta = \phi(\theta)$ for a one-to-one map ϕ. If θ has Jeffreys prior, the induced density $\tilde{\pi}$ of η satisfies*

$$\tilde{\pi}(\eta) \propto \sqrt{\det \tilde{I}(\eta)},$$

where $\tilde{I}(\eta)$ is the Fisher information of the statistical model parametrized by η. That is, $\tilde{\pi}$ is Jeffreys prior for the new parametrization.

So beliefs expressed about p^2 rather than p lead to the same prior. Lecture 18 proves this by a change of variables.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 17, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 18: Bayesian Statistics (cont.)

18.1 Invariance of Jeffreys prior

Recall from Lecture 17 that Jeffreys prior is $\pi_J(\theta) \propto \sqrt{\det I(\theta)}$. In one dimension it is proportional to $\sqrt{I(\theta)}$, the inverse of the asymptotic standard deviation of the MLE: $1/\sqrt{p(1-p)}$ for Bernoulli data, and the improper flat prior for $\mathcal{N}(\theta, 1)$ data.

Proof of reparametrization invariance (one dimension). Let $\eta = \phi(\theta)$ with ϕ one-to-one and differentiable, and let $\ell(\theta; x)$ be the log-likelihood of one observation, so that $I(\theta) = \mathbb{E}_\theta\big[(\partial_\theta \ell(\theta; X))^2\big]$. By the change of variables formula, the density of η induced by π_J is

$$\tilde{\pi}(\eta) = \pi_J(\theta)\left|\frac{d\theta}{d\eta}\right| \propto \sqrt{\mathbb{E}_\theta\Big[\Big(\frac{\partial \ell}{\partial \theta}\Big)^2\Big]\Big(\frac{d\theta}{d\eta}\Big)^2}, \qquad \theta = \phi^{-1}(\eta).$$

The factor $(d\theta/d\eta)^2$ does not depend on X, so it moves inside the expectation, and by the chain rule $\frac{\partial \ell}{\partial \theta}\frac{d\theta}{d\eta} = \frac{\partial \ell}{\partial \eta}$, the derivative of the log-likelihood in the η parametrization. Hence

$$\tilde{\pi}(\eta) \propto \sqrt{\mathbb{E}\Big[\Big(\frac{\partial \ell}{\partial \eta}\Big)^2\Big]} = \sqrt{\tilde{I}(\eta)}. \qquad \square$$

18.2 Bayesian confidence regions

A Bayesian analysis outputs the whole posterior $\pi(\cdot \mid X_1, \dots, X_n)$ rather than a single estimate. To communicate it, one summarizes where it puts most of its mass.

Definition 18.1 [Bayesian confidence region]. For $\alpha \in (0, 1)$, a Bayesian confidence region at level $1 - \alpha$ is a random subset $\mathcal{R} \subseteq \Theta$, depending on the sample $X_1, \dots, X_n$, such that

$$\mathbb{P}[\theta \in \mathcal{R} \mid X_1, \dots, X_n] = 1 - \alpha.$$

The probability is over θ drawn from the posterior, with the data held fixed, and $\mathcal{R}$ depends on the prior π. The word *region* is used because $\mathcal{R}$ need not be an interval.

A natural construction uses level sets. For a threshold $\tau > 0$ let $\mathcal{R}_\tau = \{\theta : \pi(\theta \mid X_1, \dots, X_n) \geq \tau\}$. The posterior mass of $\mathcal{R}_\tau$ decreases from 1 as $\tau \downarrow 0$ to 0 for τ above the maximum of the posterior density; choose τ so that the mass is exactly $1 - \alpha$. Because it collects the points of highest posterior density first, this gives the smallest possible region of mass $1 - \alpha$. For a Gaussian posterior it is an

interval centered at the posterior mean. For a multimodal posterior it is a union of intervals, which reflects the uncertainty between the modes, unlike a frequentist interval of the form $\bar{X}_n \pm c$.

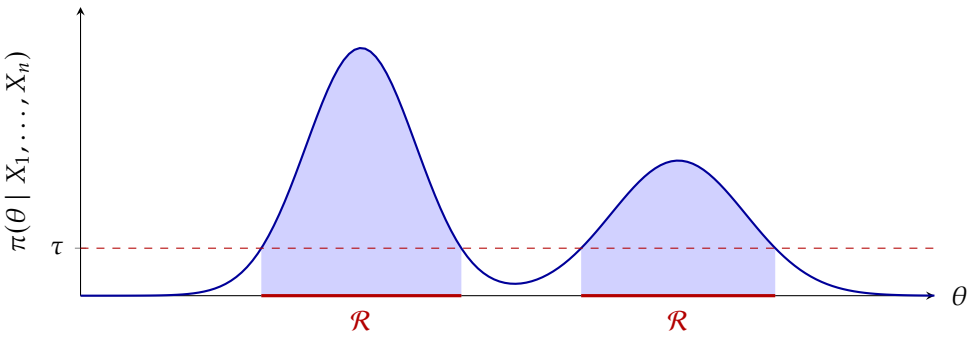


Figure 24: Level-set confidence region for an illustrative bimodal posterior, the Gaussian mixture $0.6\,\mathcal{N}(-1.2, 0.45^2) + 0.4\,\mathcal{N}(1.4, 0.55^2)$. At $\tau \approx 0.102$ the shaded area is 0.90, so $\mathcal{R}$, a union of two intervals, is a region at level 0.90.

18.3 Frequentist and Bayesian confidence

The two notions are distinct. For $\mathcal{N}(\theta, \sigma^2)$ data with σ known, the frequentist interval $\mathcal{R}_{\text{freq}} = [\bar{X}_n - 1.96\sigma/\sqrt{n},\ \bar{X}_n + 1.96\sigma/\sqrt{n}]$ satisfies $\mathbb{P}[\theta \in \mathcal{R}_{\text{freq}}] = 0.95$, where the probability is over the data and θ is fixed. Conditionally on $X_1, \ldots, X_n$ the interval is fixed, so $\mathbb{P}[\theta \in \mathcal{R}_{\text{freq}} \mid X_1, \ldots, X_n]$ is 1 if $\theta \in \mathcal{R}_{\text{freq}}$ and 0 otherwise. The 95% refers to the method: if the experiment is repeated many times, with a new sample and a new interval each time, the intervals contain θ in 95% of the repetitions. Similarly, a test run at level 5% many times is wrong in about 5% of the runs.

A Bayesian region makes a statement about the data actually observed, which is what one needs for a one-time problem, and the prior has most influence when samples are small. As the sample size grows, Bayesian and frequentist methods typically lead to the same decisions; in the Gaussian example the prior acts as a single extra observation among $n + 1$.

Example: *Two observations with sign noise.* Let $X_1 = \theta + \varepsilon_1$ and $X_2 = \theta + \varepsilon_2$, where $\varepsilon_1, \varepsilon_2$ are i.i.d. with $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = 1/2$. If $X_1 \neq X_2$, then one observation is $\theta + 1$ and the other $\theta - 1$, so $(X_1 + X_2)/2 = \theta$. If $X_1 = X_2$, then $\theta = X_1 - 1$ or $\theta = X_1 + 1$, and the data cannot tell which. Define the (frequentist) region

$$C = \begin{cases} \{X_1 - 1\} & \text{if } X_1 = X_2, \\ \{(X_1 + X_2)/2\} & \text{if } X_1 \neq X_2. \end{cases}$$

Proposition 18.2. $\mathbb{P}[\theta \in C] = 3/4$, *so C is a confidence region at level* 75%.

Proof. The two cases are disjoint. Subtracting θ throughout,

$$\mathbb{P}[\theta = X_1 - 1,\ X_1 = X_2] = \mathbb{P}[\varepsilon_1 = 1,\ \varepsilon_2 = 1] = \tfrac{1}{4},$$

and

$$\mathbb{P}\left[\theta = \tfrac{X_1+X_2}{2},\ X_1 \neq X_2\right] = \mathbb{P}[\varepsilon_1+\varepsilon_2=0] = \mathbb{P}[\varepsilon_1=1,\varepsilon_2=-1]+\mathbb{P}[\varepsilon_1=-1,\varepsilon_2=1] = \tfrac{1}{2}.$$

The total is $3/4$. The only miss is $\varepsilon_1=\varepsilon_2=-1$. □

Yet whenever $X_1 \neq X_2$ we know θ exactly, and the frequentist statement cannot express this: its 75% must account for future samples in which $X_1 = X_2$. A Bayesian statement conditions on the data. Take any prior π on the integers with $\pi(j) > 0$ for all $j \in \mathbb{Z}$, and suppose we observe $X_1 = 5$, $X_2 = 7$. The likelihood is $\mathbb{P}(5,7 \mid \theta) = \mathbb{P}(\varepsilon_1 = -1, \varepsilon_2 = 1) = 1/4$ if $\theta = 6$ and 0 for every other integer. By Bayes' formula only the term $t = 6$ survives in the normalizing sum, so

$$\pi(\theta \mid 5,7) = \frac{\pi(\theta)\mathbb{P}(5,7 \mid \theta)}{\sum_{t\in\mathbb{Z}} \pi(t)\mathbb{P}(5,7 \mid t)} = \begin{cases} \dfrac{\pi(6)\cdot\frac{1}{4}}{\pi(6)\cdot\frac{1}{4}} = 1, & \theta = 6, \\ 0, & \theta \neq 6. \end{cases}$$

So $\mathbb{P}[\theta = 6 \mid X_1 = 5, X_2 = 7] = 1$, a meaningful statement that requires almost nothing of the prior.

18.4 Bayesian estimation

The Bayesian framework can also be used to estimate the true parameter in the frequentist sense. The prior then reflects no belief; it is an artificial tool that defines a new class of estimators. Given the statistical model $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ of the sample, choose a distribution with density π on Θ (possibly improper) and compute the posterior $\pi(\cdot \mid X_1, \dots, X_n)$ as if π were a prior.

Definition 18.3 [Bayes estimator]. The Bayes estimator, or posterior mean, is

$$\hat{\theta}^{(\pi)} = \int_\Theta \theta\, \pi(\theta \mid X_1, \dots, X_n)\, d\theta.$$

The superscript records the dependence on π. Other summaries are the posterior median (not always uniquely defined), the posterior mode, which is the MAP estimator of Lecture 17 and ignores secondary modes, or a random draw from the posterior. The posterior carries more than any point summary: the posterior $\mathcal{N}(\bar{X}_n, 1/n)$ is described by two numbers, and its mean alone discards the posterior variance, which measures the confidence in the point estimate.

Lemma 18.4 [Mean of a Beta distribution]. *If $p \sim \mathcal{B}(a,b)$, then $\mathbb{E}[p] = a/(a+b)$.*

Proof. Multiplying the density by p raises the first exponent by one, and $\Gamma(x+1) = x\Gamma(x)$, so

$$\mathbb{E}[p] = \frac{1}{B(a,b)}\int_0^1 p^a(1-p)^{b-1}\,dp = \frac{B(a+1,b)}{B(a,b)} = \frac{\Gamma(a+1)\Gamma(a+b)}{\Gamma(a)\Gamma(a+b+1)} = \frac{a}{a+b}.$$ □

Example: *Bernoulli data with a $\mathcal{B}(a, a)$ prior.* The posterior is $\mathcal{B}(a + \sum_i X_i,\ a + n - \sum_i X_i)$, whose two parameters sum to $2a + n$. By Lemma 18.4,

$$\hat{p}^{(\pi)} = \frac{a + \sum_{i=1}^n X_i}{2a + n} = \frac{a/n + \bar{X}_n}{2a/n + 1}.$$

As $n \to \infty$ this behaves like $\bar{X}_n$; the effect of the prior, captured entirely by a, disappears. Jeffreys prior is $\propto p^{-1/2}(1-p)^{-1/2}$, the case $a = 1/2$, which gives

$$\hat{p}^{(\pi_J)} = \frac{1/(2n) + \bar{X}_n}{1/n + 1}.$$

The choice of a matters for n around 10, not for large n.

Example: *Gaussian data with the flat prior.* The posterior is $\mathcal{N}(\bar{X}_n, 1/n)$, so $\hat{\theta}^{(\pi_J)} = \bar{X}_n$. With $\pi \equiv 1$ the MAP estimator is the MLE; here the posterior is symmetric about its mode, so the mean coincides with it.

In each of these examples the Bayes estimator is consistent and asymptotically normal. Asymptotic normality concerns the sampling distribution of the posterior mean; the posterior itself (a Beta distribution in the first example) need not be Gaussian. In general, the asymptotic properties of the Bayes estimator do not depend on the choice of the prior: an informative prior affects the estimate at moderate n, and its effect is washed away as n grows, provided the prior puts positive weight near the true parameter. A prior with no support at the true parameter can never recover it.

More generally, one can study a Bayesian procedure from the frequentist point of view, with a fixed true θ and no prior: does the posterior concentrate around θ as $n \to \infty$, and does its width shrink like $1/\sqrt{n}$? Results of this kind are called Bernstein–von Mises theorems, and the study of such rates is called posterior concentration.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 18, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 19: Principal Component Analysis

19.1 Multivariate data

We now observe only covariates, with no response: this is unsupervised learning. The goal is to make sense of the data, to represent and visualize them. Let $\mathbf{X}$ be a d-dimensional random vector and $\mathbf{X}_1, \dots, \mathbf{X}_n$ be n independent copies of $\mathbf{X}$. Their realizations form a cloud of n points in $\mathbb{R}^d$, with d large (at least 4, so that the cloud cannot be drawn). Subscripts index observations and superscripts index coordinates: $\mathbf{X}_i = (X_i^1, \dots, X_i^d)^\top$. Stack the observations as the rows of the random $n \times d$ data matrix

$$\mathbb{X} = \begin{pmatrix} \cdots & \mathbf{X}_1^\top & \cdots \\ & \vdots & \\ \cdots & \mathbf{X}_n^\top & \cdots \end{pmatrix},$$

so that rows are observations and columns are attributes.

19.2 Mean and covariance

Everything below concerns variances, so assume $\mathbb{E}[\|\mathbf{X}\|_2^2] < \infty$. The mean of $\mathbf{X}$ is the vector $\mathbb{E}[\mathbf{X}] = (\mathbb{E}[X^1], \dots, \mathbb{E}[X^d])^\top$, the center of the distribution. Its spread in each direction is recorded by the covariance matrix $\Sigma = (\sigma_{j,k})_{j,k=1,\dots,d}$, where $\sigma_{j,k} = \mathrm{Cov}(X^j, X^k)$. It is symmetric.

Proposition 19.1 [Covariance in matrix form].

$$\Sigma = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top = \mathbb{E}\big[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top\big].$$

Proof. The outer product $\mathbf{x}\mathbf{x}^\top$ has (j,k) entry $x^j x^k$. So the (j,k) entry of the first expression is $\mathbb{E}[X^j X^k] - \mathbb{E}[X^j]\mathbb{E}[X^k] = \sigma_{j,k}$. For the second, expand the product and use linearity of expectation:

$$\mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top + \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top,$$

which equals the first expression. Matrix products do not commute, but both cross terms have expectation $\mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top$. □

The empirical counterparts replace expectations by averages. The empirical mean is $\bar{\mathbf{X}} = \frac{1}{n}\sum_{i=1}^n \mathbf{X}_i = (\bar{X}^1, \dots, \bar{X}^d)^\top$. The empirical covariance matrix (or sample covariance matrix) is $S = (s_{j,k})_{j,k=1,\dots,d}$,

where $s_{j,k}$ is the empirical covariance of X_i^j and X_i^k, $i = 1, \ldots, n$. Exactly as in Proposition 19.1,

$$S = \frac{1}{n}\sum_{i=1}^{n} \mathbf{X}_i \mathbf{X}_i^\top - \bar{\mathbf{X}}\bar{\mathbf{X}}^\top = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top.$$

19.3 Matrix form and the centering projector

The columns of $\mathbb{X}^\top$ are $\mathbf{X}_1, \ldots, \mathbf{X}_n$. With $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^n$, summing the columns gives $\bar{\mathbf{X}} = \frac{1}{n}\mathbb{X}^\top \mathbf{1}$. Multiplying $\mathbb{X}^\top$ by $\mathbb{X}$ column by row gives $\mathbb{X}^\top\mathbb{X} = \sum_{i=1}^n \mathbf{X}_i\mathbf{X}_i^\top$: writing $\mathbb{X}^\top = \sum_i M_i$, where M_i keeps only the ith column, the cross terms $M_i M_j^\top$ with $i \neq j$ vanish and $M_i M_i^\top = \mathbf{X}_i\mathbf{X}_i^\top$. Since $\bar{\mathbf{X}}\bar{\mathbf{X}}^\top = \frac{1}{n^2}\mathbb{X}^\top \mathbf{1}\mathbf{1}^\top \mathbb{X}$,

$$S = \frac{1}{n}\mathbb{X}^\top\mathbb{X} - \frac{1}{n^2}\mathbb{X}^\top \mathbf{1}\mathbf{1}^\top \mathbb{X} = \frac{1}{n}\mathbb{X}^\top H \mathbb{X}, \qquad H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top.$$

The matrix $\mathbf{1}\mathbf{1}^\top$ is the $n \times n$ matrix of ones, so H has $1 - \frac{1}{n}$ on the diagonal and $-\frac{1}{n}$ elsewhere.

Proposition 19.2 [Centering matrix]. *H is an orthogonal projector, $H^2 = H$ and $H^\top = H$. It projects onto the vectors with mean zero, the hyperplane of $\mathbb{R}^n$ orthogonal to $\mathbf{1}$.*

Proof. Symmetry is clear. Since $\mathbf{1}^\top\mathbf{1} = n$,

$$H^2 = I_n - \frac{2}{n}\mathbf{1}\mathbf{1}^\top + \frac{1}{n^2}\mathbf{1}(\mathbf{1}^\top\mathbf{1})\mathbf{1}^\top = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top = H.$$

For $\mathbf{v} \in \mathbb{R}^n$, $H\mathbf{v} = \mathbf{v} - \frac{1}{n}(\mathbf{1}^\top\mathbf{v})\mathbf{1} = \mathbf{v} - \bar{v}\mathbf{1}$, where $\bar{v}$ is the average of the entries of $\mathbf{v}$. So H subtracts the mean from each entry, and $H\mathbf{v}$ has mean zero, that is, $\mathbf{1}^\top H\mathbf{v} = 0$. If $\mathbf{v}$ already has mean zero, $H\mathbf{v} = \mathbf{v}$. □

Consequently $S = \frac{1}{n}\mathbb{X}^\top H^\top H\mathbb{X} = \frac{1}{n}(H\mathbb{X})^\top(H\mathbb{X})$: first center each column of $\mathbb{X}$, then take inner products of the centered columns.

19.4 Variance in a direction

Proposition 19.3 [Quadratic forms are variances]. *For $\mathbf{u} \in \mathbb{R}^d$,*

- *$\mathbf{u}^\top\Sigma\mathbf{u}$ is the variance of $\mathbf{u}^\top\mathbf{X}$;*
- *$\mathbf{u}^\top S\mathbf{u}$ is the sample variance of $\mathbf{u}^\top\mathbf{X}_1, \ldots, \mathbf{u}^\top\mathbf{X}_n$.*

Proof. Since $\mathbf{u}$ is deterministic, it moves inside the expectations:

$$\mathbf{u}^\top \Sigma \mathbf{u} = \mathbb{E}[\mathbf{u}^\top \mathbf{X}\mathbf{X}^\top \mathbf{u}] - \mathbb{E}[\mathbf{u}^\top \mathbf{X}]\,\mathbb{E}[\mathbf{X}^\top \mathbf{u}] = \mathbb{E}[(\mathbf{u}^\top \mathbf{X})^2] - (\mathbb{E}[\mathbf{u}^\top \mathbf{X}])^2 = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}).$$

The same computation with averages, using $\mathbf{u}^\top \bar{\mathbf{X}} = \frac{1}{n}\sum_i \mathbf{u}^\top \mathbf{X}_i$, gives $\mathbf{u}^\top S\mathbf{u} = \frac{1}{n}\sum_i (\mathbf{u}^\top \mathbf{X}_i)^2 - (\mathbf{u}^\top \bar{\mathbf{X}})^2$. □

The entries of Σ give covariances only in the coordinate system; the quadratic form gives the variance along any linear combination. If $\|\mathbf{u}\|_2 = 1$, then $\mathbf{u}^\top \mathbf{X}_i$ is the coordinate of the projection of $\mathbf{X}_i$ onto the line spanned by $\mathbf{u}$, so $\mathbf{u}^\top S\mathbf{u}$ measures how spread out (diverse) the points are in direction $\mathbf{u}$. (For a general $\mathbf{u}$ the coordinates are scaled by $\|\mathbf{u}\|_2$.)

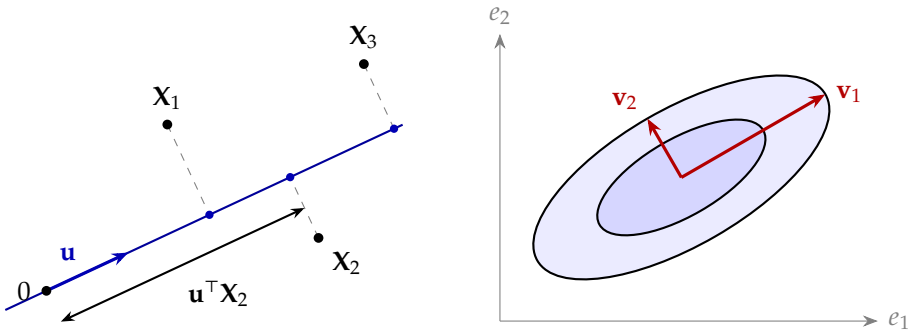


Figure 25: Schematic. Left: for a unit vector $\mathbf{u}$, the number $\mathbf{u}^\top \mathbf{X}_i$ is the coordinate of the projection of $\mathbf{X}_i$ on the line spanned by $\mathbf{u}$. Right: level sets of a two-dimensional Gaussian are ellipses whose principal axes are the eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of Σ, generally not aligned with the coordinate axes e_1, e_2.

- If $\mathbf{u}^\top S\mathbf{u} = 0$, all the projections coincide, so all the $\mathbf{X}_i$ lie in an affine subspace orthogonal to $\mathbf{u}$.
- If $\mathbf{u}^\top \Sigma \mathbf{u} = 0$, then $\mathbf{X}$ lies almost surely in an affine subspace orthogonal to $\mathbf{u}$.
- If $\mathbf{u}^\top S\mathbf{u}$ is large with $\|\mathbf{u}\|_2 = 1$, the direction $\mathbf{u}$ explains well the spread of the sample.

19.5 Why variance, and the optimization problem

The sample is a cloud of points in $\mathbb{R}^d$, and we would like to project it onto a linear subspace of dimension $d' < d$ while keeping as much information as possible. A projection loses information when points collide: projecting onto a plane orthogonal to a line on which the points lie sends them all to one point. So we want directions in which the projected points stay spread out. Spread can be measured in many ways (pairwise distances, energies), and variance is a good proxy, though not perfect: two tight clumps far apart also give large variance. Principal component analysis (PCA) keeps as much of the covariance structure as possible, by keeping orthogonal directions that discriminate the points of the cloud well.

The first goal is therefore

$$\max_{\|\mathbf{u}\|_2=1} \mathbf{u}^\top S\mathbf{u}.$$

The constraint is needed: without it, scaling $\mathbf{u}$ by 10 scales the objective by 100, and the supremum is infinite. This is the maximization of a convex function over the sphere, a non-convex problem. It is nevertheless solved exactly, not by optimization but by linear algebra.

19.6 The spectral theorem

Theorem 19.4 [Spectral decomposition]. *Any real symmetric matrix $A \in \mathbb{R}^{d\times d}$ can be written*

$$A = PDP^\top,$$

where P is a $d\times d$ orthogonal matrix, $PP^\top = P^\top P = I_d$, and D is diagonal. The diagonal elements $\lambda_1, \dots, \lambda_d$ of D are the eigenvalues *of A and the columns $\mathbf{v}_1, \dots, \mathbf{v}_d$ of P are the corresponding* eigenvectors. *A is positive semidefinite if and only if all its eigenvalues are nonnegative.*

Orthogonality of P means $\mathbf{v}_j^\top \mathbf{v}_k = 0$ for $j \neq k$ and $\|\mathbf{v}_j\|_2 = 1$. Applying A to a vector rotates it into the coordinate system of the eigenvectors (by $P^\top$), stretches each coordinate by λ_j (by D), and rotates back (by P). For a covariance matrix, think of the ellipsoidal level sets of a multivariate Gaussian: the eigenvectors are the principal axes, from the most elongated direction to the least, and the eigenvalues measure the elongation along them (Figure 25).

Both Σ and S are symmetric and positive semidefinite: by Proposition 19.3, $\mathbf{u}^\top\Sigma\mathbf{u}$ and $\mathbf{u}^\top S\mathbf{u}$ are variances, hence nonnegative. So their eigenvalues are nonnegative: the stretches never flip a sign.

For the eigenvectors, $P^\top\mathbf{v}_1 = (\mathbf{v}_1^\top\mathbf{v}_1, \mathbf{v}_2^\top\mathbf{v}_1, \dots, \mathbf{v}_d^\top\mathbf{v}_1)^\top = e_1$, so

$$\Sigma\mathbf{v}_1 = PDe_1 = P(\lambda_1 e_1) = \lambda_1\mathbf{v}_1, \qquad \mathbf{v}_1^\top\Sigma\mathbf{v}_1 = \lambda_1\mathbf{v}_1^\top\mathbf{v}_1 = \lambda_1,$$

and likewise $\mathbf{v}_j^\top\Sigma\mathbf{v}_j = \lambda_j$. The variance along the eigenvector $\mathbf{v}_j$ is the eigenvalue λ_j, which again shows that the eigenvalues are nonnegative.

The resulting PCA principle, proved in Lecture 20, is that the maximal variance over *all* unit directions is attained at an eigenvector, the one with the largest eigenvalue. PCA picks that eigenvector first, then the direction orthogonal to it with the most variance, which is the eigenvector of the second largest eigenvalue, and so on. Stopping after a few steps gives a low-dimensional representation in the coordinates given by the leading eigenvectors.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 19, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 20: Principal Component Analysis (cont.)

20.1 Decorrelating the cloud

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \in \mathbb{R}^d$ have empirical covariance matrix S, with spectral decomposition $S = PDP^\top$ as in Lecture 19: $P = (\mathbf{v}_1, \ldots, \mathbf{v}_d)$ is orthogonal and $D = \mathrm{Diag}(\lambda_1, \ldots, \lambda_d)$. The decomposition is not unique, since permuting the columns of P together with the diagonal of D gives another one, but the multiset of eigenvalues is. Fix the order

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0,$$

so that λ_1 is the largest eigenvalue.

Proposition 20.1 [Rotated data]. *Let $\mathbf{Y}_i = P^\top \mathbf{X}_i$. The empirical covariance matrix of $\mathbf{Y}_1, \ldots, \mathbf{Y}_n$ is D. In particular, λ_j is the empirical variance of $\mathbf{v}_j^\top \mathbf{X}_1, \ldots, \mathbf{v}_j^\top \mathbf{X}_n$, and different coordinates of the $\mathbf{Y}_i$ are empirically uncorrelated.*

Proof. Replacing $\mathbf{X}_i$ by $\mathbf{X}_i - \bar{\mathbf{X}}$ changes neither covariance, so assume $\bar{\mathbf{X}} = 0$. Then $\bar{\mathbf{Y}} = P^\top \bar{\mathbf{X}} = 0$ and $S = \frac{1}{n} \sum_i \mathbf{X}_i \mathbf{X}_i^\top$, so the empirical covariance of the $\mathbf{Y}_i$ is

$$\frac{1}{n} \sum_{i=1}^{n} P^\top \mathbf{X}_i \mathbf{X}_i^\top P = P^\top S P = P^\top P D P^\top P = D.$$

The jth coordinate of $\mathbf{Y}_i$ is $\mathbf{v}_j^\top \mathbf{X}_i$. □

The same holds for Σ and its eigendecomposition: the coordinates of $P^\top \mathbf{X}$ are uncorrelated with variances given by the eigenvalues, and independent if $\mathbf{X}$ is Gaussian. Geometrically, $P^\top$ rotates the coordinate axes to align them with the principal axes of the ellipsoidal cloud.

20.2 The leading eigenvector maximizes the variance

Theorem 20.2 [First principal direction]. *For every unit vector $\mathbf{u} \in \mathbb{R}^d$, $\mathbf{u}^\top S \mathbf{u} \leq \lambda_1$, with equality for $\mathbf{u} = \mathbf{v}_1$. So $\mathbf{v}_1$ maximizes the empirical variance of $\mathbf{u}^\top \mathbf{X}_1, \ldots, \mathbf{u}^\top \mathbf{X}_n$ over unit vectors $\mathbf{u}$, and the maximum is λ_1. Likewise $\mathrm{Var}(\mathbf{u}^\top \mathbf{X}) = \mathbf{u}^\top \Sigma \mathbf{u}$ is maximized by the eigenvector of Σ for its largest eigenvalue.*

Proof. Let $\mathbf{b} = P^\top \mathbf{u}$, the coordinates of $\mathbf{u}$ in the basis of eigenvectors. Then

$$\mathbf{u}^\top S \mathbf{u} = \mathbf{u}^\top P D P^\top \mathbf{u} = \mathbf{b}^\top D \mathbf{b} = \sum_{j=1}^{d} \lambda_j b_j^2 \le \lambda_1 \sum_{j=1}^{d} b_j^2 = \lambda_1 \|\mathbf{b}\|_2^2,$$

since the λ_j are nonnegative and at most λ_1. A change of orthonormal basis preserves length: $\|\mathbf{b}\|_2^2 = \mathbf{u}^\top P P^\top \mathbf{u} = \|\mathbf{u}\|_2^2 = 1$. For $\mathbf{u} = \mathbf{v}_1$, $\mathbf{b} = P^\top \mathbf{v}_1 = e_1$, so the sum is exactly λ_1. The population statement is the same argument with Σ and its eigendecomposition. □

This is the non-convex problem of Lecture 19, maximizing a convex function over the sphere, solved by the spectral decomposition with no iterative optimization. The next directions are obtained the same way.

Theorem 20.3 [Principal directions].

$$\mathbf{v}_1 \in \operatorname*{arg\,max}_{\|\mathbf{u}\|=1} \mathbf{u}^\top S \mathbf{u}, \qquad \mathbf{v}_2 \in \operatorname*{arg\,max}_{\|\mathbf{u}\|=1,\ \mathbf{u} \perp \mathbf{v}_1} \mathbf{u}^\top S \mathbf{u}, \qquad \dots, \qquad \mathbf{v}_d \in \operatorname*{arg\,max}_{\|\mathbf{u}\|=1,\ \mathbf{u} \perp \mathbf{v}_j,\ j=1,\dots,d-1} \mathbf{u}^\top S \mathbf{u}.$$

Hence the k orthogonal directions in which the cloud is most spread out are the eigenvectors associated with the k largest eigenvalues of S.

Proof. If $\mathbf{u} \perp \mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ is a unit vector, then $b_j = \mathbf{v}_j^\top \mathbf{u} = 0$ for $j < k$, so as above $\mathbf{u}^\top S \mathbf{u} = \sum_{j \ge k} \lambda_j b_j^2 \le \lambda_k \|\mathbf{b}\|_2^2 = \lambda_k$. The vector $\mathbf{v}_k$ is orthogonal to $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ and gives $\mathbf{b} = e_k$, attaining λ_k. □

So $\mathbf{v}_2$ maximizes the variance not explained by $\mathbf{v}_1$. If $\mathbf{v}_1$ were e_1, this would amount to dropping the first coordinate and repeating the problem in dimension $d - 1$; in general the data are rotated first. PCA thus combines three ingredients: statistics (variance measures spread), optimization (maximize it over unit directions), and linear algebra (the answer is the eigenvectors, in decreasing order of their eigenvalues).

20.3 The algorithm

Algorithm 20.1: Principal component analysis

Input: A cloud of n points $\mathbf{X}_1, \dots, \mathbf{X}_n$ in dimension d; a target dimension $k < d$.
Output: The projected points $\mathbf{Y}_1, \dots, \mathbf{Y}_n \in \mathbb{R}^k$.

Compute the empirical covariance matrix S
Compute $S = PDP^\top$ with $D = \mathrm{Diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_1 \ge \cdots \ge \lambda_d$, and $P = (\mathbf{v}_1, \dots, \mathbf{v}_d)$ orthogonal
$P_k \leftarrow (\mathbf{v}_1, \dots, \mathbf{v}_k) \in \mathbb{R}^{d \times k}$
for $i \leftarrow 1$ **to** n **do**
    $\mathbf{Y}_i \leftarrow P_k^\top \mathbf{X}_i \in \mathbb{R}^k$
return $\mathbf{Y}_1, \dots, \mathbf{Y}_n$

Only the diagonalization is costly; computing the k leading eigenvectors of a $d \times d$ matrix takes on the order of kd^2 operations, comparable to reading S for small k. The vector $\mathbf{Y}_i = P_k^\top \mathbf{X}_i$ holds the k coordinates of the orthogonal projection of $\mathbf{X}_i$ onto $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, the closest point of that subspace. The retained directions $\mathbf{v}_1, \dots, \mathbf{v}_k$ are the PRINCIPAL COMPONENTS.

20.4 Choosing k

There is no exact science here, as with choosing how much to compress an image.

- *Visualization.* Take $k = 2$ or 3. With larger k, one typically shows all $\binom{k}{2}$ two-dimensional plots of pairs of components.
- *Scree plot.* Take k at an inflection point (an "elbow") of the decreasing sequence $\lambda_1, \dots, \lambda_d$, after which each extra component adds much less explained variance.
- *Explained variance.* Take k such that

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d} \geq 1 - \alpha,$$

  for some $\alpha \in (0, 1)$ that sets the approximation error the practitioner accepts. The numerator is the VARIANCE EXPLAINED by the PCA and the denominator $\lambda_1 + \cdots + \lambda_d = \operatorname{Tr}(S)$ is the TOTAL VARIANCE; the ratio, between 0 and 1, is the PROPORTION OF EXPLAINED VARIANCE.

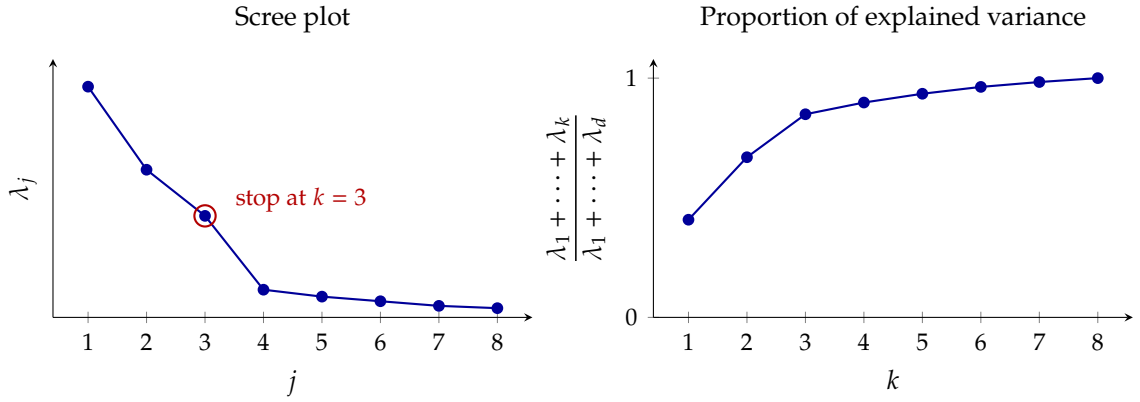


Figure 26: Schematic scree plot for hypothetical eigenvalues $\lambda_1 \geq \cdots \geq \lambda_8$, with an elbow at $k = 3$ (left), and the corresponding proportion of explained variance, whose increments shrink after the elbow (right).

Dimension reduction is useful beyond visualization. Replacing each $\mathbf{X}_i$ by its k principal coordinates compresses the data, and a regression of a response on these coordinates instead of the high-dimensional $\mathbf{X}_i$ is PRINCIPAL COMPONENT REGRESSION, used for instance with $k = 10$ when there are too few observations for the original dimension.

Example: *Genes mirror geography.* For gene expression data on 500,000 genes for about 1400

Europeans (so $n < d$, after heavy preprocessing), projecting onto the first two principal components and color-coding by country recovers roughly the map of Europe: compressed to two numbers, the genomic information gives approximately longitude and latitude.

20.5 Statistics and PCA

PCA as described is a data-analysis procedure: it applies to any cloud of points, random or not, and nothing above used independence. In statistics it can be used for estimation. If $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. in $\mathbb{R}^d$, how well does PCA on S recover the structure of the population covariance Σ?

- If d is fixed and $n \to \infty$ (so $n \gg d$), S is a consistent estimator of Σ, and the leading eigenvector of S converges to that of Σ.
- In many applications $n \ll d$, as in gene expression. In the regime studied by random matrix theory, $n, d \to \infty$ with $d/n \to \gamma > 0$. The angle between the leading eigenvectors of S and of Σ then tends to a nonzero function of γ, which tends to 0 as $\gamma \to 0$ and to a right angle as $\gamma \to \infty$, when the leading eigenvector of S carries no information. The proportion of explained variance computed from S is also biased in this regime and must be recalibrated.
- If Σ is known beforehand to have (almost) low rank, run PCA on S and keep only the k largest eigenvalues:

  $$S' = P \operatorname{Diag}(\lambda_1, \dots, \lambda_k, 0, \dots, 0) P^\top.$$

  Under the low-rank assumption S' is a better estimator of Σ than S, and a theoretical analysis leads to an optimal choice of the tuning parameter k.

20.6 Sparse PCA

When $d \gg n$, one makes more assumptions: look not for any direction of maximal variance but for a *sparse* one, with few nonzero coordinates. Sparse PCA solves

$$\max_{\mathbf{u}} \ \mathbf{u}^\top S \mathbf{u} \quad \text{subject to} \quad \|\mathbf{u}\|_2 = 1, \quad \sum_{j=1}^{d} \mathbf{1}\{u_j \neq 0\} \le s.$$

In effect d is replaced by s, so n need only be much larger than s. Sparse directions are also interpretable. A leading eigenvector in dimension 100,000 has many small coordinates, and it is hard to tell which variables matter; a sparse direction is a combination of at most s original variables (for instance a few genes). The price is computational: the sparsity constraint is very non-convex, the linear algebra solution is lost, and convex relaxations lead to semidefinite programs that scale poorly. In practice one uses heuristics such as iterative thresholding.

Example: *Survey of doctors.* About 1000 doctors rated, on a scale of 1 to 5, many aspects of their interactions with pharmaceutical sales representatives. The first three principal components were

rotated within the three-dimensional subspace they span (only the subspace matters), to sharpen the contrast between large and small coordinates. Reading off the questions with large coordinates gave three named factors: medical knowledge, quality of interaction, and fair and critical opinion. Sparse PCA produces such groups directly, with exact zeros elsewhere and a tunable number s of nonzero coordinates, instead of an ad hoc reading.

Large variance along a direction need not mean an even spread: two groups of points with different centers also produce it. This is why PCA is a standard first step for *mixtures* (clusters), where the direction through the two centers carries much variance. More generally, PCA is appropriate whenever a matrix looks like a low-rank matrix plus noise, so that most eigenvalues are small. An example is the preprocessed adjacency matrix of a network with a few communities. Under the names SPECTRAL METHODS or SVD, these techniques are used throughout.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 20, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 21: Generalized Linear Models

Generalized linear models keep the linear predictor $X^\top \beta$ of linear regression but relax two of its assumptions: the conditional distribution of the response and the way the mean is tied to $X^\top \beta$.

21.1 From linear models to GLMs

The Gaussian linear model assumes

$$Y \mid X \sim \mathcal{N}\big(\mu(X), \sigma^2\big), \qquad \mu(X) = \mathbb{E}[Y \mid X] = X^\top \beta = \sum_{j=1}^{p} \beta_j X^{(j)}.$$

It has two components.

1. Random component: given X, the response is Gaussian, and X affects only its mean $\mu(X) = \mathbb{E}[Y \mid X]$.
2. Link: the mean depends on the covariates $X = (X^{(1)}, \ldots, X^{(p)})^\top$ through $\mu(X) = X^\top \beta$.

Definition 21.1 [Generalized linear model]. A generalized linear model (GLM) assumes

1. $Y \mid X$ follows a distribution from an exponential family (Definition 21.2 below);
2. for some function g, called the link function,

$$g\big(\mu(X)\big) = X^\top \beta, \qquad \mu(X) = \mathbb{E}[Y \mid X].$$

Equivalently $\mu(X) = g^{-1}(X^\top \beta)$; the name goes to g rather than to g^{-1}. The linear form $X^\top \beta$ is never changed. The link is not a transformation of the covariates: transforming the X's and then fitting a linear model is still linear regression, whereas g transforms the mean.

The link makes two ranges compatible. If Y is Bernoulli, then $\mu(X) \in (0, 1)$, while $X^\top \beta$ can be any real number. Least squares fitted to 0/1 responses gives a line that may stay in $[0, 1]$ on the observed x's, but at a new x further out it predicts a "probability" such as 1.2. Similarly, a positive mean (Poisson, exponential) must be mapped onto all of $\mathbb{R}$.

21.2 Three examples

Example: *Disease occurrence rate.* Early in an epidemic, the expected number μ_i of new cases on day t_i may grow exponentially, $\mu_i = \gamma e^{\delta t_i}$. This is not of the form $X^\top\beta$, but the LOG LINK makes it linear:

$$\log \mu_i = \log \gamma + \delta t_i = \beta_0 + \beta_1 t_i,$$

with covariate vector $(1, t_i)$ and reparametrization $\beta_0 = \log \gamma$, $\beta_1 = \delta$. The response is a count, so a Poisson distribution with mean μ_i is a reasonable random component.

Example: *Prey capture rate.* The capture rate y_i of a predator increases with the prey density x_i but levels off when the predator catches as much as it can handle. A two-parameter model is

$$\mu_i = \frac{\alpha x_i}{h + x_i},$$

where α is the maximum capture rate and h is the density at which the rate is half the maximum (Figure 27). The model is nonlinear in its parameters, but the RECIPROCAL LINK makes it linear in the covariate $1/x_i$:

$$g(\mu_i) = \frac{1}{\mu_i} = \frac{1}{\alpha} + \frac{h}{\alpha} \cdot \frac{1}{x_i} = \beta_0 + \beta_1 \frac{1}{x_i}.$$

If the standard deviation of the capture rate is roughly proportional to its mean, a Gamma distribution (which is positive and has this property) is a natural random component.

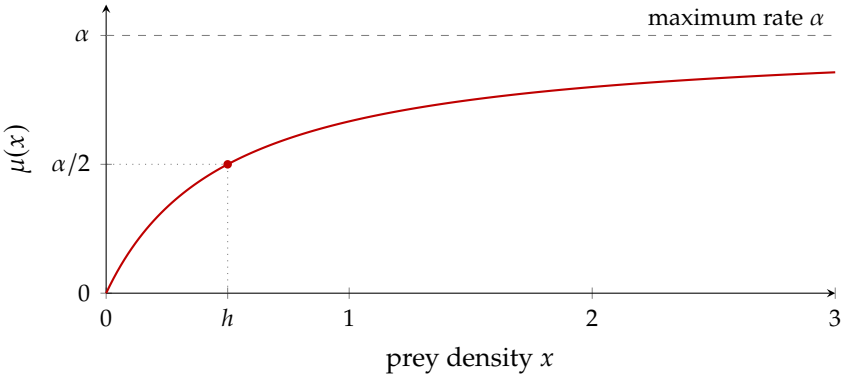


Figure 27: The capture-rate curve $\mu(x) = \alpha x/(h + x)$, plotted for $\alpha = 1$ and $h = 1/2$. It starts at 0, reaches $\alpha/2$ at $x = h$, and levels off at the maximum rate α.

Example: *Kyphosis data.* For 81 children after corrective spinal surgery, the binary response records whether a postoperative deformity (kyphosis) is present. The covariates are age in months, the number of vertebrae involved, and the first vertebra of the range operated on. A binary response leaves no modeling choice: $Y \mid X$ is Bernoulli with mean $\mu(X) \in (0, 1)$. Since $\mu(X) = X^\top\beta$ is impossible, we need an invertible f with $f(X^\top\beta) \in (0, 1)$. Every continuous, strictly increasing CDF on $\mathbb{R}$ provides one.

The normal linear model fits none of these examples. GLMs supply a broader family of response

distributions, and maximum likelihood still fits them. Unlike the Gaussian case, the likelihood has no closed-form maximizer in general. It is, however, concave with a lot of structure, and the resulting iterative algorithm solves a least-squares problem at each step.

21.3 Exponential families

Definition 21.2 [Exponential family]. A family of distributions $\{P_\theta : \theta \in \Theta\}$, $\Theta \subset \mathbb{R}^k$, is a k-parameter exponential family on $\mathbb{R}^q$ if there are real functions $\eta_1, \dots, \eta_k$ and B of θ, and $T_1, \dots, T_k$ and h of $x \in \mathbb{R}^q$, such that the density (PDF or PMF) of P_θ is

$$p_\theta(x) = \exp\Big[\sum_{i=1}^{k} \eta_i(\theta) T_i(x) - B(\theta)\Big] h(x).$$

The definition restricts how θ and x may interact. In one dimension the interaction is the product θx inside an exponential, times factors depending only on θ or only on x. In higher dimensions, θ and x are first mapped to vectors $\eta(\theta)$ and $T(x)$ of the same length, and the interaction is the inner product $\langle \eta(\theta), T(x) \rangle$. A factor $c(\theta)$ can be absorbed as $B(\theta) = \log\big(1/c(\theta)\big)$; thus B is a normalizing constant that makes p_θ integrate to one. The factor $h(x)$ amounts to a change of reference measure: p_θ / h is the density with respect to the measure with density h. This lets discrete and continuous distributions be treated together.

Example: *Gaussian.* Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$. Expanding the square in the exponent,

$$p_\theta(x) = \exp\Big(\frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2 - \frac{\mu^2}{2\sigma^2}\Big) \frac{1}{\sigma\sqrt{2\pi}},$$

a two-parameter exponential family with

$$\eta_1 = \frac{\mu}{\sigma^2}, \quad \eta_2 = -\frac{1}{2\sigma^2}, \quad T_1(x) = x, \quad T_2(x) = x^2, \quad B(\theta) = \frac{\mu^2}{2\sigma^2} + \log\big(\sigma\sqrt{2\pi}\big), \quad h(x) = 1.$$

Constants can go into either B or h; for instance $B = \mu^2/(2\sigma^2) + \log\sigma$ and $h \equiv 1/\sqrt{2\pi}$ also work. If σ^2 is known, then $\theta = \mu$ and the term in x^2 no longer involves the parameter. The result is a one-parameter family on $\mathbb{R}$:

$$\eta = \frac{\mu}{\sigma^2}, \quad T(x) = x, \quad B(\theta) = \frac{\mu^2}{2\sigma^2}, \quad h(x) = \frac{e^{-x^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}.$$

Here h is the $\mathcal{N}(0, \sigma^2)$ density, so $\exp(\eta x - B)$ is the density of $\mathcal{N}(\mu, \sigma^2)$ relative to $\mathcal{N}(0, \sigma^2)$.

Example: *Bernoulli.* For $x \in \{0, 1\}$, write $p^x (1-p)^{1-x}$ in exponential form and collect the terms in x:

$$p^x (1-p)^{1-x} = \exp\big(x \log p + (1-x)\log(1-p)\big) = \exp\Big(x \log \frac{p}{1-p} + \log(1-p)\Big).$$

So $T(x) = x$, $\eta(p) = \log \frac{p}{1-p}$, $B(p) = \log \frac{1}{1-p}$, and $h(x) = 1$. The Poisson PMF $\lambda^x e^{-\lambda}/x!$, $x = 0, 1, \ldots$, is also a discrete exponential family.

Example: *Continuous families.* The Gamma(a, b) density with shape a and scale b,

$$\frac{1}{\Gamma(a) b^a} x^{a-1} e^{-x/b} = \frac{1}{\Gamma(a) b^a} \exp\Big((a-1)\log x - \frac{x}{b}\Big), \qquad x > 0,$$

has $T_1(x) = \log x$, $T_2(x) = x$, $\eta_1 = a - 1$, $\eta_2 = -1/b$. In terms of the mean $\mu = ab$ it reads $\frac{1}{\Gamma(a)} \big(\frac{a}{\mu}\big)^a x^{a-1} e^{-ax/\mu}$. Other members are the inverse Gamma(α, β) density $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}$, the inverse Gaussian$(\mu, \sigma^2)$ density $\sqrt{\sigma^2/(2\pi x^3)}\, e^{-\sigma^2(x-\mu)^2/(2\mu^2 x)}$, and the chi-square, Beta, binomial, and negative binomial distributions.

Example: *A non-member.* The uniform distribution on $[0, \theta]$ is not an exponential family. More generally, a family whose support depends on the parameter is not one. In an exponential family the exponential factor is positive, so the support $\{x : h(x) > 0\}$ is the same for every θ. The indicator $\mathbf{1}\{x \le \theta\}$ is an interaction between x and θ that cannot be written in this form.

21.4 The canonical exponential family

For the response of a GLM we restrict to a one-dimensional parameter and response, and to $T(y) = y$. Any one-dimensional family can be reparametrized so that $\eta(\theta)$ is the parameter. Requiring $T(y) = y$ is a genuine restriction, because the response cannot be transformed.

Definition 21.3 [Canonical exponential family]. A canonical exponential family for $k = 1$, $y \in \mathbb{R}$, has density

$$f_\theta(y) = \exp\Big(\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)\Big)$$

for known functions $b(\cdot)$ and $c(\cdot, \cdot)$. If ϕ is known, this is a one-parameter exponential family with canonical parameter θ. The constant ϕ is the dispersion parameter; if it is unknown, the family may or may not be a two-parameter exponential family. Throughout, $\phi > 0$ is assumed known.

With ϕ known, $e^{c(y,\phi)}$ plays the role of $h(y)$, and the interaction is $y\theta$ scaled by a known constant. Keeping ϕ separate avoids reparametrizing by known quantities, such as using μ/σ^2 in the Gaussian case. Everything specific to the distribution sits in $b(\theta)$. It is called the cumulant generating function, because its successive derivatives give the cumulants of Y, beginning with the mean and the variance.

Example: *Gaussian with known variance.*

$$f_\theta(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(y-\mu)^2/(2\sigma^2)} = \exp\Big\{\frac{y\mu - \frac{1}{2}\mu^2}{\sigma^2} - \frac{1}{2}\Big(\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2)\Big)\Big\},$$

so $\theta = \mu$, $\phi = \sigma^2$, $b(\theta) = \theta^2/2$, and $c(y, \phi) = -\frac{1}{2}\big(y^2/\phi + \log(2\pi\phi)\big)$.

The Bernoulli example above becomes canonical with $\theta = \log \frac{p}{1-p}$. The canonical form of the Poisson distribution follows in the same way.

	Normal	Poisson	Bernoulli
Notation	$\mathcal{N}(\mu, \sigma^2)$	$\mathcal{P}(\mu)$	$\mathcal{B}(p)$
Range of y	$(-\infty, \infty)$	$\{0, 1, 2, \ldots\}$	$\{0, 1\}$
ϕ	σ^2	1	1
$b(\theta)$	$\theta^2/2$	e^θ	$\log(1 + e^\theta)$
$c(y, \phi)$	$-\frac{1}{2}\left(\frac{y^2}{\phi} + \log(2\pi\phi)\right)$	$-\log y!$	0

Table 1: Three canonical exponential families.

Not every exponential family is canonical. In the Gamma family the term $(a-1)\log y$ cannot be removed unless $a = 1$. In that case the density $\frac{1}{b}e^{-y/b}$ is the exponential distribution with rate $\lambda = 1/b$, which is canonical. The range of y shows which range the mean $\mathbb{E}[Y]$ can take, and so indicates which link function is suitable.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 21, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 22: Generalized Linear Models (cont.)

Throughout, Y has a density in the canonical exponential family

$$f_\theta(y) = \exp\Big(\frac{y\theta - b(\theta)}{\phi} + c(y,\phi)\Big),$$

with $\phi > 0$ known. With ϕ fixed, the function b determines the distribution. This lecture extracts the mean, the variance, and a natural link from b, and then writes the likelihood of a GLM in terms of β.

22.1 Two likelihood identities

Let $\ell(\theta) = \log f_\theta(Y)$ be the log-likelihood of one observation.

Lemma 22.1 [Score identities]. *If differentiation in θ and integration in y can be interchanged, then*

$$\mathbb{E}_\theta\Big[\frac{\partial \ell}{\partial \theta}\Big] = 0, \qquad \mathbb{E}_\theta\Big[\frac{\partial^2 \ell}{\partial \theta^2}\Big] + \mathbb{E}_\theta\Big[\Big(\frac{\partial \ell}{\partial \theta}\Big)^2\Big] = 0.$$

Proof. Both follow from $\int f_\theta(y)\,dy \equiv 1$ for every θ (a sum for a PMF), whose derivatives in θ vanish. First, $\partial_\theta \log f_\theta = \partial_\theta f_\theta / f_\theta$, so

$$\mathbb{E}_\theta[\partial_\theta \ell] = \int \frac{\partial_\theta f_\theta(y)}{f_\theta(y)} f_\theta(y)\,dy = \frac{\partial}{\partial\theta}\int f_\theta(y)\,dy = 0.$$

Second,

$$\partial_\theta^2 \log f_\theta = \frac{\partial_\theta^2 f_\theta}{f_\theta} - \Big(\frac{\partial_\theta f_\theta}{f_\theta}\Big)^2.$$

The expectation of the first term is $\int \partial_\theta^2 f_\theta\,dy = \partial_\theta^2 \int f_\theta\,dy = 0$, and the second term is $(\partial_\theta \ell)^2$. □

The second identity gives the two expressions for the Fisher information: $-\mathbb{E}[\ell''] = \mathbb{E}[(\ell')^2] \geq 0$.

22.2 Mean and variance from b

For the canonical family,

$$\ell(\theta) = \frac{Y\theta - b(\theta)}{\phi} + c(Y,\phi), \qquad \frac{\partial \ell}{\partial \theta} = \frac{Y - b'(\theta)}{\phi}, \qquad \frac{\partial^2 \ell}{\partial \theta^2} = -\frac{b''(\theta)}{\phi}.$$

Theorem 22.2 [Mean and variance]. *If Y has density f_θ in the canonical exponential family, then*

$$\boxed{\mathbb{E}[Y] = \mu = b'(\theta)}, \qquad \boxed{\operatorname{Var}(Y) = V(Y) = b''(\theta)\,\phi}.$$

Proof. By the first identity, $0 = \mathbb{E}[\partial_\theta \ell] = (\mathbb{E}[Y] - b'(\theta))/\phi$, so $\mathbb{E}[Y] = b'(\theta)$. Hence $\partial_\theta \ell = (Y - \mathbb{E}[Y])/\phi$, whose square has expectation $\operatorname{Var}(Y)/\phi^2$. The second derivative is deterministic, so the second identity reads

$$0 = -\frac{b''(\theta)}{\phi} + \frac{\operatorname{Var}(Y)}{\phi^2},$$

which gives $\operatorname{Var}(Y) = b''(\theta)\phi$. □

Computing the mean then needs only a derivative of b, instead of integrating y against a density whose normalization may be messy.

Example: *Poisson.* For $Y \sim \mathcal{P}(\mu)$,

$$f(y) = \frac{\mu^y}{y!} e^{-\mu} = \exp\bigl(y \log \mu - \mu - \log y!\bigr).$$

To be canonical, the coefficient of y must be the parameter itself: $\theta = \log \mu$, so $\mu = e^\theta$ and

$$f(y) = \exp\bigl(y\theta - e^\theta - \log y!\bigr), \qquad b(\theta) = e^\theta, \quad \phi = 1, \quad c(y, \phi) = -\log y!.$$

Then $\mathbb{E}[Y] = b'(\theta) = e^\theta = \mu$ and $\operatorname{Var}(Y) = b''(\theta) = e^\theta = \mu$, with no series manipulations.

A known numerical factor in front of y is absorbed into θ. A known but named quantity such as σ^2 is kept in ϕ; when ϕ is known this makes no difference.

Corollary 22.3 [Convexity of b]. *If Y is nondegenerate, then $b''(\theta) = \operatorname{Var}(Y)/\phi > 0$. Hence b is strictly convex and b' is strictly increasing.*

For example, $b(\theta) = \log \theta$ is impossible, since it is concave. A computed b that turns out concave signals a sign error.

22.3 Link functions

A GLM models the conditional mean, which is the regression function $\mu(x) = \mathbb{E}[Y \mid X = x]$. It assumes

$$g\bigl(\mu(X)\bigr) = X^\top \beta,$$

because $X^\top \beta$ ranges over all of $\mathbb{R}$, whereas μ may be confined to $(0, \infty)$ or $(0, 1)$. The parameter of interest is now β, and it enters the likelihood through g.

Definition 22.4 [Link function]. A link function g is continuously differentiable and strictly

increasing, and maps the set of possible means onto all of $\mathbb{R}$. Then g^{-1} is defined on $\mathbb{R}$, is increasing, and

$$\mu = g^{-1}(X^\top \beta).$$

- *Linear model.* g is the identity.
- *Poisson data.* $Y \mid X \sim \mathcal{P}(\mu(X))$ with $\mu(X) > 0$, so g must map $(0, \infty)$ onto $\mathbb{R}$. The log link $\log \mu(X) = X^\top \beta$ is the natural choice, though not the only one ($2 \log$ also works).
- *Bernoulli or binomial data.* $0 < \mu < 1$, so g must map $(0, 1)$ onto $\mathbb{R}$. Equivalently, g^{-1} maps $\mathbb{R}$ into $(0, 1)$, and the inverse of any continuous CDF that is strictly increasing on $\mathbb{R}$ works. Three standard choices:
  1. LOGIT: $\log \dfrac{\mu(X)}{1 - \mu(X)} = X^\top \beta$, with inverse $f_1(x) = e^x/(1 + e^x)$;
  2. PROBIT: $\Phi^{-1}(\mu(X)) = X^\top \beta$, where Φ is the standard normal CDF;
  3. COMPLEMENTARY LOG-LOG: $\log\bigl(-\log(1 - \mu(X))\bigr) = X^\top \beta$.

Logistic regression uses the logit link; with the probit link the method is called probit regression. Neither shape is clearly preferable on the data alone: the choice is part of the modeling assumption, and it can be compared on held-out data, for instance by cross-validation. At $\mu = 1/2$ the logit has slope $\frac{1}{\mu} + \frac{1}{1-\mu} = 4$, while the probit has slope $1/\varphi(0) = \sqrt{2\pi} \approx 2.51$, where $\varphi = \Phi'$.

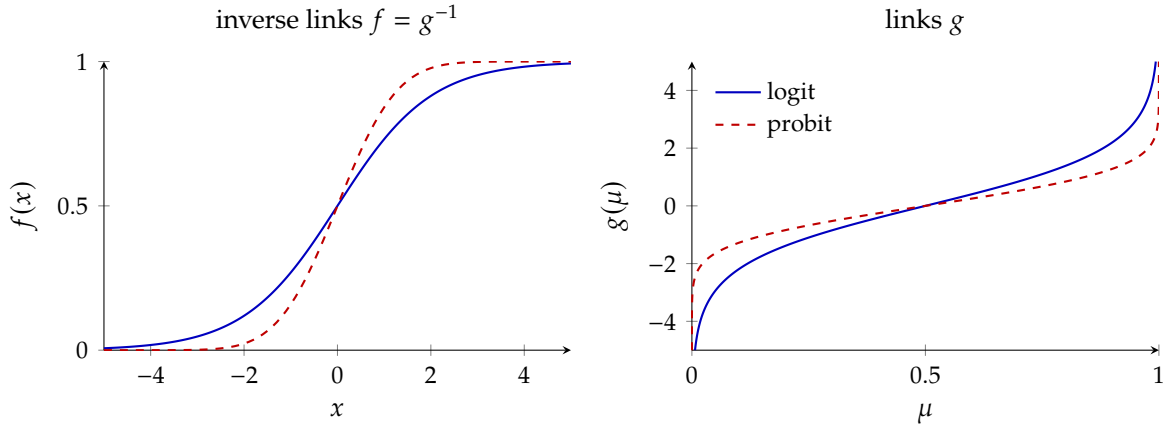


Figure 28: Left: the inverse links $f_1(x) = e^x/(1 + e^x)$ (solid) and $\Phi(x)$ (dashed). Right: the logit link $\log \frac{\mu}{1-\mu}$ (solid) and the probit link $\Phi^{-1}(\mu)$ (dashed), both mapping $(0, 1)$ onto $\mathbb{R}$.

22.4 The canonical link

The canonical parameter θ already ranges over $\mathbb{R}$, so one natural choice is to let θ itself equal $X^\top \beta$.

Definition 22.5 [Canonical link]. The canonical link is the g with $g(\mu) = \theta$. Since $\mu = b'(\theta)$,

$$\boxed{g(\mu) = (b')^{-1}(\mu)}.$$

The canonical link is a valid link when $\phi > 0$. By Corollary 22.3, b' is strictly increasing. It maps $\mathbb{R}$ onto the set of possible means, so its inverse is strictly increasing and maps the means onto $\mathbb{R}$.

Example: *Bernoulli.* Grouping the terms in y,

$$p^y(1-p)^{1-y} = \exp\Big(y \log \frac{p}{1-p} + \log(1-p)\Big).$$

Canonical form requires $\theta = \log \frac{p}{1-p}$, so $p = e^\theta/(1+e^\theta)$ and $\log(1-p) = -\log(1+e^\theta)$. Thus

$$f_\theta(y) = \exp\big(y\theta - \log(1+e^\theta)\big), \qquad b(\theta) = \log(1+e^\theta), \qquad \phi = 1.$$

Then $b'(\theta) = e^\theta/(1+e^\theta) = p$ is the mean, and solving $b'(\theta) = \mu$ gives $\theta = \log \frac{\mu}{1-\mu}$. The canonical link of the Bernoulli distribution is the logit. This could be read off directly once $\theta = \log \frac{p}{1-p}$ and $\mathbb{E}[Y] = p$ are known.

	$b(\theta)$	canonical $g(\mu)$	name
Normal	$\theta^2/2$	μ	identity
Poisson	e^θ	$\log \mu$	log
Bernoulli	$\log(1+e^\theta)$	$\log \frac{\mu}{1-\mu}$	logit
Gamma	$-\log(-\theta)$	$-1/\mu$	reciprocal

Table 2: Canonical links. In the Gaussian case the GLM is the linear model.

So b gives the mean through b', the variance through b'', and the canonical link through $(b')^{-1}$.

22.5 From one observation to a GLM

Three parameters appear: θ, μ, and β. The map $\mu = b'(\theta)$ is invertible, so θ and μ are reparametrizations of each other. The model ties μ to β through g. Once β is fixed, everything else is determined (Figure 29). With infinitely many pairs (x, y) one would know the regression function $\mu(x)$, and hence θ and β. With finitely many, the model reduces the problem to estimating the vector β rather than an entire function.

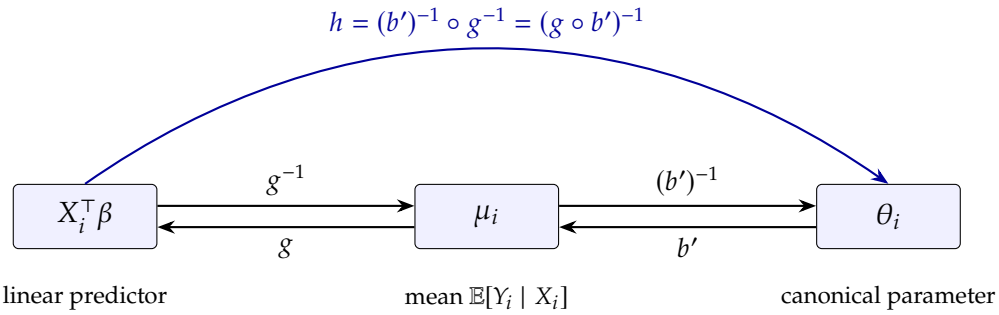


Figure 29: The chain of parameters for observation i. The link connects the linear predictor to the mean, and b' connects the canonical parameter to the mean. Their composition h maps $X_i^\top \beta$ to θ_i.

Observe independent pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ with $X_i \in \mathbb{R}^p$, $Y_i \in \mathbb{R}$. Given $X_i = x_i$, the density of Y_i is

$$f_{\theta_i}(y_i) = \exp\Big(\frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi)\Big), \qquad \mu_i = b'(\theta_i), \qquad g(\mu_i) = x_i^\top \beta.$$

The observations are independent but not identically distributed, since θ_i depends on x_i; the same β links all of them. Inverting both maps,

$$\theta_i = (b')^{-1}(\mu_i) = (b')^{-1}\big(g^{-1}(X_i^\top \beta)\big) = h(X_i^\top \beta), \qquad h = (b')^{-1} \circ g^{-1} = (g \circ b')^{-1}.$$

If g is the canonical link, then $g = (b')^{-1}$, h is the identity, and $\theta_i = X_i^\top \beta$. Up to the constant $\sum_i c(Y_i, \phi)$, the log-likelihood is

$$\ell_n(\beta) = \sum_{i=1}^n \frac{Y_i h(X_i^\top \beta) - b\big(h(X_i^\top \beta)\big)}{\phi}, \qquad \text{canonical link: } \ell_n(\beta) = \sum_{i=1}^n \frac{Y_i X_i^\top \beta - b(X_i^\top \beta)}{\phi}.$$

With the canonical link, ℓ_n is a linear function of β minus a sum of convex functions of β, so it is concave. There is no closed-form maximizer in general, so ℓ_n is maximized iteratively. Each step makes a local quadratic (Newton-Raphson) approximation, which can be carried out as a weighted least-squares problem.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 22, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Generalized Linear Models (cont.)

The setting is the GLM of Lecture 22. Independent pairs $(X_i, Y_i) \in \mathbb{R}^p \times \mathbb{R}$ have

$$\mu_i = \mathbb{E}[Y_i \mid X_i] = b'(\theta_i), \qquad g(\mu_i) = X_i^\top \beta, \qquad \theta_i = h(X_i^\top \beta), \quad h = (g \circ b')^{-1}.$$

Stack the responses into $\mathbf{Y} = (Y_1, \dots, Y_n)^\top \in \mathbb{R}^n$ and the covariates into the design matrix $\mathbb{X} = (X_1^\top, \dots, X_n^\top)^\top \in \mathbb{R}^{n \times p}$. The same β links every pair, so information accumulates as n grows. The MLE maximizes

$$\ell_n(\beta) = \sum_{i=1}^{n} \frac{Y_i h(X_i^\top \beta) - b\big(h(X_i^\top \beta)\big)}{\phi}$$

over β; the omitted term $\sum_i c(Y_i, \phi)$ does not affect the maximizer.

23.1 Strict concavity under the canonical link

Proposition 23.1 [Concavity of the log-likelihood]. *Under the canonical link, with $\phi > 0$ and nondegenerate responses,*

$$\ell_n(\beta) = \frac{1}{\phi} \sum_{i=1}^{n} \big(Y_i X_i^\top \beta - b(X_i^\top \beta)\big)$$

is concave, with Hessian $-\frac{1}{\phi} \sum_i b''(X_i^\top \beta) X_i X_i^\top$. If $\mathbb{X}$ has full column rank, ℓ_n is strictly concave, so the MLE, when it exists, is unique.

Proof. The term $Y_i X_i^\top \beta$ is linear in β and does not affect concavity. By Lecture 22, $b''(\theta) = \mathrm{Var}(Y)/\phi > 0$, so b is strictly convex and $-b(X_i^\top \beta)$ is concave. Differentiating twice,

$$\nabla^2 \ell_n(\beta) = -\frac{1}{\phi} \sum_{i=1}^{n} b''(X_i^\top \beta)\, X_i X_i^\top.$$

For $u \in \mathbb{R}^p$, $u^\top \nabla^2 \ell_n u = -\frac{1}{\phi} \sum_i b''(X_i^\top \beta)(X_i^\top u)^2 \le 0$. The sum is zero only if $X_i^\top u = 0$ for all i, that is, $\mathbb{X}u = 0$, which forces $u = 0$ under full rank. With $p = 1$, this is the statement that the second derivative $-b''(X_i \beta) X_i^2/\phi$ of each term is negative unless $X_i = 0$. □

Strict concavity in $X_i^\top \beta$ alone does not identify β. If, for example, every $X_i = (1, \dots, 1)^\top$, then ℓ_n depends only on $\sum_j \beta_j$. As in linear regression, a condition on $\mathbb{X}$ is needed. With a noncanonical link, ℓ_n need not be strictly concave and may have several local maxima.

23.2 Newton-Raphson

The goal is to find x^* with $f(x^*) \geq f(x)$ for all $x \in \mathcal{X} \subset \mathbb{R}^m$. Three methods follow, from fully general to specific to GLMs: Newton-Raphson, Fisher scoring, and iteratively reweighted least squares.

Let $f : \mathbb{R}^m \to \mathbb{R}$ have two continuous derivatives. Its gradient and Hessian at x_0 are

$$\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_m}\right)^\top, \qquad (H_f)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j},$$

and the Hessian is a symmetric $m \times m$ matrix. The map $x \mapsto H_f(x)$ is the Hessian map. If f is strictly concave, $H_f(x)$ is negative definite: $u^\top H_f(x) u < 0$ for all $u \neq 0$.

Near x_0, the second-order Taylor expansion gives

$$f(x) \approx f(x_0) + \nabla f(x_0)^\top (x - x_0) + \tfrac{1}{2}(x - x_0)^\top H_f(x_0)(x - x_0).$$

Setting $\nabla f = 0$ directly requires inverting ∇f, which is rarely possible. A concave quadratic, however, has an explicit maximizer. The gradient of the right-hand side is $\nabla f(x_0) + H_f(x_0)(x - x_0)$, and setting it to zero gives the linear system $H_f(x_0) x^* = H_f(x_0) x_0 - \nabla f(x_0)$, so

$$x^* = x_0 - H_f(x_0)^{-1} \nabla f(x_0).$$

Definition 23.2 [Newton-Raphson method]. Starting from an arbitrary $x^{(0)}$, iterate

$$x^{(k+1)} = x^{(k)} - H_f(x^{(k)})^{-1} \nabla f(x^{(k)}),$$

so that $x^{(k+1)}$ maximizes the quadratic approximation of f at $x^{(k)}$.

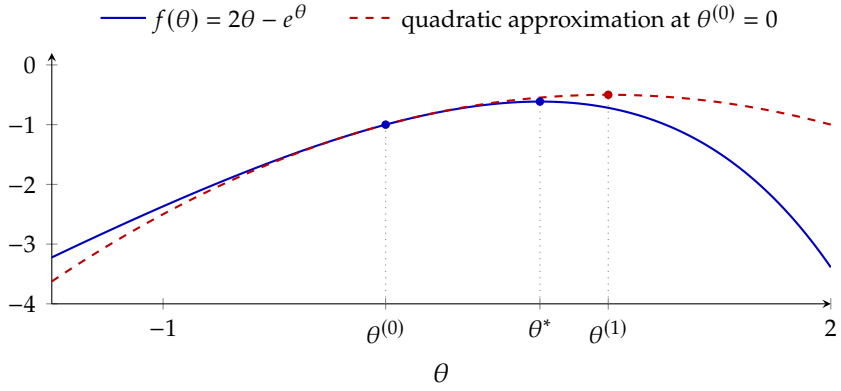


Figure 30: One Newton step for the concave function $f(\theta) = 2\theta - e^\theta$ (a Poisson log-likelihood with $y = 2$). From $\theta^{(0)} = 0$, the quadratic approximation $-1 + \theta - \theta^2/2$ is maximized at $\theta^{(1)} = 1$; the maximizer of f is $\theta^* = \log 2$.

For concave f the Hessian is negative, so $-H_f^{-1}$ acts like a positive step size. The update is gradient

ascent with the direction rescaled by the local geometry encoded in the Hessian. In large problems, inverting H_f is too costly, and quasi-Newton methods approximate it, for example by a diagonal matrix or from past gradients. Under mild conditions on f, which hold for the log-likelihoods here, the method converges from any starting point and, near the maximizer, at a quadratic rate: each step roughly doubles the number of correct digits (stated without proof).

23.3 Fisher scoring

Newton-Raphson ignores that ℓ_n is a log-likelihood, and its Hessian can be complicated. Maximum likelihood minimizes an empirical version of the KL divergence, obtained by replacing an expectation with an average. For independent observations,

$$\mathbb{E}_\theta\Big[\frac{1}{n}\sum_{i=1}^{n}\log f_{\theta_i}(Y_i)\Big] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_\theta\big[\log f_{\theta_i}(Y_i)\big] = -\mathrm{KL} + \text{const},$$

where KL is the divergence between the products of the true and the model distributions (identically distributed or not). The Hessian can therefore be replaced by its expectation, computed as if the current iterate were the true parameter. By the second identity of Lecture 22, $\mathbb{E}_\theta[H_{\ell_n}(\theta)] = -I(\theta)$, the negative Fisher information.

Definition 23.3 [Fisher scoring]. The Fisher-scoring algorithm iterates

$$\theta^{(k+1)} = \theta^{(k)} + I(\theta^{(k)})^{-1}\nabla\ell_n(\theta^{(k)}).$$

The sign is + because $I = -\mathbb{E}[H]$ is positive definite. The method has essentially the same convergence properties as Newton-Raphson, and I is often easier to compute than H_{ℓ_n}. The Hessian of a product uv has two terms, $u'v + uv'$. In a GLM one of them has conditional expectation zero, so the expected Hessian is shorter. Taking the expectation adds no information, since it pretends that the current iterate is the truth; the gain is only computational.

23.4 Example: logistic regression

Let $Y_i \mid X_i \sim \text{Bernoulli}(p_i)$ independently, with $b(\theta) = \log(1 + e^\theta)$, $\phi = 1$, and the canonical link $\theta_i = \log\frac{p_i}{1-p_i} = X_i^\top\beta$. Then

$$\ell_n(\beta) = \sum_{i=1}^{n}\Big(Y_i X_i^\top\beta - \log\big(1 + e^{X_i^\top\beta}\big)\Big).$$

By the chain rule, with $\nabla(X_i^\top\beta) = X_i$,

$$\nabla\ell_n(\beta) = \sum_{i=1}^{n}\Big(Y_i - \frac{e^{X_i^\top\beta}}{1 + e^{X_i^\top\beta}}\Big)X_i, \qquad H_{\ell_n}(\beta) = -\sum_{i=1}^{n}\frac{e^{X_i^\top\beta}}{\big(1 + e^{X_i^\top\beta}\big)^2}X_i X_i^\top,$$

using $\frac{d}{dx}\frac{e^x}{1+e^x} = \frac{e^x}{(1+e^x)^2}$, and the outer product $X_i X_i^\top$ to form a matrix. The Newton-Raphson update is

$$\beta^{(k+1)} = \beta^{(k)} - H_{\ell_n}(\beta^{(k)})^{-1}\nabla \ell_n(\beta^{(k)}),$$

which requires H_{ℓ_n} to be invertible (typically true once n is large); in practice one solves the linear system.

The score is a weighted combination of the X_i, and the Hessian is a weighted combination of the $X_i X_i^\top$. This is typical of GLMs, because every derivative of $X_i^\top \beta$ produces X_i. The Hessian is negative definite, so there is a unique local maximizer, which is global. Moreover, Y_i does not appear in H_{ℓ_n}. Conditionally on the X_i, the Hessian is deterministic, so $H_{\ell_n}(\beta) = \mathbb{E}[H_{\ell_n}(\beta)] = -I(\beta)$, and Fisher scoring coincides with Newton-Raphson. The same holds for every canonical link: Y_i appears only in the linear term $Y_i X_i^\top \beta$, which vanishes after two derivatives. With a noncanonical h, the term $Y_i h(X_i^\top \beta)$ keeps Y_i in the Hessian.

23.5 Toward IRLS: the gradient

Iteratively reweighted least squares (IRLS) carries out the Newton-Raphson or Fisher-scoring step with a weighted least-squares solver. Solving $Ax = b$ is the same as minimizing $\|Ax - b\|^2$, so each linear system can be posed as a least-squares problem. Write

$$\ell_n = \sum_{i=1}^n \frac{Y_i\theta_i - b(\theta_i)}{\phi} + c(Y_i, \phi), \qquad \mu_i = b'(\theta_i), \quad X_i^\top \beta = g(\mu_i), \quad \frac{d\mu_i}{d\theta_i} = b''(\theta_i) \equiv V_i,$$

and $\theta_i = h(X_i^\top \beta)$; for $\phi = 1$, $V_i = \mathrm{Var}(Y_i)$. By the chain rule,

$$\frac{\partial \ell_n}{\partial \beta_j} = \sum_{i=1}^n \frac{\partial \ell_i}{\partial \theta_i}\frac{\partial \theta_i}{\partial \beta_j} = \sum_{i=1}^n \frac{Y_i - \mu_i}{\phi} h'(X_i^\top \beta) X_{ij} = \sum_{i=1}^n (\tilde{Y}_i - \tilde{\mu}_i) W_i X_{ij},$$

where the second step multiplies and divides by $g'(\mu_i)$, and

$$W_i = \frac{h'(X_i^\top \beta)}{g'(\mu_i)\,\phi}, \qquad \tilde{Y}_i = g'(\mu_i) Y_i, \qquad \tilde{\mu}_i = g'(\mu_i)\mu_i.$$

With $W = \mathrm{diag}(W_1, \dots, W_n)$, the vector form is

$$\boxed{\nabla \ell_n(\beta) = \mathbb{X}^\top W(\tilde{\mathbf{Y}} - \tilde{\mu})}.$$

The factor $g'(\mu_i)$ looks artificial here; it makes the same W reappear in the expected Hessian.

Source. Philippe Rigollet, MIT 18.650, *Statistics for Applications*, Lecture 23, Fall 2016. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Generalized Linear Models (cont.)

The Fisher-scoring iteration $\beta^{(k+1)} = \beta^{(k)} + I(\beta^{(k)})^{-1}\nabla\ell_n(\beta^{(k)})$ is a gradient-ascent step, since I is positive semidefinite. This lecture computes $I(\beta)$ and shows that each step is a weighted least-squares problem. In the notation of Lecture 23, $\mu_i = b'(\theta_i)$, $g(\mu_i) = X_i^\top\beta$, $\theta_i = h(X_i^\top\beta)$ with $h = (g \circ b')^{-1}$, and

$$\frac{\partial \ell_n}{\partial \beta_j} = \sum_{i=1}^{n} \frac{Y_i - \mu_i}{\phi}\, h'(X_i^\top\beta) X_{ij}, \qquad \nabla\ell_n(\beta) = \mathbb{X}^\top W(\tilde{\mathbf{Y}} - \tilde{\mu}),$$

where $W = \operatorname{diag}(W_i)$, $W_i = h'(X_i^\top\beta)/\big(g'(\mu_i)\phi\big)$, $\tilde{Y}_i = g'(\mu_i)Y_i$, and $\tilde{\mu}_i = g'(\mu_i)\mu_i$.

24.1 The expected Hessian

Proposition 24.1 [Fisher information of a GLM]. *Conditionally on $X_1, \dots, X_n$,*

$$H_{\ell_n}(\beta) = \frac{1}{\phi}\sum_{i=1}^{n}\Big[(Y_i - \mu_i)h''(X_i^\top\beta) - \frac{h'(X_i^\top\beta)}{g'(\mu_i)}\Big]X_iX_i^\top,$$

and therefore

$$\boxed{I(\beta) = -\mathbb{E}\big[H_{\ell_n}(\beta)\big] = \mathbb{X}^\top W\mathbb{X}}, \qquad W_i = \frac{h'(X_i^\top\beta)}{g'(\mu_i)\,\phi}.$$

Proof. Differentiate $\partial\ell_n/\partial\beta_j$ in β_k; both μ_i and $h'(X_i^\top\beta)$ depend on β, the data do not. Since $\mu_i = g^{-1}(X_i^\top\beta)$ and $(g^{-1})' = 1/(g' \circ g^{-1})$,

$$\frac{\partial \mu_i}{\partial \beta_k} = \frac{X_{ik}}{g'(\mu_i)}.$$

By the product rule,

$$\frac{\partial^2 \ell_n}{\partial\beta_j\partial\beta_k} = \frac{1}{\phi}\sum_{i=1}^{n}\Big[-\frac{h'(X_i^\top\beta)}{g'(\mu_i)} + (Y_i - \mu_i)h''(X_i^\top\beta)\Big]X_{ij}X_{ik},$$

and $X_{ij}X_{ik}$ is the (j,k) entry of $X_iX_i^\top$. Taking the conditional expectation given the X_i removes the second term, because $\mathbb{E}[Y_i - \mu_i \mid X_i] = 0$ and the rest depends on X_i only, leaving $-\mathbb{X}^\top W\mathbb{X}$. □

The same weights arise from the chain rule applied to $g^{-1} = b' \circ h$: with $\eta = X_i^\top\beta$, $b''(\theta_i)h'(\eta) =$

$d\mu_i/d\eta = 1/g'(\mu_i)$, so

$$\mathbb{E}[H_{\ell_n}] = -\frac{1}{\phi}\sum_{i=1}^{n} b''(\theta_i)\, h'(X_i^\top\beta)^2 X_i X_i^\top = -\sum_{i=1}^{n} \frac{h'(X_i^\top\beta)}{g'(\mu_i)\,\phi} X_i X_i^\top.$$

Thus the factor $g'(\mu_i)$, which was inserted into the gradient by multiplying and dividing, appears naturally in the Hessian. It was placed in the gradient so that the gradient and the information share the same W. For logistic regression, h is the identity and $g'(\mu) = 1/\big(\mu(1-\mu)\big)$. Then $W_i = p_i(1-p_i)$, and $\mathbb{X}^\top W \mathbb{X}$ is minus the Hessian of Lecture 23.

Corollary 24.2 [Canonical link]. *With the canonical link, h is the identity and $h'' = 0$. The Hessian is then already deterministic, $H_{\ell_n} = -\mathbb{X}^\top W \mathbb{X}$, and Fisher scoring coincides with Newton-Raphson.*

Write $W^{(k)}$, $\mu_i^{(k)} = g^{-1}(X_i^\top \beta^{(k)})$, $\tilde{\mathbf{Y}}^{(k)}$, $\tilde{\mu}^{(k)}$ for these quantities at $\beta^{(k)}$. The Fisher-scoring update becomes

$$\beta^{(k+1)} = \beta^{(k)} + \big(\mathbb{X}^\top W^{(k)} \mathbb{X}\big)^{-1} \mathbb{X}^\top W^{(k)} \big(\tilde{\mathbf{Y}}^{(k)} - \tilde{\mu}^{(k)}\big) = \big(\mathbb{X}^\top W^{(k)} \mathbb{X}\big)^{-1} \mathbb{X}^\top W^{(k)} \big(\tilde{\mathbf{Y}}^{(k)} - \tilde{\mu}^{(k)} + \mathbb{X}\beta^{(k)}\big).$$

With $W^{(k)} = I_n$, this would be the least-squares formula $(\mathbb{X}^\top\mathbb{X})^{-1}\mathbb{X}^\top\mathbf{Y}$ applied to a modified response.

24.2 Weighted least squares

Definition 24.3 [Heteroskedastic linear model]. In the linear model $\mathbf{Y} = \mathbb{X}\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}_n(0, W^{-1})$ and W^{-1} a known diagonal matrix with positive, finite entries, the noise variances differ across observations. The regression is then called heteroskedastic.

Observations with small variance should count more. If a line is fitted to points that are very noisy on one side and precise on the other, ordinary least squares follows the noisy points too much. The Gaussian log-likelihood is

$$-\frac{1}{2}(\mathbf{Y} - \mathbb{X}\beta)^\top W (\mathbf{Y} - \mathbb{X}\beta) - \frac{1}{2}\log\det\big(2\pi W^{-1}\big),$$

and with W known, the second term is a constant. For $n = 1$ the exponent is $-(y-\mu)^2/(2\sigma^2)$ and the normalization is $\sigma\sqrt{2\pi}$, as usual. So the MLE solves the weighted least squares problem

$$\min_{\beta} (\mathbf{Y} - \mathbb{X}\beta)^\top W (\mathbf{Y} - \mathbb{X}\beta).$$

Proposition 24.4 [Weighted least-squares solution]. *If $\mathbb{X}^\top W \mathbb{X}$ is invertible, the minimizer is*

$$\boxed{\hat{\beta} = (\mathbb{X}^\top W \mathbb{X})^{-1} \mathbb{X}^\top W \mathbf{Y}}.$$

Proof. Let $W^{1/2} = \operatorname{diag}(\sqrt{W_i})$, so that $W = W^{1/2}W^{1/2}$ and $W^{1/2}$ is symmetric. The objective is $\|\mathbf{Y}' - \mathbb{X}'\beta\|^2$ with $\mathbf{Y}' = W^{1/2}\mathbf{Y}$ and $\mathbb{X}' = W^{1/2}\mathbb{X}$. By ordinary least squares, the minimizer is $(\mathbb{X}'^{\top}\mathbb{X}')^{-1}\mathbb{X}'^{\top}\mathbf{Y}' = (\mathbb{X}^{\top}W\mathbb{X})^{-1}\mathbb{X}^{\top}W\mathbf{Y}$. □

The argument works for any positive definite W, using a matrix square root. In particular, the diagonal assumption is not needed. For a general covariance Σ of correlated observations, $W = \Sigma^{-1}$. Weighted least squares is routinely implemented in statistical software.

24.3 Iteratively reweighted least squares

The Fisher-scoring step has exactly the form of Proposition 24.4. It is a weighted least-squares fit with weight matrix $W^{(k)}$ and design $\mathbb{X}$. The response can be written in two equivalent ways. One option is to regress $\tilde{\mathbf{Y}}^{(k)} - \tilde{\mu}^{(k)}$ on $\mathbb{X}$ and add the solution $\hat{u}^{(k+1)}$ to $\beta^{(k)}$. The other is to regress the ADJUSTED DEPENDENT RESPONSE

$$Z_i^{(k)} = X_i^{\top}\beta^{(k)} + g'(\mu_i^{(k)})\big(Y_i - \mu_i^{(k)}\big)$$

directly, which gives $\beta^{(k+1)}$ in one step. Any weighted least-squares routine therefore computes the MLE.

Algorithm 24.1: Iteratively reweighted least squares (IRLS)

Input: Design $\mathbb{X}$, responses $\mathbf{Y}$, link g, $h = (g \circ b')^{-1}$, dispersion ϕ, initial $\beta^{(0)}$, tolerance ε.
Output: Approximate MLE $\hat{\beta}$.

for $k = 0, 1, 2, \ldots$ **do**
  $\mu_i^{(k)} \leftarrow g^{-1}(X_i^{\top}\beta^{(k)})$ for $i = 1, \ldots, n$
  $Z_i^{(k)} \leftarrow X_i^{\top}\beta^{(k)} + g'(\mu_i^{(k)})\,(Y_i - \mu_i^{(k)})$
  $W^{(k)} \leftarrow \operatorname{diag}\Big(\dfrac{h'(X_i^{\top}\beta^{(k)})}{g'(\mu_i^{(k)})\,\phi}\Big)$
  $\beta^{(k+1)} \leftarrow \big(\mathbb{X}^{\top}W^{(k)}\mathbb{X}\big)^{-1}\mathbb{X}^{\top}W^{(k)}\mathbf{Z}^{(k)}$ *// weighted LS of $\mathbf{Z}^{(k)}$ on $\mathbb{X}$*
  if $\|\beta^{(k+1)} - \beta^{(k)}\|_2 < \varepsilon$ **then**
    return $\beta^{(k+1)}$

The procedure needs only $\mathbb{X}$, $\mathbf{Y}$, the link g, the function b (through h), and ϕ. Each iteration recomputes the means $\mu_i^{(k)}$, then from them the weights and the adjusted responses, and calls the weighted least-squares solver.

- *Stopping.* The iterates converge but never reach the limit exactly. Machine precision is unnecessary, since the estimate has statistical error anyway. A rule of thumb with n observations is $\varepsilon = 1/n$.
- *Starting value.* Starting close to the solution helps. The observations are the best available guesses of their own means, so take $\mu^{(0)} = \mathbf{Y}$. Then $g(\mu^{(0)}) \approx \mathbb{X}\beta$, and the least-squares solution of $g(\mathbf{Y})$ on $\mathbb{X}$ gives $\beta^{(0)}$; this requires $g(Y_i)$ to be finite.

- *Cost.* Fisher scoring computed directly forms and inverts the full information matrix at every step. IRLS updates only the n diagonal weights and two vectors and then calls a highly optimized least-squares solver, so it is one of the fastest ways to fit a GLM, especially for large problems.
- *Canonical link.* By Corollary 24.2, IRLS then performs exact Newton-Raphson steps, and the weights reduce to $W_i = 1/\big(g'(\mu_i)\phi\big) = b''(\theta_i)/\phi$.

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